

Workbook

Chapters 0–5

Minimal theory, taken from the book's own definitions, strategies and results, followed immediately by the book's own exercises: every exercise of Chapters 0–5 (sections 0.1–5.2 and the chapter exercises 0.E–5.E) is either worked in the text or solved in Appendix S.

- **Source.** C. Newstead, *An Infinite Descent into Pure Mathematics*, Utrecht adaptation (version of 29 July 2026), pages 1–180. All numbers (Definition 0.36, Strategy 1.1.28, Exercise 3.2.40, Theorem 5.2.26, ...) are the book's numbers; nothing here comes from outside the book except the standard algebra and calculus the book itself takes for granted.
- **Which exercises are worked where.** Exercises flagged in the course planning (★ essential, ★ recommended, bonus) are worked in full in the chapter text. All other book exercises (◦) appear in NOW YOU boxes and are solved in Appendix S, so that you can attempt *any* question in the book and check your work.
- **How to use it.** Read one theory box, cover the solution of the worked exercise that follows, attempt it, then compare. Do the NOW YOU exercises before looking at Appendix S. The one-page toolbox in Appendix T lists the first line of every kind of proof.

How this workbook works

Every section is a sequence of short cycles: **one piece of theory, then the exercises that use it.**

Definition n.m

The book's definition, condensed. Learn it word for word: every proof starts by unfolding one of these.

Theorem / Proposition / Axiom

A result you may quote by number in a proof, with the idea of the book's proof.

Strategy n.m

The book's proof strategies. Each one is a rule for what to write next.

Exercise n.m badge

A book exercise worked in full, in the style the book uses; the badge gives its planning status. Cover the solution and try first.

NOW YOU

The remaining book exercises for that piece of theory. Do them; every one is solved in Appendix S.

Watch out

The book's own warnings ("Common error", asides) and the places where marks are lost.

Flags from the course planning

★ = **essential** exercise in the planning ★ = **recommended** ◦ = in the book but not in the planning (still solved here). Section 4.1 (Peano's axioms) is not in the planning and is covered briefly; the polynomial and complex-number part of Chapter 0 is outside the course but its exercises are solved.

Quiz 1 (23 Sep): numbers and §1.1. Quiz 2 (7 Oct): sets and functions. Quiz 3 (21 Oct): relations and modular arithmetic. Exam 4 Nov.

Conventions of this book

- $\mathbb{N} = \{0, 1, 2, \dots\}$: 0 is a natural number (Definition 0.6). $[n] = \{k \in \mathbb{N} \mid k < n\}$ (Definition 0.25), so $[0] = \emptyset$ and $[3] = \{0, 1, 2\}$.
- Truth values in truth tables are $\checkmark$ (true) and $\times$ (false), as in the book.
- “ b divides a ” means $a = qb$ for some $q \in \mathbb{Z}$ (Definition 0.36); it is about multiplication, never about the fraction a/b .
- “Odd” means “not even” (Definition 0.41); that a is odd iff $a = 2b + 1$ is a *proposition* (1.1.58), not the definition.
- $f[U]$ and $f^{-1}[V]$ are images and preimages (Definitions 3.1.31, 3.1.37); $[a]_{\sim}$ and $[a]_n$ are equivalence and congruence classes (Definitions 5.2.12, 5.2.18).
- $\square$ ends a proof. {grey text in braces} names the definition, strategy or rule used for a step.

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Chapter 0

Getting started

Contents

- A** Propositions, proofs, number sets
- B** Sets, notation, intervals
- C** Number bases
- D** Division of integers
- E** Irrational numbers
- F** Polynomials and complex numbers ◦
- G** Chapter 0 exercises (0.E)

Planning

Planning (Mon 7 Sep): Chapter 0 up to and including irrational numbers.

★ 0.2, 0.21, 0.24, 0.40; 0.E.3, 4, 22, 27, 28

★ 0.14, 0.30, 0.38, 0.45, 0.51; 0.E.17, 18, 29, 30

Section F and the 0.E questions on polynomials, complex and algebraic numbers are not in the planning.

A PROPOSITIONS, PROOFS, NUMBER SETS

Definitions 0.1 and 0.3 · proposition, proof

A **proposition** is a statement to which it makes sense to assign a truth value (true or false, not both). A **proof** of a true proposition is an argument that demonstrates its truth: it starts from what is known or assumed, uses agreed logical steps, and can be verified by its intended reader. Results are called propositions, theorems (key results), lemmas (stepping stones) or corollaries (immediate consequences).

NOW YOU 0.2 ★ · solutions in Appendix S

0.2 Give (a) a true proposition, (b) a false one, (c) one whose truth value you don't know.

Definitions 0.4–0.10 · the number line and the number types

Real numbers are the points of the number line (0.4). **Natural numbers** $0, 1, 2, \dots$ are reached from 0 by unit steps to the right (0.6); **integers** by unit steps in either direction (0.8).

Closure (Assumptions 0.5, 0.7, 0.9): $\mathbb{R}$ is closed under $+$, $-$, $\times$ and $\div$ by non-zero; $\mathbb{N}$ under $+$ and $\times$ only; $\mathbb{Z}$ under $+$, $-$, $\times$. Between an integer n and $n + 1$ there is no integer.

Floor and ceiling (0.10): for $x \in \mathbb{R}$, $\lfloor x \rfloor$ is the integer n with $n \leq x < n + 1$; $\lceil x \rceil$ is the integer n with $n - 1 < x \leq n$.

Proposition 0.12

Let n be an integer. Then $\lfloor n \rfloor = n = \lceil n \rceil$. *Proof idea:* n itself satisfies $n \leq n < n + 1$, and the floor is the (only) integer with that property.

The first proofs in the book all have the same shape: *name the integer that the definition asks for, then check the two inequalities.*

Exercise 0.13

from the book

Suppose that x is a real number that is not an integer. Prove that $\lceil x \rceil = \lfloor x \rfloor + 1$.

Let $n = \lfloor x \rfloor$. By Definition 0.10, n is an integer with $n \leq x < n + 1$. Since x is not an integer, $x \neq n$, so in fact $n < x < n + 1$.

Now $n + 1$ is an integer, and $(n + 1) - 1 < x \leq n + 1$. By Definition 0.10, $\lceil x \rceil$ is the integer m with $m - 1 < x \leq m$; so $\lceil x \rceil = n + 1 = \lfloor x \rfloor + 1$. $\square$

Exercise 0.14

recommended

Suppose that x is a real number. Prove that $\lfloor -x \rfloor = -\lceil x \rceil$ and $\lceil -x \rceil = -\lfloor x \rfloor$.

Let $n = \lceil x \rceil$, so $n \in \mathbb{Z}$ and $n - 1 < x \leq n$ (Definition 0.10). Multiplying through by -1 reverses the inequalities: $-n \leq -x < -n + 1 = (-n) + 1$. So $-n$ is an integer m with $m \leq -x < m + 1$, which is exactly the defining property of $\lfloor -x \rfloor$. Hence $\lfloor -x \rfloor = -n = -\lceil x \rceil$.

Similarly let $n = \lfloor x \rfloor$, so $n \leq x < n + 1$. Then $-n - 1 < -x \leq -n$, i.e. $(-n) - 1 < -x \leq -n$, so $\lceil -x \rceil = -n = -\lfloor x \rfloor$. $\square$

Definition 0.15 · rational number

$x \in \mathbb{R}$ is **rational** if there are integers a, b with $b \neq 0$ and $x = \frac{a}{b}$.

Theorem 0.16 · closure of $\mathbb{Q}$

If $x, y \in \mathbb{Q}$ then $x + y, x - y, xy \in \mathbb{Q}$, and $x/y \in \mathbb{Q}$ if $y \neq 0$.

Proof pattern (the book proves $x + y$): write $x = \frac{a}{b}, y = \frac{c}{d}$ with $b, d \neq 0$; compute; check that numerator and denominator are integers and the denominator is non-zero.

Exercise 0.17

from the book

Complete the proof of Theorem 0.16.

Let $x = \frac{a}{b}$ and $y = \frac{c}{d}$ with $a, b, c, d \in \mathbb{Z}$ and $b, d \neq 0$ (Definition 0.15).

$x - y = \frac{ad - bc}{bd}$: here $ad - bc \in \mathbb{Z}$ and $bd \in \mathbb{Z}$ by closure of $\mathbb{Z}$ (Assumption 0.9), and $bd \neq 0$ since $b \neq 0$ and $d \neq 0$. So $x - y \in \mathbb{Q}$.

$xy = \frac{ac}{bd}$: $ac, bd \in \mathbb{Z}$ and $bd \neq 0$. So $xy \in \mathbb{Q}$.

If $y \neq 0$ then $c \neq 0$ (otherwise $y = 0$), and $\frac{x}{y} = \frac{a}{b} \cdot \frac{d}{c} = \frac{ad}{bc}$ with $ad, bc \in \mathbb{Z}$ and $bc \neq 0$. So $x/y \in \mathbb{Q}$. $\square$

B SETS, NOTATION, INTERVALS
Definitions 0.18, 0.19 and the two notations

A **set** is a collection of objects, its **elements**; $x \in X$ asserts that x is an element of X (revised in §2.1). The **number sets** are $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ (and $\mathbb{C}$, 0.62).

List notation $\{0, 1, 2, 3\}$, implied lists $\{0, 1, 4, 9, \dots\}$.

Set-builder notation. $a \in \{x \mid p(x)\}$ iff $p(a)$ is true. Two variants:

- $\{x \in X \mid p(x)\} = \{x \mid x \in X \text{ and } p(x)\}$: the elements of X satisfying p ;

- $\{\text{expr}(x) \mid p(x)\}$: all objects of the form $\text{expr}(x)$ for some x satisfying p . E.g. $\{n^2 \mid n \in \mathbb{N}\} = \{0, 1, 4, 9, \dots\}$.

Exercise 0.21**essential**

Determine the truth values: (a) $2 \in \mathbb{N}$; (b) $-2 \in \mathbb{N}$; (c) $-2 \in \mathbb{Q}$; (d) $\frac{2}{5} \in \mathbb{Z}$; (e) $\frac{\sqrt{8} + \sqrt{2}}{\sqrt{8} - \sqrt{2}} \in \mathbb{Q}$; (f) $\pi^\pi \in \mathbb{Q}$; (g) $\sqrt{-7} \in \mathbb{R}$; (h) $\mathbb{N} \in \mathbb{Z}$.

(a) True: 2 is a non-negative whole number. (b) False: $-2 < 0$. (c) True: $-2 = \frac{-2}{1}$ with $-2, 1 \in \mathbb{Z}$, $1 \neq 0$. (d) False: $0 < \frac{2}{5} < 1$ and there is no integer strictly between 0 and 1. (e) True: $\sqrt{8} = 2\sqrt{2}$, so the number is $\frac{3\sqrt{2}}{\sqrt{2}} = 3 = \frac{3}{1} \in \mathbb{Q}$. (f) This is a proposition, but its truth value is not known: nobody has proved π^π rational or irrational. (g) False: $\sqrt{-7} = \sqrt{7}i$ is not a point of the real number line (0.62). (h) False: the elements of $\mathbb{Z}$ are numbers, and $\mathbb{N}$ is a set, not a number. $\square$

Exercise 0.24**essential**

A dyadic rational is a rational number that can be expressed as an integer divided by a power of 2. Express the set of all dyadic rationals in set-builder notation in at least two ways.

Filter form: $\{x \in \mathbb{Q} \mid \exists a \in \mathbb{Z}, \exists n \in \mathbb{N}, x = \frac{a}{2^n}\}$. Expression form: $\{\frac{a}{2^n} \mid a \in \mathbb{Z}, n \in \mathbb{N}\}$.

A third: $\{x \in \mathbb{R} \mid 2^n x \in \mathbb{Z} \text{ for some } n \in \mathbb{N}\}$. $\square$

Definitions 0.25 and 0.28 · $[n]$ and intervals

For $n \in \mathbb{N}$, $[n] = \{k \in \mathbb{N} \mid k < n\}$, so $[0] = \emptyset$, $[1] = \{0\}$, $[3] = \{0, 1, 2\}$: $[n]$ has n elements.

For $a < b$: $(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$, $[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$, $[a, b)$, $(a, b]$ likewise; $(-\infty, b) = \{x \in \mathbb{R} \mid x < b\}$, $[a, \infty) = \{x \in \mathbb{R} \mid x \geq a\}$, etc.; $(-\infty, \infty) = \mathbb{R}$. Intervals consist of *real* numbers: $\{x \in \mathbb{Z} \mid 2 \leq x \leq 5\}$ is not $[2, 5]$.

Exercise 0.27**from the book**

Let $A = \{x \in [10] \mid x \text{ is a power of } 2\}$, $B = \{2x \mid x \in [10]\}$ and $C = \{-99, -97, \dots, 97, 99\}$. (a) List A . (b) List B . (c) Write C in set-builder notation using some $[n]$.

(a) $[10] = \{0, 1, \dots, 9\}$; the powers of 2 in it are 1, 2, 4, 8. So $A = \{1, 2, 4, 8\}$.

(b) $B = \{0, 2, 4, 6, 8, 10, 12, 14, 16, 18\}$.

(c) The elements are $2k - 99$ for $k = 0, 1, \dots, 99$: $C = \{2k - 99 \mid k \in [100]\}$. $\square$

Exercise 0.30**recommended**

Express each illustrated set in the most appropriate notation. (The figures show, on a number line: (a) filled dot at -2 , hollow at 4 , segment between; (b) filled dots at -2 and 4 , segment between; (c) filled dots at $-2, -1, 0, 1, 2, 3, 4$ only; (d) hollow dot at 4 , shaded to the left without end; (e) filled dot at -2 , shaded to the right without end; (f) the whole line shaded.)

(a) $[-2, 4)$. (b) $[-2, 4]$. (c) $\{-2, -1, 0, 1, 2, 3, 4\} = \{n \in \mathbb{Z} \mid -2 \leq n \leq 4\}$ (not an interval: only integers). (d) $(-\infty, 4)$. (e) $[-2, \infty)$. (f) $(-\infty, \infty) = \mathbb{R}$. $\square$

C NUMBER BASES

Definition 0.31, Notation 0.34 · base- b expansion

Let $b > 1$. The **base- b expansion** of $n \in \mathbb{N}$ is the string $d_r d_{r-1} \dots d_0$ with (i) $n = d_r b^r + \dots + d_1 b + d_0$; (ii) $0 \leq d_i < b$; (iii) $d_r \neq 0$ if $n > 0$ (no leading zeros; the expansion of 0 is 0). We write $d_r \dots d_0(b)$ for the number with that expansion; no subscript means base 10. Digits beyond 9: $A = 10, B = 11, \dots$

The algorithm (page 17): repeated division

Divide n by b : the remainder is d_0 , the quotient is n_0 . Divide n_0 by b : remainder d_1 , quotient n_1 . Repeat until the quotient is 0. The expansion is $d_i d_{i-1} \dots d_0$ (last remainder first). Check by evaluating $\sum d_k b^k$.

Exercise 0.33

from the book

Find the binary, ternary, octal, decimal, hexadecimal and base-36 expansions of 21127.

Repeated division (for base 16: $21127 = 1320 \cdot 16 + 7$, $1320 = 82 \cdot 16 + 8$, $82 = 5 \cdot 16 + 2$, $5 = 0 \cdot 16 + 5$, read upwards: 5287). The others the same way:

$$21127 = 101001010000111_{(2)} = 1001222111_{(3)} = 51207_{(8)} = 21127_{(10)} = 5287_{(16)} = GAV_{(36)}.$$

Check of the last one: $G = 16, A = 10, V = 31$: $16 \cdot 36^2 + 10 \cdot 36 + 31 = 20736 + 360 + 31 = 21127$. $\square$

NOW YOU 0.47 · solutions in Appendix S

0.47 Find the base-17 expansion of 408 735 787 and the base-36 expansion of 1 442 151 747. (Both answers spell something.)

D DIVISION OF INTEGERS

Definition 0.36 · divides

Let $a, b \in \mathbb{Z}$. b **divides** a if $a = qb$ for some integer q . Also said: a is divisible by b , b is a divisor (factor) of a , a is a multiple of b . **No division is involved**: it is a statement about multiplication, so 0 divides 0 ($0 = q \cdot 0$).

How divisibility is used in proofs

To prove “ b divides a ”: write a as (an explicit integer) $\cdot b$ and say why the integer is an integer. **To use** “ b divides a ”: introduce a new letter, “ $a = qb$ for some $q \in \mathbb{Z}$ ”, and compute with it. (This is Proposition 0.39’s proof, which the book dissects in Example 1.1.1.)

Proposition 0.39

Let $a, b, c \in \mathbb{Z}$. If c divides b and b divides a , then c divides a . *Proof*: $b = qc$, $a = rb$ for some $q, r \in \mathbb{Z}$; then $a = r(qc) = (rq)c$ with $rq \in \mathbb{Z}$.

Exercise 0.38

recommended

Prove that 1 divides every integer, and that every integer divides 0.

Let $a \in \mathbb{Z}$. Then $a = a \cdot 1$ with $a \in \mathbb{Z}$, so 1 divides a by Definition 0.36. Let $b \in \mathbb{Z}$. Then $0 = 0 \cdot b$ with $0 \in \mathbb{Z}$, so b divides 0. $\square$

Exercise 0.40

essential

Let $a, b, d \in \mathbb{Z}$. Suppose that d divides a and d divides b . Given integers u and v , prove that d divides $au + bv$.

Since d divides a and d divides b , there are integers q, r with $a = qd$ and $b = rd$ (Definition 0.36). Then

$$au + bv = (qd)u + (rd)v = (qu + rv)d.$$

Since $qu + rv$ is an integer (closure of $\mathbb{Z}$), d divides $au + bv$ by Definition 0.36. $\square$

Definition 0.41 and Theorem 0.42 · even, odd, the division theorem

An integer n is **even** if it is divisible by 2; otherwise n is **odd**.

Division theorem. Let $a, b \in \mathbb{Z}$ with $b \neq 0$. There is exactly one way to write $a = qb + r$ with $q, r \in \mathbb{Z}$ and $0 \leq r < |b|$. q is the **quotient**, r the **remainder**. (“Exactly one” is the clause you use to prove that two expressions give the *same* remainder.)

Proposition 0.44

If a leaves remainder $r > 0$ when divided by b , then $-a$ leaves remainder $b - r$. *Proof:* $a = qb + r$ gives $-a = -(q+1)b + (b-r)$ with $0 < b-r < b$.

Exercise 0.45

recommended

Prove that if an integer a leaves a remainder of r when divided by an integer b , then a leaves a remainder of r when divided by $-b$.

Suppose $a = qb + r$ with $q \in \mathbb{Z}$ and $0 \leq r < |b|$. Then $a = (-q)(-b) + r$, where $-q \in \mathbb{Z}$ and $0 \leq r < |b| = |-b|$. By the uniqueness clause of Theorem 0.42, $-q$ is the quotient and r the remainder of a when divided by $-b$. $\square$

Watch out

- “ b divides a ” is not “ a/b is an integer”. Writing $\frac{a}{b} \in \mathbb{Z}$ in a proof about divisibility is marked wrong; produce the integer q .
- A remainder is never negative: $-7 = (-2) \cdot 4 + 1$, so -7 leaves remainder 1 on division by 4.
- Using the division theorem gives a *case split*: $n = 2k$ or $n = 2k + 1$; $n = 3k$, $3k + 1$ or $3k + 2$. Cover every case.

E IRRATIONAL NUMBERS

Definition 0.48, Assumption 0.49

An **irrational number** is a real number that is not rational. $\sqrt{2}$ is irrational (assumed here, proved in Proposition 4.3.12).

Proposition 0.50

There exist irrational numbers a and b such that ab is rational. *Proof:* $a = b = \sqrt{2}$ gives $ab = 2 = \frac{2}{1}$.

“It is possible that” means: *exhibit one example*. That is all Exercise 0.51 asks for.

Exercise 0.51**recommended**

Let r be a rational number and a an irrational number. Prove that it is possible that ra be rational, and possible that ra be irrational.

Take $r = 0$ and $a = \sqrt{2}$: then $ra = 0 = \frac{0}{1}$ is rational. Take $r = 1$ and $a = \sqrt{2}$: then $ra = \sqrt{2}$ is irrational (Assumption 0.49). Both possibilities occur. $\square$

F POLYNOMIALS AND COMPLEX NUMBERS

Not in the planning; included so that the 0.E questions that use it can be done.

Definitions 0.52–0.62

A **polynomial over S** (S one of $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$) is $a_0 + a_1x + \cdots + a_nx^n$ with all $a_k \in S$; its **degree** is the largest k with $a_k \neq 0$. Degrees $\leq 0, 1, 2, 3, 4, 5$: constant, linear, quadratic, cubic, quartic, quintic. A **root** is a number α with $p(\alpha) = 0$ (0.56). For $p(x) = ax^2 + bx + c$ the **discriminant** is $\Delta = b^2 - 4ac$ (0.59). The **imaginary unit** i satisfies $i^2 = -1$; for $a < 0$, $\sqrt{a} = \sqrt{|a|}i$; a **complex number** is $a + bi$ with $a, b \in \mathbb{R}$; $\mathbb{C}$ is the set of them (0.62). Compute with i as a variable, replacing i^2 by -1 ; $\frac{1}{a+bi} = \frac{a-bi}{a^2+b^2}$.

Theorem 0.60 · quadratic formula

Let $p(x) = ax^2 + bx + c$ with $a, b, c \in \mathbb{R}$, $a \neq 0$, and suppose $\Delta \geq 0$. Then the roots of $p(x)$ are precisely $\alpha = \frac{-b+\sqrt{\Delta}}{2a}$ and $\beta = \frac{-b-\sqrt{\Delta}}{2a}$.

Proof shape: complete the square, $p(x) = a(x + \frac{b}{2a})^2 - \frac{\Delta}{4a}$; check $p(\alpha) = p(\beta) = 0$; then show any root γ equals α or β (“precisely” means both directions).

Exercise 0.58**from the book**

Let $p(x) = ax + b$ be a linear polynomial over $\mathbb{Z}$. (a) Find a root. (b) Prove it is the only root. (c) When is the root an integer?

(a) Linear means degree 1, so $a \neq 0$. Put $\gamma = -\frac{b}{a}$; then $p(\gamma) = a \cdot (-\frac{b}{a}) + b = 0$.

(b) Let δ be any root: $a\delta + b = 0$. Then $a\delta = -b$ and, since $a \neq 0$, $\delta = -\frac{b}{a} = \gamma$.

(c) $-\frac{b}{a} \in \mathbb{Z}$ iff $b = qa$ for some $q \in \mathbb{Z}$ (take $q = -\frac{b}{a}$), i.e. iff a divides b (Definition 0.36). $\square$

Exercise 0.61

from the book

Let $p(x) = ax^2 + bx + c$ over $\mathbb{R}$ ($a \neq 0$) with discriminant Δ . Prove that $p(x)$ has exactly two real roots if $\Delta > 0$, exactly one if $\Delta = 0$, and none if $\Delta < 0$.

$\Delta > 0$. By Theorem 0.60 the roots are exactly α and β , and $\alpha - \beta = \frac{\sqrt{\Delta}}{a} \neq 0$, so they are two distinct real roots.

$\Delta = 0$. Then $\sqrt{\Delta} = 0$, so $\alpha = \beta = -\frac{b}{2a}$: by Theorem 0.60 this is a root and the only root.

$\Delta < 0$. Suppose $\gamma \in \mathbb{R}$ is a root. Completing the square, $0 = p(\gamma) = a(\gamma + \frac{b}{2a})^2 - \frac{\Delta}{4a}$, so $(\gamma + \frac{b}{2a})^2 = \frac{\Delta}{4a^2} < 0$. But the square of a real number is ≥ 0 : contradiction. So there is no real root. $\square$

NOW YOU 0.55, 0.64–0.68 · solutions in Appendix S

0.55 Find a quartic $s(w)$ over $\mathbb{R}$ that is not over $\mathbb{Q}$ but with $s(3) \in \mathbb{Q}$. **0.64** Express i^n in the form $a + bi$ for each $n \in \mathbb{Z}$. **0.65** Verify the conclusion of Theorem 0.60 for every discriminant. **0.66** Show $\alpha = a + bi$ is a root of a quadratic over $\mathbb{R}$ and find the other root. **0.67** State and prove the quadratic formula over $\mathbb{C}$. **0.68** Interpret subtraction and division geometrically on the number line.

G CHAPTER 0 EXERCISES (0.E)

Worked here: the essential and recommended ones. All others are in Appendix S. The True–False and Always–Sometimes–Never formats recur in every chapter: *true/always* needs a proof for all cases; *false/never* needs a counterexample (or a proof that it fails for all cases); *sometimes* needs one example and one counterexample.

Exercise 0.E.3

essential

Suppose m leaves remainder i and n leaves remainder j when divided by 3 ($0 \leq i, j < 3$). Prove that $m + n$ and $i + j$ leave the same remainder when divided by 3.

By Theorem 0.42, $m = 3q + i$ and $n = 3q' + j$ for some $q, q' \in \mathbb{Z}$. Divide $i + j$ by 3: $i + j = 3s + t$ with $s, t \in \mathbb{Z}$ and $0 \leq t < 3$, so the remainder of $i + j$ is t . Then

$$m + n = 3q + 3q' + i + j = 3(q + q' + s) + t, \quad 0 \leq t < 3,$$

and by the uniqueness clause of Theorem 0.42 the remainder of $m + n$ is t as well. $\square$

Exercise 0.E.4

essential

What are the possible remainders of n^2 when divided by 3, where $n \in \mathbb{Z}$?

By Theorem 0.42, $n = 3k, 3k + 1$ or $3k + 2$ for some $k \in \mathbb{Z}$. Then $n^2 = 3(3k^2)$, or $9k^2 + 6k + 1 = 3(3k^2 + 2k) + 1$, or $9k^2 + 12k + 4 = 3(3k^2 + 4k + 1) + 1$. So the remainder is 0 or 1 (and both occur: $n = 0, n = 1$); it is never 2. $\square$ (This is Proposition 1.1.24.)

Exercise 0.E.17 and 0.E.18

recommended

True or false? **0.E.17** Every integer is a natural number. **0.E.18** Every integer is a rational number.

0.E.17 False: $-1 \in \mathbb{Z}$ but $-1 \notin \mathbb{N}$ since $-1 < 0$ (Definition 0.6).

0.E.18 True: let $n \in \mathbb{Z}$; then $n = \frac{n}{1}$ with $n, 1 \in \mathbb{Z}$ and $1 \neq 0$, so $n \in \mathbb{Q}$ (Definition 0.15). $\square$

Exercise 0.E.22**essential**

True or false? The square root of every positive rational number is a rational number.

False. 2 is a positive rational number, and its square root $\sqrt{2}$ is irrational by Assumption 0.49. $\square$

Exercise 0.E.27 and 0.E.28**essential**

Let $a, b, c \in \mathbb{Z}$ with a dividing c and b dividing c . Always, sometimes or never: **0.E.27** ab divides c ; **0.E.28** ab divides c^2 .

0.E.27 Sometimes. $a = b = c = 1$: 1 divides 1. $a = b = c = 2$: 2 divides 2, but $ab = 4$ does not divide 2 (if $2 = 4q$ then $q = \frac{1}{2} \notin \mathbb{Z}$).

0.E.28 Always. Write $c = qa$ and $c = rb$ with $q, r \in \mathbb{Z}$. Then $c^2 = (qa)(rb) = (qr)(ab)$ with $qr \in \mathbb{Z}$, so ab divides c^2 . $\square$

Exercise 0.E.29 and 0.E.30**recommended**

0.E.29 Let $x, y \in \mathbb{Q}$ and $a, b, c, d \in \mathbb{Z}$ with $cy + d \neq 0$. Then $\frac{ax+b}{cy+d} \in \mathbb{Q}$: always, sometimes, never? **0.E.30** Let $\frac{a}{b}$ be a rational number. Then $a \in \mathbb{Z}$ and $b \in \mathbb{Z}$: always, sometimes, never?

0.E.29 Always. Integers are rational (0.E.18), so $a, b, c, d \in \mathbb{Q}$; by Theorem 0.16, $ax + b \in \mathbb{Q}$ and $cy + d \in \mathbb{Q}$, and since $cy + d \neq 0$ the quotient is rational.

0.E.30 Sometimes. $\frac{1}{2}$: $a = 1, b = 2$ are integers. $\frac{\sqrt{2}}{\sqrt{2}} = 1$ is rational, but $a = \sqrt{2} \notin \mathbb{Z}$. $\square$

NOW YOU 0.E, the rest · solutions in Appendix S

◦ 0.E.1 (base 64), 0.E.2, 0.E.6–0.E.11 (closure), 0.E.13–0.E.16 (algebraic numbers), 0.E.19–0.E.21, 0.E.23–0.E.26, 0.E.31–0.E.33.

Chapter 1

Logical structure

Contents

- A** Propositional logic (§1.1)
- B** Variables and quantifiers (§1.2)
- C** Logical equivalence (§1.3)
- D** Chapter 1 exercises (1.E)

Planning

★ 1.1: 21, 25, 32, 33, 38, 47, 56, 57 · 1.2: 6, 15, 18, 39 · 1.3: 5, 13, 14, 22, 27, 44ace · 1.E: 18
 ★ 1.1: 37, 43, 54, 55, 65, 66 · 1.2: 21, 22, 28, 37ab · 1.3: 32, 35 · 1.E: 11, 28, 29, 30
 Bonus: 1.2.14. §1.1 is in **Quiz 1**.

The chapter's one idea: a proposition's *logical structure* (which operators and quantifiers it is built from) tells you what to write. Each operator has **introduction rules** (how to prove a goal of that shape) and **elimination rules** (how to use an assumption of that shape); each rule is a **Strategy**.

A PROPOSITIONAL LOGIC

Example 1.1.1 · assumptions and goals

At every point of a proof you have a list of **assumptions** (what is known or supposed) and **goals** (what remains to be shown). Every sentence you write changes the lists: “Suppose P ” adds P to the assumptions; unfolding a definition (“ $b = qc$ for some $q \in \mathbb{Z}$ ”) replaces an assumption by its definition and introduces a new variable; “it suffices to show ...” replaces the goal. Keep both lists in your head (or on scrap paper) at all times.

Definitions 1.1.2–1.1.3

A **propositional variable** (p, q, r, s) is a symbol representing a proposition. A **propositional formula** is a propositional variable, or is built from simpler formulae using **logical operators** ($\wedge, \vee, \Rightarrow, \neg, \Leftrightarrow$). Its truth value is determined by the truth values of its subformulae.

Conjunction $\wedge$ (“and”)

Definition 1.1.4 · rules for $\wedge$

($\wedge I$) if p and q are true then $p \wedge q$ is true; ($\wedge E_1$) if $p \wedge q$ is true then p is true; ($\wedge E_2$) if $p \wedge q$ is true then q is true. Watch for hidden “and”s: “7 is a prime factor of 28” is “7 is prime” $\wedge$ “7 divides 28”.

Strategies 1.1.7 and 1.1.10

Proving $p \wedge q$: prove p , then prove q (two proofs tied together). **Assuming $p \wedge q$:** you may assume p and assume q separately.

Rules are written in natural-deduction style, premises above a line and conclusion below; stacking rules gives a **proof tree**, which is a plan for a proof (page 36).

NOW YOU 1.1.6, 1.1.13, 1.1.14 · solutions in Appendix S

1.1.6 Express “John is a mathematician who lives in Pittsburgh” as $p \wedge q$. **1.1.13** Given $(p \wedge q) \wedge r$, prove q using Strategy 1.1.10. **1.1.14** Proof tree for $(p \wedge q) \wedge (r \wedge s)$; apply it to “2 is an even prime number and 3 is an odd prime number”.

Disjunction $\vee$ (“or”)**Definition 1.1.15 · rules for $\vee$**

($\vee I_1$) if p is true then $p \vee q$ is true; ($\vee I_2$) if q is true then $p \vee q$ is true; ($\vee E$) if $p \vee q$ is true, and r can be derived from p and also from q , then r is true. (“Or” is inclusive.)

Strategies 1.1.16 and 1.1.19

Proving $p \vee q$: it suffices to prove just one of p or q . **Assuming $p \vee q$ (proof by cases):** to derive r , first derive r from the temporary assumption p , then derive r from the temporary assumption q . Write “Case 1 ... Case 2 ... In both cases, r .”

The division theorem produces disjunctions to split on: $n = 3k$ or $n = 3k + 1$ or $n = 3k + 2$ (Proposition 1.1.24: n^2 leaves remainder 0 or 1 on division by 3).

Exercise 1.1.21**essential**

Let p, q, r be propositions and suppose $q \vee r$ is true. Prove that $(p \vee q) \vee (p \vee r)$ is true, using Strategies 1.1.19 and 1.1.16.

We split into cases on the assumption $q \vee r$ (Strategy 1.1.19).

Case 1: q is true. Then $p \vee q$ is true by Strategy 1.1.16 (rule $\vee I_2$), and hence $(p \vee q) \vee (p \vee r)$ is true by Strategy 1.1.16 ($\vee I_1$).

Case 2: r is true. Then $p \vee r$ is true ($\vee I_2$), and hence $(p \vee q) \vee (p \vee r)$ is true ($\vee I_2$).

In both cases $(p \vee q) \vee (p \vee r)$ is true. □

Exercise 1.1.25**essential**

Let n be an integer. Prove that n^2 leaves a remainder of 0, 1 or 4 when divided by 5.

By the division theorem (Theorem 0.42), $n = 5k + r$ for some $k \in \mathbb{Z}$ and $r \in \{0, 1, 2, 3, 4\}$. Then

$$n^2 = 25k^2 + 10kr + r^2 = 5(5k^2 + 2kr) + r^2.$$

We split into cases on r . If $r = 0$: $n^2 = 5(5k^2)$, remainder 0. If $r = 1$: $n^2 = 5(5k^2 + 2k) + 1$, remainder 1. If $r = 2$: $n^2 = 5(5k^2 + 4k) + 4$, remainder 4. If $r = 3$: $r^2 = 9 = 5 + 4$, so $n^2 = 5(5k^2 + 6k + 1) + 4$, remainder 4. If $r = 4$: $r^2 = 16 = 15 + 1$, so $n^2 = 5(5k^2 + 8k + 3) + 1$, remainder 1.

In all cases the number in brackets is an integer and the remainder is 0, 1 or 4 (uniqueness in Theorem 0.42). □

NOW YOU 1.1.23, 1.1.26 · solutions in Appendix S

1.1.23 Let n be 8 or 9. Prove that n is even or a perfect square. **1.1.26** Let $a, b \in \mathbb{R}$ with $a^2 - 4b \neq 0$ and let α, β be the roots of $x^2 + ax + b$. Prove there is $c \in \mathbb{R}$ with $\alpha - \beta = c$ or $\alpha - \beta = ci$.

Implication $\Rightarrow$ (“if ... then”)

Definition 1.1.27 · rules for $\Rightarrow$

($\Rightarrow$ I) if q can be derived from the assumption that p is true, then $p \Rightarrow q$ is true; ($\Rightarrow$ E, *modus ponens*) if $p \Rightarrow q$ and p are true, then q is true.

Strategies 1.1.28 and 1.1.34

Proving $p \Rightarrow q$: assume p , derive q . First words: “Suppose p .” **Assuming $p \Rightarrow q$:** if you also have p , conclude q . Used to reduce a goal: if $p \Rightarrow q$ is known and p is easy, prove p instead of q (Example 1.1.36).

Proposition 1.1.30

Let $x, y \in \mathbb{R}$. If x and $x + y$ are rational, then y is rational. *Proof:* suppose $x = \frac{a}{b}$, $x + y = \frac{c}{d}$; then $y = (x + y) - x = \frac{bc - ad}{bd} \in \mathbb{Q}$.

Exercise 1.1.32

essential

Let r and s be propositions. Prove that $r \Rightarrow (s \Rightarrow s)$ is true, using the strategies for implications.

By Strategy 1.1.28 it suffices to assume r and derive $s \Rightarrow s$. So assume r . To prove $s \Rightarrow s$, again by Strategy 1.1.28 it suffices to assume s and derive s . So assume s . Then s is true, since it is assumed. Hence $s \Rightarrow s$ is true, and therefore $r \Rightarrow (s \Rightarrow s)$ is true. $\square$

Exercise 1.1.33

essential

Let a, b, c be propositions. Prove that $(a \vee b) \Rightarrow (c \Rightarrow ((c \wedge a) \vee b))$ is true.

Assume $a \vee b$ (Strategy 1.1.28). We must derive $c \Rightarrow ((c \wedge a) \vee b)$, so assume c as well. Now split into cases on $a \vee b$ (Strategy 1.1.19).

Case 1: a is true. Since c and a are both true, $c \wedge a$ is true (Strategy 1.1.7), and so $(c \wedge a) \vee b$ is true (Strategy 1.1.16).

Case 2: b is true. Then $(c \wedge a) \vee b$ is true (Strategy 1.1.16).

In both cases $(c \wedge a) \vee b$ holds. Hence $c \Rightarrow ((c \wedge a) \vee b)$, and hence the whole implication, is true. $\square$

Exercise 1.1.37

recommended

Let p, q, r be propositions. Prove that $(p \Rightarrow (q \Rightarrow r)) \Rightarrow (q \Rightarrow (p \Rightarrow r))$ is true.

Assume $p \Rightarrow (q \Rightarrow r)$. To prove $q \Rightarrow (p \Rightarrow r)$, assume q ; to prove $p \Rightarrow r$, assume p . By modus ponens (Strategy 1.1.34) applied to $p \Rightarrow (q \Rightarrow r)$ and p , we get $q \Rightarrow r$; by modus ponens again with q , we get r . Hence $p \Rightarrow r$, hence $q \Rightarrow (p \Rightarrow r)$, hence the whole formula. $\square$

Exercise 1.1.38

essential

Let p, q, r be propositions. Prove that $(p \Rightarrow p \wedge q) \Rightarrow ((p \vee q) \Rightarrow q)$ is true.

Assume $p \Rightarrow (p \wedge q)$. To prove $(p \vee q) \Rightarrow q$, assume $p \vee q$ and split into cases.

Case 1: p is true. By modus ponens, $p \wedge q$ is true, so q is true by Strategy 1.1.10 ($\wedge E_2$).

Case 2: q is true. Then q is true.

In both cases q holds, so $(p \vee q) \Rightarrow q$ is true, and hence the whole formula is true. $\square$

NOW YOU 1.1.31 · solutions in Appendix S

1.1.31 Let $p(x)$ be a polynomial over $\mathbb{C}$. Prove that if α is a root of $p(x)$ and $a \in \mathbb{C}$, then α is a root of $(x - a)p(x)$.

Converse and biconditional $\Leftrightarrow$ **Definitions 1.1.39 and 1.1.40**

The **converse** of $p \Rightarrow q$ is $q \Rightarrow p$. Ways of saying $p \Rightarrow q$: “if p then q ”, “ p only if q ”, “ p is sufficient for q ”. Ways of saying $q \Rightarrow p$: “ p if q ”, “ p is necessary for q ”.

$p \Leftrightarrow q$ means $(p \Rightarrow q) \wedge (q \Rightarrow p)$, read “ p if and only if q ”. So: **to prove $p \Leftrightarrow q$, prove both implications**, labelled $(\Rightarrow)$ and $(\Leftarrow)$; **to use $p \Leftrightarrow q$** , use either implication.

Examples 1.1.44 and 1.1.45 · solving an equation is a biconditional

Rearranging an equation only proves “if x is a solution then $x = \dots$ ” (it narrows the candidates). You must also check the converse by substituting back: for $x + \sqrt{x} = 0$ the rearrangement gives $x = 0$ or $x = 1$, but 1 is not a solution.

Example 1.1.46 · divisibility tests

To prove “ n is divisible by 8 iff its last three digits form a number divisible by 8”: write $n = n' + n''$ with n' the last three digits and $n'' = n - n' = 1000 \cdot (\dots)$, note 8 divides n'' , and use Exercise 0.40 in both directions ($n' = 1 \cdot n + (-1) \cdot n''$ and $n = 1 \cdot n' + 1 \cdot n''$).

Exercise 1.1.43**recommended**

Let p and q be propositions. Prove that $(p \Leftrightarrow q) \Leftrightarrow (q \Leftrightarrow p)$ is true.

By Definition 1.1.40 we prove both implications.

$(\Rightarrow)$ Assume $p \Leftrightarrow q$, i.e. $(p \Rightarrow q) \wedge (q \Rightarrow p)$. By Strategy 1.1.10 we have $q \Rightarrow p$ and $p \Rightarrow q$, so by Strategy 1.1.7, $(q \Rightarrow p) \wedge (p \Rightarrow q)$ is true, which is $q \Leftrightarrow p$.

$(\Leftarrow)$ Assume $q \Leftrightarrow p$, i.e. $(q \Rightarrow p) \wedge (p \Rightarrow q)$. The same two steps give $(p \Rightarrow q) \wedge (q \Rightarrow p)$, i.e. $p \Leftrightarrow q$. $\square$

Exercise 1.1.47**essential**

Prove that a natural number n is divisible by 3 if and only if the sum of its base-10 digits is divisible by 3.

Let $d_r \dots d_1 d_0$ be the base-10 expansion of n , so $n = \sum_{i=0}^r d_i 10^i$, and let $s = \sum_{i=0}^r d_i$ be the digit sum. Put $n'' = n - s = \sum_{i=0}^r d_i (10^i - 1)$.

Claim: 3 divides n'' . For each i , $10^i - 1 = (10 - 1)(10^{i-1} + 10^{i-2} + \dots + 1) = 9m_i$ with $m_i \in \mathbb{N}$ (for $i = 0$, $10^0 - 1 = 0 = 9 \cdot 0$), so $10^i - 1 = 3 \cdot (3m_i)$. Hence $n'' = 3 \sum_i d_i \cdot 3m_i$ is divisible by 3.

$(\Rightarrow)$ Suppose 3 divides n . Since 3 divides n'' , Exercise 0.40 gives that 3 divides $1 \cdot n + (-1) \cdot n'' = n - n'' = s$.

$(\Leftarrow)$ Suppose 3 divides s . Since 3 divides n'' , Exercise 0.40 gives that 3 divides $1 \cdot s + 1 \cdot n'' = n$. $\square$

Negation $\neg$ (“not”) and contradiction $\perp$ **Definitions 1.1.48 and 1.1.50**

A **contradiction** is a proposition known or assumed to be false; $\perp$ denotes an arbitrary one (e.g. $0 = 1$, $\sqrt{2} \in \mathbb{Q}$). Rules for $\neg$: ($\neg$ I) if a contradiction can be derived from the assumption that p is true, then $\neg p$ is true; ($\neg$ E) if $\neg p$ and p are both true, a contradiction may be derived.

Strategies 1.1.51 and 1.1.59 · proof by contradiction

Proving $\neg p$: assume p and derive a contradiction. First words: “Suppose p (towards a contradiction).”

Assuming $\neg p$: any derivation of p is a contradiction; this is how the contradiction is usually reached. Statements that are negations in disguise: “ x is irrational” ($= \neg(x \in \mathbb{Q})$), “ b does not divide a ”, “there is no ...”, “ X is empty”.

Propositions 1.1.53 and 1.1.58

1.1.53 If $r \in \mathbb{Q}$ and a is irrational, then $r + a$ is irrational. *Proof:* suppose $r + a \in \mathbb{Q}$; then $a = (r + a) - r \in \mathbb{Q}$ by Proposition 1.1.30, contradicting the assumption.

1.1.58 An integer a is odd iff $a = 2b + 1$ for some $b \in \mathbb{Z}$. *Proof:* $(\Rightarrow)$ by the division theorem $a = 2b$ or $a = 2b + 1$; the first contradicts “odd”. $(\Leftarrow)$ $a = 2b + 1$ leaves remainder 1, and even numbers leave remainder 0, so a is not even.

Exercise 1.1.54**recommended**

Let p and q be propositions. Prove that $(p \Rightarrow \neg q) \Rightarrow (q \Rightarrow \neg p)$ is true.

Assume $p \Rightarrow \neg q$. To prove $q \Rightarrow \neg p$, assume q . To prove $\neg p$, we use Strategy 1.1.51: assume p and derive a contradiction. By modus ponens from $p \Rightarrow \neg q$ and p we obtain $\neg q$. But q is also assumed, so by $(\neg E)$ we have a contradiction. Hence $\neg p$ is true, so $q \Rightarrow \neg p$ is true, so the whole formula is true. $\square$

Exercise 1.1.55**recommended**

Let p and q be propositions. Prove that $\neg p \Rightarrow (\neg q \Rightarrow \neg(p \vee q))$ is true.

Assume $\neg p$ and assume $\neg q$. To prove $\neg(p \vee q)$, assume $p \vee q$ and derive a contradiction, by cases. *Case 1:* p . This contradicts $\neg p$. *Case 2:* q . This contradicts $\neg q$. In both cases we have a contradiction, so $\neg(p \vee q)$ is true. Hence $\neg q \Rightarrow \neg(p \vee q)$, and hence the whole formula, is true. $\square$

Exercise 1.1.56**essential**

Let $x \in \mathbb{R}$. Prove by contradiction that if x is irrational then $-x$ and $\frac{1}{x}$ are irrational.

Suppose x is irrational. (Note $x \neq 0$, since $0 = \frac{0}{1}$ is rational.)

$-x$ is irrational. Suppose, towards a contradiction, that $-x$ is rational, say $-x = \frac{a}{b}$ with $a, b \in \mathbb{Z}$, $b \neq 0$. Then $x = -\frac{a}{b}$ with $-a \in \mathbb{Z}$, so x is rational, contradicting the assumption. Hence $-x$ is irrational.

$\frac{1}{x}$ is irrational. Suppose $\frac{1}{x} = \frac{a}{b}$ with $a, b \in \mathbb{Z}$, $b \neq 0$. Since $\frac{1}{x} \neq 0$ we have $a \neq 0$, and then $x = \frac{b}{a}$ with $b, a \in \mathbb{Z}$ and $a \neq 0$, so x is rational: contradiction. Hence $\frac{1}{x}$ is irrational. $\square$

Exercise 1.1.57**essential**

Prove by contradiction that there is no least positive real number: there is no positive real a such that $a \leq b$ for all positive real b .

Suppose, towards a contradiction, that a is a positive real number such that $a \leq b$ for every positive real number b . Then $b = \frac{a}{2}$ is a positive real number, so $a \leq \frac{a}{2}$. Multiplying by 2 gives $2a \leq a$, hence $a \leq 0$, contradicting $a > 0$. So no such a exists. $\square$

Two logical axioms

Axioms 1.1.60 and 1.1.67

Law of excluded middle (LEM): for every propositional formula p , the formula $p \vee (\neg p)$ is true.

Principle of explosion: from a contradiction, any consequence may be derived.

Strategy 1.1.61 · using LEM

To prove q , you may split into the cases “ p true” and “ p false” for any proposition p you like, and prove q in each case. (Proposition 1.1.63: if ab is even then a or b is even; split on “ a even”.)

Exercise 1.1.65**recommended**

Let p and q be propositions. Prove that $(\neg q \Rightarrow \neg p) \Rightarrow (p \Rightarrow q)$ is true.

Assume $\neg q \Rightarrow \neg p$, and assume p ; we must derive q . By the law of excluded middle, q or $\neg q$.

Case 1: q is true. Done.

Case 2: $\neg q$ is true. By modus ponens, $\neg p$ is true. Together with p this is a contradiction ($\neg E$), from which q follows by the principle of explosion.

In both cases q holds. Hence $p \Rightarrow q$, and hence the whole formula, is true. $\square$

Exercise 1.1.66**recommended**

Let a and b be irrational. By considering $\sqrt{2}^{\sqrt{2}}$, prove that it is possible that a^b be rational.

By the law of excluded middle, $\sqrt{2}^{\sqrt{2}}$ is rational or it is not.

Case 1: $\sqrt{2}^{\sqrt{2}}$ is rational. Take $a = b = \sqrt{2}$, both irrational (Assumption 0.49); then $a^b = \sqrt{2}^{\sqrt{2}}$ is rational.

Case 2: $\sqrt{2}^{\sqrt{2}}$ is irrational. Take $a = \sqrt{2}^{\sqrt{2}}$ and $b = \sqrt{2}$, both irrational. Then $a^b = (\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = \sqrt{2}^{\sqrt{2} \cdot \sqrt{2}} = \sqrt{2}^2 = 2$, which is rational.

In both cases we found irrational a, b with a^b rational. $\square$ (We never learn which case holds: a *non-constructive* proof, which is why LEM is called non-constructive.)

NOW YOU 1.1.64 · solutions in Appendix S

1.1.64 In the proof of Proposition 1.1.63, where is LEM used, where is contradiction used, and what is the contradiction? Prove the proposition again twice, once without LEM and once without contradiction.

Watch out

- Keep the direction straight: to prove $p \Rightarrow q$ assume p ; proving $q \Rightarrow p$ instead (the converse) proves a different proposition (1.E.12).
- When you use the law of excluded middle or proof by cases, the cases must be exhaustive and *every* case must reach the goal.
- In a proof by contradiction, say what the contradiction is: “this contradicts the assumption that x is irrational”.

B VARIABLES AND QUANTIFIERS

Definitions 1.2.1, 1.2.2, 1.2.4

In a statement, a variable x (understood to range over a set X) is **free** if it makes sense to substitute elements of X for it, and **bound** otherwise (“there exist integers a, b with $x = a^2 + b^2$ ”: x free, a, b bound). A **predicate** $p(x_1, \dots, x_n)$ is a symbol with a list of free variables, each with a **domain of discourse**; it becomes a proposition when values are substituted. A **logical formula** is built from predicates with logical operators and the **quantifiers** $\forall$ (“for all $x \in X$ ”), $\exists$ (“there exists $x \in X$ such that”), $\exists!$ (“there is exactly one”). Quantifiers go *in front*: “ $n = 2k$ for some $k \in \mathbb{Z}$ ” is written $\exists k \in \mathbb{Z}, n = 2k$.

Translating (Examples 1.2.5, 1.2.7)

English $\rightarrow$ symbols: introduce a variable for each noun phrase, unfold definitions (“ n is even” is $\exists k \in \mathbb{Z}, n = 2k$), and restrict with an implication inside $\forall$ (“every positive real has a square root”: $\forall a \in \mathbb{R}, (a > 0 \Rightarrow \exists b \in \mathbb{R}, b^2 = a)$). Symbols $\rightarrow$ English: read symbol by symbol, then remove bound variables by naming the concept they encode (“ $\exists b \in \mathbb{R}, a = b^2$ ” is “ a has a real square root”).

Exercise 1.2.6

essential

Find logical formulae representing: (a) there is an integer that is divisible by every integer; (b) there is no greatest odd integer; (c) between any two distinct rational numbers is a third distinct rational number; (d) if an integer has a rational square root, then that root is an integer.

- (a) $\exists n \in \mathbb{Z}, \forall m \in \mathbb{Z}, \exists q \in \mathbb{Z}, n = qm.$ { m divides n unfolded }
 (b) $\neg \exists n \in \mathbb{Z}, ([\exists k \in \mathbb{Z}, n = 2k + 1] \wedge \forall m \in \mathbb{Z}, ([\exists \ell \in \mathbb{Z}, m = 2\ell + 1] \Rightarrow m \leq n));$ equivalently $\forall n \in \mathbb{Z}, (n \text{ odd} \Rightarrow \exists m \in \mathbb{Z}, (m \text{ odd} \wedge m > n)).$
 (c) $\forall a \in \mathbb{Q}, \forall b \in \mathbb{Q}, (a < b \Rightarrow \exists c \in \mathbb{Q}, (a < c \wedge c < b)).$
 (d) $\forall n \in \mathbb{Z}, \forall x \in \mathbb{Q}, (x^2 = n \Rightarrow x \in \mathbb{Z}).$ □

NOW YOU 1.2.8 · solutions in Appendix S

1.2.8 Plain English, as few variables as possible: (a) $\exists q \in \mathbb{Z}, a = qb$; (b) $\exists a \in \mathbb{Z}, \exists b \in \mathbb{Z}, (b \neq 0 \wedge bx = a)$; (c) $\forall d \in \mathbb{N}, [(\exists q \in \mathbb{Z}, n = qd) \Rightarrow (d = 1 \vee d = n)]$; (d) $\forall a \in \mathbb{R}, [a > 0 \Rightarrow \exists b \in \mathbb{R}, (b > 0 \wedge a < b)]$.

Universal quantifier $\forall$

Definition 1.2.9 · rules for $\forall$

($\forall$ I) if $p(x)$ can be derived from the assumption that x is an *arbitrary* element of X , then $\forall x \in X, p(x)$;
 ($\forall$ E) if $\forall x \in X, p(x)$ is true and $a \in X$, then $p(a)$ is true.

Strategies 1.2.10 and 1.2.19

Proving $\forall x \in X, p(x)$: “Let $x \in X$ ” (or “fix”, “take”), then prove $p(x)$ assuming nothing about x except $x \in X$. **Assuming** $\forall x \in X, p(x)$: you may use $p(a)$ for any particular $a \in X$ you need.

Common error (page 54)

“Let n be an arbitrary integer, say $n = 17$ ” proves $\exists n$, not $\forall n$. The *reader* chooses n , not you. A correct proof of $\forall n \in \mathbb{Z}, n^2 \geq 0$: let $n \in \mathbb{Z}$; either $n \geq 0$ or $n < 0$; in each case $n^2 \geq 0$.

Propositions 1.2.11 and 1.2.12

The square of every odd integer is odd ($n = 2k + 1 \Rightarrow n^2 = 2(2k^2 + 2k) + 1$). The last decimal digit of a square is 0, 1, 4, 5, 6 or 9 (write $n = 10m + d_0$; then $n^2 = 10m(10m + 2d_0) + d_0^2$ ends in the last digit of d_0^2).

Exercise 1.2.15**essential**

Prove that for every integer n , n is even if and only if n^2 is even.

Let $n \in \mathbb{Z}$. We prove both directions.

($\Rightarrow$) Suppose n is even, so $n = 2k$ for some $k \in \mathbb{Z}$. Then $n^2 = 4k^2 = 2(2k^2)$ with $2k^2 \in \mathbb{Z}$, so n^2 is even.

($\Leftarrow$) Suppose n^2 is even. By the division theorem, $n = 2k$ or $n = 2k + 1$ for some $k \in \mathbb{Z}$. If $n = 2k + 1$, then n^2 is odd by Proposition 1.2.11, contradicting that n^2 is even. So $n = 2k$, i.e. n is even. $\square$

Exercise 1.2.18**essential**

Prove that for all real numbers x and y , if x is irrational, then $x + y$ and $x - y$ are not both rational.

Let $x, y \in \mathbb{R}$ and suppose x is irrational. Suppose, towards a contradiction, that $x + y$ and $x - y$ are both rational. Then by Theorem 0.16 their sum $(x + y) + (x - y) = 2x$ is rational, and hence $x = \frac{2x}{2}$ is rational (Theorem 0.16 again, dividing by $2 \neq 0$). This contradicts the assumption that x is irrational. Hence $x + y$ and $x - y$ are not both rational. $\square$

Exercise 1.2.21 and 1.2.22**recommended**

1.2.21 Suppose $\forall x \in \mathbb{R}, P(x)$. Show that we can conclude $P(21)$. **1.2.22** Let $P(x), Q(x)$ be predicates with $x \in \mathbb{R}$. Prove that $(\forall x \in \mathbb{R}, P(x) \Rightarrow Q(x)) \Rightarrow (\forall x \in \mathbb{R}, P(x)) \Rightarrow (\forall x \in \mathbb{R}, Q(x))$.

1.2.21 Since $21 \in \mathbb{R}$ and $\forall x \in \mathbb{R}, P(x)$ holds, the rule $(\forall E)$ (Strategy 1.2.19) gives $P(21)$. $\square$

1.2.22 Assume $\forall x \in \mathbb{R}, (P(x) \Rightarrow Q(x))$ and assume $\forall x \in \mathbb{R}, P(x)$. To prove $\forall x \in \mathbb{R}, Q(x)$, let $x \in \mathbb{R}$ be arbitrary. By $(\forall E)$, $P(x) \Rightarrow Q(x)$ and $P(x)$ are both true, so $Q(x)$ is true by modus ponens. Since x was arbitrary, $\forall x \in \mathbb{R}, Q(x)$. $\square$

NOW YOU 1.2.14 (bonus), 1.2.16, 1.2.17 · solutions in Appendix S

1.2.14 Prove $\forall x \in \mathbb{R}, x = x$. **1.2.16** Prove that every integer is rational. **1.2.17** Prove that every linear polynomial over $\mathbb{Q}$ has a rational root.

Existential quantifier $\exists$ **Definition 1.2.23 · rules for $\exists$**

$(\exists I)$ if $a \in X$ and $p(a)$ is true, then $\exists x \in X, p(x)$; $(\exists E)$ if $\exists x \in X, p(x)$ is true, and q can be derived from the assumption that $p(a)$ holds for some fixed $a \in X$, then q is true.

Strategies 1.2.24 and 1.2.30

Proving $\exists x \in X, p(x)$: define a specific a (“Define $n = 9$ ”, “Take $z = \frac{x+y}{2}$ ”), check $a \in X$, and prove $p(a)$.

Assuming $\exists x \in X, p(x)$: “Take $a \in X$ such that $p(a)$ ” with a letter not yet in use. Divisibility assumptions are existential: from “3 divides n^3 ” take q with $n^3 = 3q$ (Proposition 1.2.31).

Exercise 1.2.28**recommended**

Prove that there is an integer which is divisible by zero.

Define $n = 0$. Then $n \in \mathbb{Z}$, and $0 = 0 \cdot 0$ with $0 \in \mathbb{Z}$, so 0 divides n by Definition 0.36. Hence $\exists n \in \mathbb{Z}$ such that 0 divides n . $\square$

Exercise 1.2.29**from the book**

Prove that for all $x, y \in \mathbb{Q}$, if $x < y$ then there is some $z \in \mathbb{Q}$ with $x < z < y$.

Let $x, y \in \mathbb{Q}$ and suppose $x < y$. Define $z = \frac{x+y}{2}$. Then $z \in \mathbb{Q}$ by Theorem 0.16 ($x+y \in \mathbb{Q}$ and $2 \neq 0$). Moreover $x < z$ because $2x < x+y$ (add x to $x < y$), and $z < y$ because $x+y < 2y$. So $x < z < y$, and z is the required element. $\square$

NOW YOU 1.2.27 · solutions in Appendix S

1.2.27 Prove that there is a real number that is irrational but whose square is rational.

Unique existence $\exists!$ **Definition 1.2.32 and Proposition 1.2.38**

$\exists! x \in X$, $p(x)$ is shorthand for

$$\underbrace{(\exists x \in X, p(x))}_{\text{existence}} \wedge \underbrace{(\forall a \in X, \forall b \in X, [(p(a) \wedge p(b)) \Rightarrow a = b])}_{\text{uniqueness}},$$

equivalently (1.2.38) $\exists x \in X, (p(x) \wedge \forall y \in X, (p(y) \Rightarrow y = x))$.

Strategy 1.2.35

Existence: exhibit a with $p(a)$. **Uniqueness:** “Let $a, b \in X$ with $p(a)$ and $p(b)$ ” ... “so $a = b$ ”. (Or: with the a from existence, show $\forall y, p(y) \Rightarrow y = a$.) Example 1.2.36: for $a > 0$ the positive square root is unique because $y^2 = z^2$ gives $(y-z)(y+z) = 0$ and $y+z > 0$.

Exercise 1.2.37 (a), (b)**recommended**

Write as a formula with $\exists!$ and prove: (a) for each real a , the equation $x^2 + 2ax + a^2 = 0$ has exactly one real solution; (b) there is a unique real a for which $x^2 + a^2 = 0$ has a real solution.

(a) $\forall a \in \mathbb{R}, \exists! x \in \mathbb{R}, x^2 + 2ax + a^2 = 0$. Let $a \in \mathbb{R}$. *Existence:* $x = -a$ gives $a^2 - 2a^2 + a^2 = 0$. *Uniqueness:* let $x, y \in \mathbb{R}$ both be solutions. Since $x^2 + 2ax + a^2 = (x+a)^2$, we get $(x+a)^2 = 0$, so $x+a = 0$, so $x = -a$; likewise $y = -a$; hence $x = y$. $\square$

(b) $\exists! a \in \mathbb{R}, \exists x \in \mathbb{R}, x^2 + a^2 = 0$. *Existence:* $a = 0$ works, with $x = 0$. *Uniqueness:* let $a, b \in \mathbb{R}$ both have the property; take $x \in \mathbb{R}$ with $x^2 + a^2 = 0$. Since $x^2 \geq 0$ and $a^2 \geq 0$, the sum is 0 only if $a^2 = 0$, so $a = 0$. The same argument gives $b = 0$, so $a = b$. $\square$

NOW YOU 1.2.37 (c), 1.2.34 · solutions in Appendix S

1.2.37 (c) There is a unique natural number with exactly one positive divisor. **1.2.34 (Discussion)** Why does Definition 1.2.32 capture “exactly one”? Other possible definitions?

Quantifier alternation**Theorem 1.2.40**

$\exists y \in Y, \forall x \in X, p(x, y) \Rightarrow \forall x \in X, \exists y \in Y, p(x, y)$. *Proof:* fix $a \in X$; choose b with $\forall x, p(x, b)$; then $p(a, b)$.

The converse fails in general: in $\forall x \exists y$ the y may depend on x (“every door has a key”); in $\exists y \forall x$ one y works for all x (“a master key”).

Exercise 1.2.39**essential**

Let $p(x, y)$ be “ $x + y$ is even”. Prove that $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, p(x, y)$ is true, and that $\exists y \in \mathbb{Z}, \forall x \in \mathbb{Z}, p(x, y)$ is false.

First statement. Let $x \in \mathbb{Z}$. Define $y = x$. Then $y \in \mathbb{Z}$ and $x + y = 2x$ is even (it is $2 \cdot x$ with $x \in \mathbb{Z}$). So $\exists y \in \mathbb{Z}, p(x, y)$, and as x was arbitrary the statement holds.

Second statement. Suppose, towards a contradiction, that there is $y \in \mathbb{Z}$ with $x + y$ even for every $x \in \mathbb{Z}$. Take $x = 1 - y \in \mathbb{Z}$. Then $x + y = 1$, so 1 is even, i.e. $1 = 2k$ for some $k \in \mathbb{Z}$; but then $k = \frac{1}{2} \notin \mathbb{Z}$, a contradiction. Hence the statement is false. $\square$

C LOGICAL EQUIVALENCE**Definition 1.3.2 · logically equivalent**

Logical formulae p and q are **logically equivalent**, $p \equiv q$, if q can be derived from p and p can be derived from q . Equivalent formulae mean the same thing but suggest different proof strategies (Example 1.3.1: “every prime > 2 is odd” versus “there is no even prime > 2 ”).

Proving $p \equiv q$ from the definition (Examples 1.3.3, 1.3.4)

Two derivations: “First assume $p \dots$ hence q .” then “Now assume $q \dots$ hence p .” Inside each, use the strategies of §1.1 (cases for $\vee$, assume-and-derive for $\Rightarrow$, LEM and explosion where needed).

Exercise 1.3.5**essential**

Let p, q, r be propositional variables. Prove that $(p \vee q) \Rightarrow r$ is logically equivalent to $(p \Rightarrow r) \wedge (q \Rightarrow r)$.

First assume $(p \vee q) \Rightarrow r$. To prove $p \Rightarrow r$: assume p ; then $p \vee q$ by $(\vee I_1)$, so r by modus ponens. To prove $q \Rightarrow r$: assume q ; then $p \vee q$ by $(\vee I_2)$, so r by modus ponens. Hence $(p \Rightarrow r) \wedge (q \Rightarrow r)$ by $(\wedge I)$. Now assume $(p \Rightarrow r) \wedge (q \Rightarrow r)$, so $p \Rightarrow r$ and $q \Rightarrow r$ by $(\wedge E)$. To prove $(p \vee q) \Rightarrow r$, assume $p \vee q$ and split into cases: if p , then r by modus ponens; if q , then r by modus ponens. In both cases r , so $(p \vee q) \Rightarrow r$. Each formula can be derived from the other, so they are logically equivalent. $\square$

Truth tables

Theorem 1.3.6 and Definition 1.3.7

Two *propositional* formulae are logically equivalent iff they have the same truth value under every assignment of truth values to their variables. The **truth table** of a formula has one row per assignment (2^n rows for n variables) and one column per subformula. Basic tables: $\neg p$ is $\times$ iff p is $\checkmark$; $p \wedge q$ is $\checkmark$ only in row $\checkmark\checkmark$; $p \vee q$ is $\times$ only in row $\times\times$; $p \Rightarrow q$ is $\times$ only in row $\checkmark\times$.

Strategy 1.3.11

To prove propositional formulae are equivalent, show that their columns in a (combined) truth table are identical. Layout: variables first (p : four $\checkmark$ then four $\times$; q : two and two; r : alternating), then subformulae, then the main formulae.

Exercise 1.3.13**essential**

Use a truth table to prove that $p \Rightarrow q \equiv (\neg p) \vee q$.

p	q	$p \Rightarrow q$	$\neg p$	$(\neg p) \vee q$
$\checkmark$	$\checkmark$	$\checkmark$	$\times$	$\checkmark$
$\checkmark$	$\times$	$\times$	$\times$	$\times$
$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
$\times$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$

The columns for $p \Rightarrow q$ and $(\neg p) \vee q$ are identical, so by Strategy 1.3.11 the formulae are logically equivalent. $\square$

Exercise 1.3.14**essential**

Use a truth table to prove that $(p \vee q) \Rightarrow r \equiv (p \Rightarrow r) \wedge (q \Rightarrow r)$.

p	q	r	$p \vee q$	$(p \vee q) \Rightarrow r$	$p \Rightarrow r$	$q \Rightarrow r$	$(p \Rightarrow r) \wedge (q \Rightarrow r)$
$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
$\checkmark$	$\checkmark$	$\times$	$\checkmark$	$\times$	$\times$	$\times$	$\times$
$\checkmark$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
$\checkmark$	$\times$	$\times$	$\checkmark$	$\times$	$\times$	$\checkmark$	$\times$
$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
$\times$	$\checkmark$	$\times$	$\checkmark$	$\times$	$\checkmark$	$\times$	$\times$
$\times$	$\times$	$\checkmark$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
$\times$	$\times$	$\times$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$

The fifth and eighth columns agree in every row, so the formulae are logically equivalent. $\square$

NOW YOU 1.3.9 · solutions in Appendix S

1.3.9 Use the definitions of $\wedge$, $\vee$, $\Rightarrow$ to justify the basic truth tables.

Strategies from equivalences

Theorem 1.3.15 and Strategy 1.3.16 · double negation, indirect contradiction

$p \equiv \neg\neg p$ (two-row truth table). Hence **to prove p is true, you may assume p is false and derive a contradiction**. Compare Strategy 1.1.51 (to prove $\neg p$, assume p): that one is direct, this one is indirect. Example 1.3.17: $a, b, c \geq 0$ with $a^2 + b^2 = c^2$ implies $a + b \geq c$; assume $a + b < c$ and square.

Definition 1.3.18, Theorem 1.3.19, Strategy 1.3.20 · contraposition

The **contrapositive** of $p \Rightarrow q$ is $\neg q \Rightarrow \neg p$, and $p \Rightarrow q \equiv (\neg q) \Rightarrow (\neg p)$ (truth table). **Proof by contraposition:** to prove $p \Rightarrow q$, assume $\neg q$ and derive $\neg p$. Example 1.3.21: $mn > 64 \Rightarrow (m > 8 \vee n > 8)$; assume $m \leq 8$ and $n \leq 8$, get $mn \leq 64$. Announce it: “By contraposition, it suffices to ...”.

Exercise 1.3.22**essential**

Use the law of contraposition to prove that $p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge ((\neg p) \Rightarrow (\neg q))$, and use the technique this suggests to prove that an integer is even if and only if its square is even.

By Definition 1.1.40, $p \Leftrightarrow q$ is $(p \Rightarrow q) \wedge (q \Rightarrow p)$. By the law of contraposition (Theorem 1.3.19 applied to $q \Rightarrow p$), $q \Rightarrow p \equiv (\neg p) \Rightarrow (\neg q)$. Replacing the second conjunct by this equivalent formula gives $p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge ((\neg p) \Rightarrow (\neg q))$.

Technique: to prove “ p iff q ”, prove $p \Rightarrow q$ and then prove $\neg p \Rightarrow \neg q$.

Application. Let $n \in \mathbb{Z}$. ($p \Rightarrow q$) If n is even, $n = 2k$, then $n^2 = 2(2k^2)$ is even. ($\neg p \Rightarrow \neg q$) If n is odd, then n^2 is odd by Proposition 1.2.11. Hence n is even iff n^2 is even. $\square$

Exercise 1.3.23**from the book**

Prove that $p \vee q \equiv (\neg p) \Rightarrow q$. Use the strategy this suggests to prove that if two integers sum to an even number, then either both are even or both are odd.

By Exercise 1.3.13 (with $\neg p$ in place of p), $(\neg p) \Rightarrow q \equiv (\neg\neg p) \vee q \equiv p \vee q$ by the law of double negation.

Strategy: to prove $p \vee q$, assume $\neg p$ and derive q .

Application. Let $a, b \in \mathbb{Z}$ with $a + b$ even, say $a + b = 2m$. We prove “both even $\vee$ both odd” by assuming they are not both even and deriving that both are odd. So suppose a and b are not both even; then a is odd or b is odd (de Morgan, Theorem 1.3.24). The two cases are symmetric, so suppose a is odd: $a = 2k + 1$ (Proposition 1.1.58). Then $b = 2m - a = 2m - 2k - 1 = 2(m - k - 1) + 1$ is odd. So both are odd. $\square$

Negation: de Morgan’s laws and maximal negation**Theorems 1.3.24 and 1.3.29 · de Morgan’s laws**

Operators: (a) $\neg(p \wedge q) \equiv (\neg p) \vee (\neg q)$; (b) $\neg(p \vee q) \equiv (\neg p) \wedge (\neg q)$.

Quantifiers: (a) $\neg\forall x \in X, p(x) \equiv \exists x \in X, \neg p(x)$; (b) $\neg\exists x \in X, p(x) \equiv \forall x \in X, \neg p(x)$.

Also $\neg(p \Rightarrow q) \equiv p \wedge (\neg q)$ (Exercise 1.3.27) and $\neg\neg p \equiv p$. Quantifier laws cannot be checked with truth tables; the book derives each side from the other.

Strategy 1.3.30 · proof by counterexample

To prove $\forall x \in X, p(x)$ is false, exhibit one $a \in X$ with $p(a)$ false (that is $\exists x, \neg p(x)$). Example 1.3.31: not every integer is divisible by a prime; counterexample 1.

Exercise 1.3.27**essential**

Prove that $\neg(p \Rightarrow q) \equiv p \wedge (\neg q)$ twice: with a truth table, and using Exercise 1.3.13 with de Morgan's laws and double negation.

Truth table.

p	q	$p \Rightarrow q$	$\neg(p \Rightarrow q)$	$\neg q$	$p \wedge (\neg q)$
✓	✓	✓	×	×	×
✓	×	×	✓	✓	✓
×	✓	✓	×	×	×
×	×	✓	×	✓	×

The fourth and sixth columns agree, so the formulae are equivalent.

Algebraically. $\neg(p \Rightarrow q) \equiv \neg((\neg p) \vee q)$ {Exercise 1.3.13} $\equiv (\neg\neg p) \wedge (\neg q)$ {de Morgan (b)} $\equiv p \wedge (\neg q)$ {double negation}. $\square$

Exercise 1.3.28**from the book**

Prove that $\neg(p \wedge q) \equiv p \Rightarrow (\neg q)$. What strategy does this suggest for proving a conjunction false, and how does it compare with de Morgan's?

$p \Rightarrow (\neg q) \equiv (\neg p) \vee (\neg q)$ by Exercise 1.3.13, and $(\neg p) \vee (\neg q) \equiv \neg(p \wedge q)$ by de Morgan (a).

Strategy: to prove $p \wedge q$ is false, assume p and prove that q is false. De Morgan's law instead asks you to prove $\neg p$ or to prove $\neg q$, which requires deciding which one fails in advance; the new strategy lets you use p as an extra assumption while refuting q . $\square$

Exercise 1.3.32**recommended**

Prove by counterexample that not every rational number can be expressed as $\frac{a}{b}$ with $a \in \mathbb{Z}$ even and $b \in \mathbb{Z}$ odd.

Counterexample: $\frac{1}{3}$. Suppose $\frac{1}{3} = \frac{a}{b}$ with a even and b odd. Then $b = 3a$, and $3a$ is even (it is $3 \cdot 2k = 2(3k)$), so b is even and odd at once, contradicting Definition 0.41. Hence $\frac{1}{3}$ has no such expression. $\square$

NOW YOU 1.3.25 · solutions in Appendix S

1.3.25 Prove de Morgan (b) three times: from the definition of $\equiv$, with a truth table, and from (a) plus double negation.

Definition 1.3.33 and Theorem 1.3.36 · maximally negated

A formula is **maximally negated** if every $\neg$ stands immediately before a predicate (or proposition) that contains no operators or quantifiers. Every formula is equivalent to a maximally negated one: push $\neg$ inwards with the table

$$\begin{aligned} \neg(p \wedge q) &\equiv (\neg p) \vee (\neg q) & \neg\forall x \in X, p(x) &\equiv \exists x \in X, \neg p(x) \\ \neg(p \vee q) &\equiv (\neg p) \wedge (\neg q) & \neg\exists x \in X, p(x) &\equiv \forall x \in X, \neg p(x) \\ \neg(p \Rightarrow q) &\equiv p \wedge (\neg q) & \neg\neg p &\equiv p \end{aligned}$$

one step at a time, ignoring the “stuff” inside (Example 1.3.37 negates the definition of convergence this way). Finally replace $\neg(x < y)$ by $x \geq y$, $\neg(a = b)$ by $a \neq b$.

Exercise 1.3.35**recommended**

Which of the following are maximally negated? (a) $\forall x \in X, (\neg p(x)) \Rightarrow \forall y \in X, \neg(r(x,y) \wedge s(x,y))$; (b) $\forall x \in X, (\neg p(x)) \Rightarrow \forall y \in X, (\neg r(x,y)) \vee (\neg s(x,y))$; (c) $\forall x \in \mathbb{R}, [x > 1 \Rightarrow (\exists y \in \mathbb{R}, [x < y \wedge \neg(x^2 \leq y)])]$; (d) $\neg \exists x \in \mathbb{R}, [x > 1 \wedge (\forall y \in \mathbb{R}, [x < y \Rightarrow x^2 \leq y])]$.

(a) No: $\neg(r \wedge s)$ has a negation in front of a conjunction. (b) Yes: every $\neg$ is directly before a predicate. (c) Yes: the only negation is $\neg(x^2 \leq y)$, in front of a predicate without operators (one may still rewrite it as $x^2 > y$). (d) No: the negation is in front of a quantified formula. $\square$

Exercise 1.3.39**from the book**

Maximally negate $\exists x \in \mathbb{R}, [x > 1 \wedge (\forall y \in \mathbb{R}, [x < y \Rightarrow x^2 \leq y])]$, then prove it true or prove it false.

Negation: $\forall x \in \mathbb{R}, [x \leq 1 \vee (\exists y \in \mathbb{R}, [x < y \wedge x^2 > y])]$ { $\neg \exists \rightarrow \forall \neg$; de Morgan (a); $\neg \forall \rightarrow \exists \neg$; $\neg(\Rightarrow) \rightarrow \wedge \neg$ }. The original statement is false; we prove the negation. Let $x \in \mathbb{R}$. If $x \leq 1$, the first disjunct holds. Otherwise $x > 1$; define $y = \frac{x+x^2}{2}$. Since $x > 1$ we have $x^2 > x$ (multiply $x > 1$ by $x > 0$), so y is strictly between x and x^2 : $x < y$ and $x^2 > y$. Hence the second disjunct holds. In both cases the negation holds. $\square$

NOW YOU 1.3.38 · solutions in Appendix S

1.3.38 Find a maximally negated formula equivalent to $\neg(p \Leftrightarrow q)$.

Tautologies**Definition 1.3.40, Strategy 1.3.41, Theorem 1.3.45**

A **tautology** is a formula that is true under every assignment of truth values to its variables and predicates. Any tautology may be assumed at any point of any proof (LEM says $p \vee \neg p$ is one). **Theorem 1.3.45:** q can be derived from p iff $p \Rightarrow q$ is a tautology; $p \equiv q$ iff $p \Leftrightarrow q$ is a tautology. (So “ $\equiv$ ” is a statement about formulae; “ $\Leftrightarrow$ ” is part of a formula.)

Proving that a formula is a tautology

Either give a derivation from no assumptions (assume the antecedents, derive the consequent), which proves it by Theorem 1.3.45(a); or, for a propositional formula, show that its truth-table column is all $\checkmark$ (Example 1.3.43 does both for $p \Rightarrow (q \Rightarrow p)$).

Exercise 1.3.44 (a), (c), (e)**essential**

Prove that each is a tautology and interpret it: (a) $[(p \Rightarrow q) \wedge (q \Rightarrow r)] \Rightarrow (p \Rightarrow r)$; (c) $\exists y \in Y, \forall x \in X, p(x,y) \Rightarrow \forall x \in X, \exists y \in Y, p(x,y)$; (e) $(\neg \forall x \in X, p(x)) \Leftrightarrow (\exists x \in X, \neg p(x))$.

(a) Assume $(p \Rightarrow q) \wedge (q \Rightarrow r)$, so $p \Rightarrow q$ and $q \Rightarrow r$. Assume p . Modus ponens gives q , then r . Hence $p \Rightarrow r$; this derivation from no assumptions shows the formula is a tautology (Theorem 1.3.45(a)). *Meaning:* implications chain; in a proof you may pass from p to r through an intermediate q .

(c) This is Theorem 1.2.40, whose proof is a derivation of the consequent from the antecedent; by Theorem 1.3.45(a) the implication is a tautology. *Meaning:* a “one size fits all” y gives a y for each x .

(e) By Theorem 1.3.29(a), $\neg \forall x \in X, p(x) \equiv \exists x \in X, \neg p(x)$, so by Theorem 1.3.45(b) the biconditional is a tautology. *Meaning:* to refute a universal statement, give a counterexample (Strategy 1.3.30). $\square$

NOW YOU 1.3.44 (b), (d) · solutions in Appendix S

1.3.44 (b) $[p \Rightarrow (q \Rightarrow r)] \Rightarrow [(p \Rightarrow q) \Rightarrow (p \Rightarrow r)]$; **(d)** $[\neg(p \wedge q)] \Leftrightarrow [(\neg p) \vee (\neg q)]$.

D CHAPTER 1 EXERCISES (1.E)**Exercise 1.E.11****recommended**

Prove that $p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge ((\neg p) \Rightarrow (\neg q))$. How does this help to prove “ p if and only if q ”?

This is Exercise 1.3.22: replace $q \Rightarrow p$ in $(p \Rightarrow q) \wedge (q \Rightarrow p)$ by its contrapositive $(\neg p) \Rightarrow (\neg q)$ (Theorem 1.3.19). It helps when $\neg p$ is easier to work with than q : prove $p \Rightarrow q$, then assume $\neg p$ and derive $\neg q$ instead of proving $q \Rightarrow p$ (as in “ n even iff n^2 even”). $\square$

Exercise 1.E.18**essential**

Let X be $\mathbb{Z}$ or $\mathbb{Q}$ and let p be $\forall x \in X, \exists y \in X, (x < y \wedge [\forall z \in X, \neg(x < z \wedge z < y)])$. Write $\neg p$ maximally negated. Prove that p is true for $X = \mathbb{Z}$ and false for $X = \mathbb{Q}$.

$\neg p \equiv \exists x \in X, \forall y \in X, (x \geq y \vee [\exists z \in X, (x < z \wedge z < y)])$. $\{\neg \forall \rightarrow \exists \neg, \neg \exists \rightarrow \forall \neg, \text{de Morgan (a) on the } \wedge, \neg \forall \neg \rightarrow \exists, \neg(x < y) \rightarrow x \geq y\}$

(p says: every x has a “next” element y with nothing strictly between.)

$X = \mathbb{Z}$: p is true. Let $x \in \mathbb{Z}$ and define $y = x + 1 \in \mathbb{Z}$. Then $x < y$. Let $z \in \mathbb{Z}$; there is no integer strictly between x and $x + 1$ (page 6), so $\neg(x < z \wedge z < y)$. Hence y has the required property.

$X = \mathbb{Q}$: p is false. We prove $\neg p$. Take $x = 0 \in \mathbb{Q}$. Let $y \in \mathbb{Q}$. If $y \leq 0$ then $x \geq y$. Otherwise $0 < y$; define $z = \frac{0+y}{2} = \frac{y}{2} \in \mathbb{Q}$, which satisfies $0 < z < y$ (Exercise 1.2.29), so the second disjunct holds. Hence $\neg p$ is true. $\square$

Exercise 1.E.28, 1.E.29, 1.E.30**recommended**

True or false? Let ϕ be $\neg \forall x \geq 0, \exists y \in \mathbb{R}, y^2 = x$. **1.E.28** $\phi \equiv \forall x < 0, \exists y \notin \mathbb{R}, y^2 \neq x$. **1.E.29** $\phi \equiv \exists x \geq 0, \forall y \in \mathbb{R}, y^2 \neq x$. **1.E.30** $\phi \equiv \exists x < 0, \forall y \in \mathbb{R}, y^2 = x$.

Here “ $\forall x \geq 0$ ” means $\forall x \in \mathbb{R}, (x \geq 0 \Rightarrow \dots)$. Maximally negating: $\phi \equiv \exists x \in \mathbb{R}, (x \geq 0 \wedge \forall y \in \mathbb{R}, y^2 \neq x)$, i.e. $\exists x \geq 0, \forall y \in \mathbb{R}, y^2 \neq x$.

1.E.29 True: this is exactly the maximally negated form.

1.E.28 False: negation swaps the quantifiers and negates the predicate, but it never changes the domain of a bound variable ($x \geq 0$ stays, $y \in \mathbb{R}$ stays). The proposed formula is not obtained by any of the rules; and indeed ϕ is false (every $x \geq 0$ has the square root $y = \sqrt{x}$), while the proposed formula is true (for $x < 0$ take $y = 2i \notin \mathbb{R}$ if $x \neq -4$, and $y = i$ if $x = -4$). Different truth values, so not equivalent.

1.E.30 False: it negates the domain condition ($x < 0$) and keeps the predicate un-negated. As formulae they are not equivalent: treat “ $y^2 = x$ ” as an abstract predicate $q(x, y)$ and the domains as abstract sets; the correct negation is $\exists x \in A, \forall y, \neg q(x, y)$, which differs from $\exists x \in B, \forall y, q(x, y)$ (take A non-empty, B empty: then the first may be true while the second is false). $\square$

NOW YOU 1.E, the rest · solutions in Appendix S

1.E.1–1.E.10, 1.E.12–1.E.17, 1.E.19–1.E.27, 1.E.31–1.E.35.

 The three most common losses of marks in this chapter

- Picking a value when the goal is $\forall$ (the Common error on page 54).
- Rearranging an equation and stopping: the converse must be checked (Examples 1.1.44–45).
- Negating by flipping everything: domains of quantifiers never change; $\neg(p \Rightarrow q)$ is $p \wedge \neg q$, not $\neg p \Rightarrow \neg q$.

Chapter 2

Sets

Contents

- A** Sets (§2.1)
- B** Set operations (§2.2)
- C** Tuples and cartesian products (§2.2)
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Planning

- ★ 2.1: 10, 27, 31, 35
 - ★ 2.2: 5, 24, 26, 28, 35, 36, 37, 40, 51
 - ★ 2.E: 6, 33, 38
 - ★ 2.1: 17, 18, 23, 24, 32, 36
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 - ★ 2.E: 4, 5, 16, 28, 32, 41; bonus 18
- Tested in **Quiz 2** (7 Oct).

A SETS

Definitions 2.1.2 and 2.1.4 · sets, the universe, set-builder notation

A **set** is a collection of elements from a specified **universe of discourse** $\mathcal{U}$; $\mathcal{U}$ contains every mathematical object of interest but $\mathcal{U} \notin \mathcal{U}$. Unrestricted quantifiers range over $\mathcal{U}$: $\forall x, p(x)$ means $\forall x \in \mathcal{U}, p(x)$, and

$$\forall x \in X, p(x) \equiv \forall x, (x \in X \Rightarrow p(x)), \quad \exists x \in X, p(x) \equiv \exists x, (x \in X \wedge p(x)).$$

Set-builder notation: $a \in \{x \mid p(x)\} \Leftrightarrow a \in \mathcal{U} \wedge p(a)$; $\{x \in X \mid p(x)\} = \{x \mid x \in X \wedge p(x)\}$; $\{\text{expr}(x) \mid p(x)\} = \{y \mid \exists x, y = \text{expr}(x) \wedge p(x)\}$. A set written in set-builder notation need not be an element of $\mathcal{U}$ (e.g. $\{x \mid x = x\} = \mathcal{U}$); this is what dissolves Russell's paradox (Exercises 2.1.1, 2.1.5).

NOW YOU 2.1.1, 2.1.3, 2.1.5 · solutions in Appendix S

2.1.1 With $S = \{X \mid X \text{ is a set}\}$ and $R = \{X \in S \mid X \notin X\}$, determine the truth value of $R \in R$. **2.1.3** Why does “ $\forall x \in X$ ” expand with $\Rightarrow$ but “ $\exists x \in X$ ” with $\wedge$? Verify de Morgan's laws for the expanded forms. **2.1.5** Repeat 2.1.1 with the updated definitions.

Subsets

Definition 2.1.6

U is a **subset** of X , written $U \subseteq X$, if $\forall a \in U, a \in X$ (equivalently $\forall a, (a \in U \Rightarrow a \in X)$). $U \not\subseteq X$: not a subset. $U \subsetneq X$ (**proper subset**): $U \subseteq X$ and $U \neq X$. Example 2.1.11: $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$.

Strategy 2.1.7 · proving $U \subseteq X$

“Let $a \in U$.” Unfold what that means. ... “So $a \in X$.” (It is a $\forall$ -proof, Strategy 1.2.10.) Example 2.1.9: $[c, d] \subseteq (a, b)$ when $a < c < d < b$, because $c \leq x \leq d$ gives $a < x < b$.

Proposition 2.1.12 · transitivity

If $X \subseteq Y$ and $Y \subseteq Z$ then $X \subseteq Z$. *Proof:* let $a \in X$; then $a \in Y$ (as $X \subseteq Y$); then $a \in Z$ (as $Y \subseteq Z$).

Exercise 2.1.10**essential**

Let $a, b, c, d \in \mathbb{R}$ with $a < b$ and $c < d$. Prove that $[a, b] \subseteq (c, d]$ if and only if $a > c$ and $b \leq d$.

($\Leftarrow$) Suppose $a > c$ and $b \leq d$. Let $x \in [a, b]$, so $a \leq x < b$. Then $c < a \leq x$ gives $c < x$, and $x < b \leq d$ gives $x \leq d$. So $x \in (c, d]$. Hence $[a, b] \subseteq (c, d]$.

($\Rightarrow$) Suppose $[a, b] \subseteq (c, d]$. Since $a < b$, we have $a \in [a, b]$, so $a \in (c, d]$, which gives $c < a$ (and $a \leq d$). For $b \leq d$, suppose towards a contradiction that $b > d$. Define $x = \frac{d+b}{2}$. Then $d < x < b$, and $x > d \geq a$, so $x \in [a, b]$; but $x > d$ means $x \notin (c, d]$, contradicting $[a, b] \subseteq (c, d]$. Hence $b \leq d$. $\square$

Set equality**Axiom 2.1.14 · set extensionality**

$X = Y$ if and only if $\forall a, (a \in X \Leftrightarrow a \in Y)$, equivalently $X \subseteq Y$ and $Y \subseteq X$. Consequence: order and repetition in list notation do not matter.

Strategy 2.1.15 · proof by double containment

($\subseteq$) Let $a \in X \dots$ so $a \in Y$. ($\supseteq$) Let $a \in Y \dots$ so $a \in X$. Example 2.1.16: $\{x \in \mathbb{R} \mid x^2 \leq 1\} = [-1, 1]$, using $(1-a)(1+a) \geq 0$ for $\subseteq$ and $|a| \leq 1 \Rightarrow a^2 \leq 1$ for $\supseteq$.

Exercise 2.1.17**recommended**

Prove that $\{x \in \mathbb{R} \mid x^2 < x\} = (0, 1)$.

($\subseteq$) Let $a \in \{x \in \mathbb{R} \mid x^2 < x\}$, so $a \in \mathbb{R}$ and $a^2 < a$, i.e. $a(a-1) = a^2 - a < 0$. A product of two real numbers is negative only if one factor is positive and the other negative. If $a > 0$ and $a-1 < 0$ then $0 < a < 1$. If $a < 0$ and $a-1 > 0$ then $a < 0$ and $a > 1$, impossible. So $0 < a < 1$, i.e. $a \in (0, 1)$.

($\supseteq$) Let $a \in (0, 1)$, so $0 < a < 1$. Multiplying $a < 1$ by $a > 0$ gives $a^2 < a$; and $a \in \mathbb{R}$. So $a \in \{x \in \mathbb{R} \mid x^2 < x\}$. $\square$

Exercise 2.1.18**recommended**

Prove by double containment that $\{0, 1\} = \{1, 0\}$ and $\{0, 0\} = \{0\}$.

($\subseteq$) Let $a \in \{0, 1\}$. Then $a = 0$ or $a = 1$; in either case a is one of the listed elements $1, 0$, so $a \in \{1, 0\}$.

($\supseteq$) Let $a \in \{1, 0\}$; then $a = 1$ or $a = 0$, so $a \in \{0, 1\}$. Hence $\{0, 1\} = \{1, 0\}$.

($\subseteq$) Let $a \in \{0, 0\}$. Then $a = 0$ or $a = 0$, so $a = 0$, so $a \in \{0\}$. ($\supseteq$) Let $a \in \{0\}$; then $a = 0$, so $a \in \{0, 0\}$. Hence $\{0, 0\} = \{0\}$. $\square$

Empty and non-empty sets**Definition 2.1.19, Strategy 2.1.20, Theorem 2.1.25, Notation 2.1.26**

X is **non-empty** if it has at least one element ($\exists a, a \in X$); otherwise **empty** ($\neg \exists a, a \in X$). **To prove non-empty:** exhibit an element. **To prove empty:** assume there is $a \in X$ and derive a contradiction. Any two empty sets are equal (2.1.25), so there is one **empty set**, $\emptyset$. Examples 2.1.21–22: $\{x \in \mathbb{Q} \mid x^2 = 2\}$ and $[0]$ are empty.

Exercise 2.1.23**recommended**

Let $a, b \in \mathbb{R}$. Prove that $[a, b]$ is empty iff $a > b$, and that (a, b) is empty iff $a \geq b$.

$[a, b]$. ($\Leftarrow$) Suppose $a > b$. If $x \in [a, b]$ then $a \leq x \leq b$, so $a \leq b$, contradiction. So $[a, b]$ is empty. ($\Rightarrow$) By contraposition: suppose $a \leq b$. Then $a \leq a \leq b$, so $a \in [a, b]$ and $[a, b]$ is non-empty.

(a, b) . ($\Leftarrow$) Suppose $a \geq b$. If $x \in (a, b)$ then $a < x < b$, so $a < b$, contradiction. ($\Rightarrow$) By contraposition: suppose $a < b$. Then $\frac{a+b}{2} \in (a, b)$ (as in Exercise 1.2.29), so (a, b) is non-empty. $\square$

Exercise 2.1.24**recommended**

Let E be empty and $p(x)$ a predicate on E . Show that $\forall x \in E, p(x)$ is true and $\exists x \in E, p(x)$ is false. What does $\forall x \in E, x \neq x$ mean, and is it true?

$\forall x \in E, p(x)$ means $\forall x, (x \in E \Rightarrow p(x))$ (Definition 2.1.2). Let x be arbitrary and suppose $x \in E$. Since E is empty this is a contradiction, so $p(x)$ follows by the principle of explosion (Axiom 1.1.67). Hence the implication holds for every x , and the statement is true (“vacuously”).

$\exists x \in E, p(x)$ means $\exists x, (x \in E \wedge p(x))$; if it were true there would be some $x \in E$, contradicting emptiness. So it is false.

$\forall x \in E, x \neq x$ says “every element of the empty set is different from itself”. It is true, by the first part with $p(x)$ being $x \neq x$. $\square$

Exercise 2.1.27**essential**

Is it true that $\emptyset \in X$ for all sets X ? Is it true that $\emptyset \subseteq X$ for all sets X ?

$\emptyset \in X$ for all X is **false**: take $X = \emptyset$; it has no elements, so $\emptyset \notin \emptyset$.

$\emptyset \subseteq X$ for all X is **true**: by Definition 2.1.6 we need $\forall a \in \emptyset, a \in X$, which holds by Exercise 2.1.24 (vacuously). $\square$

NOW YOU 2.1.28 · solutions in Appendix S

2.1.28 Why are $\emptyset$, $\{\emptyset\}$, $\{\{\emptyset\}\}$ different? Truth values of $\emptyset \in \emptyset$, $\emptyset \subseteq \emptyset$, $\emptyset \in \{\emptyset\}$, $\emptyset \in \{\{\emptyset\}\}$, $\{\emptyset\} \subseteq \{\emptyset\}$, $\{\emptyset\} \subseteq \{\{\emptyset\}\}$; find more.

Power sets**Definition 2.1.29, Example 2.1.34**

$\mathcal{P}(X)$ is the set of all subsets of X : $U \in \mathcal{P}(X) \Leftrightarrow U \subseteq X$. $\mathcal{P}(\{1, 2\}) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$. Confusions come from mixing $\in$ (element) with $\subseteq$ (subset): $U \subseteq X$ is the same as $U \in \mathcal{P}(X)$.

Exercise 2.1.31**essential**

Write out the elements of $\mathcal{P}(\{1, 2, 3\})$.

$\mathcal{P}(\{1, 2, 3\}) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$: eight subsets, one for each choice of which of 1, 2, 3 to include. $\square$

Exercise 2.1.32**recommended**

Let X be a set. Show that $\emptyset \in \mathcal{P}(X)$ and $X \in \mathcal{P}(X)$.

By Exercise 2.1.27, $\emptyset \subseteq X$, so $\emptyset \in \mathcal{P}(X)$ by Definition 2.1.29. By Example 2.1.8, $X \subseteq X$, so $X \in \mathcal{P}(X)$. $\square$

Exercise 2.1.35**essential**

Let X and Y be sets. Prove that if $X \subseteq Y$, then $\mathcal{P}(X) \subseteq \mathcal{P}(Y)$.

Suppose $X \subseteq Y$. Let $U \in \mathcal{P}(X)$. Then $U \subseteq X$ (Definition 2.1.29). Since $U \subseteq X$ and $X \subseteq Y$, transitivity (Proposition 2.1.12) gives $U \subseteq Y$, so $U \in \mathcal{P}(Y)$. Hence $\mathcal{P}(X) \subseteq \mathcal{P}(Y)$ by Strategy 2.1.7. $\square$

Exercise 2.1.36**recommended**

True or false? (a) $\mathcal{P}(\emptyset) \in \{\emptyset\}$; (b) $\mathcal{P}(\emptyset) \in \mathcal{P}(\mathcal{P}(\emptyset))$; (c) $\mathcal{P}(\mathcal{P}(\emptyset)) \in \{\emptyset, \{\emptyset, \{\emptyset\}\}\}$. Repeat with $\in$ replaced by $\subseteq$.

First compute: $\mathcal{P}(\emptyset) = \{\emptyset\}$ and $\mathcal{P}(\mathcal{P}(\emptyset)) = \mathcal{P}(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}$.

(a) False: the only element of $\{\emptyset\}$ is $\emptyset$, and $\{\emptyset\} \neq \emptyset$ (it has an element).

(b) True: $\mathcal{P}(\emptyset) \subseteq \mathcal{P}(\emptyset)$ (Example 2.1.8), so $\mathcal{P}(\emptyset) \in \mathcal{P}(\mathcal{P}(\emptyset))$ by Definition 2.1.29.

(c) True: $\mathcal{P}(\mathcal{P}(\emptyset)) = \{\emptyset, \{\emptyset\}\}$ is the second listed element.

With $\subseteq$: (a') $\{\emptyset\} \subseteq \{\emptyset\}$: true (Example 2.1.8). (b') $\{\emptyset\} \subseteq \{\emptyset, \{\emptyset\}\}$: true, its only element $\emptyset$ is listed.

(c') $\{\emptyset, \{\emptyset\}\} \subseteq \{\emptyset, \{\emptyset, \{\emptyset\}\}\}$: false, because $\{\emptyset\}$ is an element of the left set but not of the right one (the right set's elements are $\emptyset$ and $\{\emptyset, \{\emptyset\}\}$, and $\{\emptyset\}$ differs from both). $\square$

NOW YOU 2.1.33 · solutions in Appendix S

2.1.33 Write out $\mathcal{P}(\emptyset)$, $\mathcal{P}(\mathcal{P}(\emptyset))$ and $\mathcal{P}(\mathcal{P}(\mathcal{P}(\emptyset)))$.

B SET OPERATIONS**Intersection****Definitions 2.2.1, 2.2.2, 2.2.10, Notation 2.2.13**

The **intersection** of a collection of sets is the set of objects that are elements of all of them. $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$, so $x \in A \cap B \Leftrightarrow x \in A \wedge x \in B$. A and B are **disjoint** if $A \cap B$ is empty. $A_0 \cap \cdots \cap A_{n-1} = \{x \mid x \in A_i \text{ for all } i \in [n]\}$.

Exercise 2.2.4**recommended**

Prove that $[0, \infty) \cap \mathbb{Z} = \mathbb{N}$.

($\subseteq$) Let $x \in [0, \infty) \cap \mathbb{Z}$. Then $x \in [0, \infty)$ and $x \in \mathbb{Z}$, so x is an integer with $x \geq 0$, i.e. a non-negative whole number, so $x \in \mathbb{N}$ (Definition 0.6). ($\supseteq$) Let $n \in \mathbb{N}$. Then n is a whole number, so $n \in \mathbb{Z}$, and $n \geq 0$, so $n \in [0, \infty)$. Hence $n \in [0, \infty) \cap \mathbb{Z}$. $\square$

Exercise 2.2.5

essential

Write down the elements of $\{0, 1, 4, 7\} \cap \{1, 2, 3, 4, 5\}$.

The elements common to both lists are 1 and 4: $\{0, 1, 4, 7\} \cap \{1, 2, 3, 4, 5\} = \{1, 4\}$. □

Exercise 2.2.6

recommended

Express $[-2, 5) \cap [4, 7)$ as a single interval.

x lies in both iff $-2 \leq x < 5$ and $4 \leq x < 7$, iff $4 \leq x < 5$. So $[-2, 5) \cap [4, 7) = [4, 5)$. □

Proposition 2.2.8

$X \subseteq Y$ if and only if $X \cap Y = X$. *Proof:* $(\Rightarrow)$ double containment, using $X \subseteq Y$ for $\supseteq$; $(\Leftarrow)$ if $a \in X = X \cap Y$ then $a \in Y$.

Exercise 2.2.9

recommended

Let X be a set. Prove that $X \cap \emptyset = \emptyset$ and $X \cap X = X$.

$X \cap \emptyset = \emptyset$. Suppose $X \cap \emptyset$ is non-empty, say $a \in X \cap \emptyset$. Then $a \in \emptyset$, contradicting that $\emptyset$ has no elements. So $X \cap \emptyset$ is empty, hence equal to $\emptyset$ (Theorem 2.1.25).

$X \cap X = X$. For any a : $a \in X \cap X \Leftrightarrow a \in X \wedge a \in X \Leftrightarrow a \in X$. So the sets are equal by extensionality (Axiom 2.1.14). □

NOW YOU 2.2.7, 2.2.12, 2.2.15 · solutions in Appendix S

2.2.7 For $A \subseteq \mathbb{R}$, express with $\cap$: (a) the positive elements of A ; (b) the integer elements of A . **2.2.12** Prove that (a, b) and (c, d) are disjoint iff $b \leq c$ or $d \leq a$. **2.2.15** Express the set of non-negative rational elements of A using $\cap$.

Indexed families**Definition 2.2.16, Notation 2.2.18**

An **indexed family** over a set I assigns a set A_i to each $i \in I$: $\{A_i \mid i \in I\}$. $\bigcap_{i \in I} A_i = \{x \mid x \in A_i \text{ for all } i \in I\}$, so $x \in \bigcap_{i \in I} A_i \Leftrightarrow \forall i \in I, x \in A_i$. For $I = \{n \in \mathbb{N} \mid n \geq 1\}$ one writes $\bigcap_{n \geq 1} A_n$ or $\bigcap_{n=1}^{\infty} A_n$.

Example 2.2.19 · the technique for intersections over $n \geq 1$

To prove $\bigcap_{n \geq 1} [0, 1 + \frac{1}{n}] = [0, 1]$: $(\supseteq)$ let $x \in [0, 1]$ and let $n \geq 1$ be arbitrary; check $x < 1 + \frac{1}{n}$. $(\subseteq)$ let x be in every $[0, 1 + \frac{1}{n}]$ and suppose $x > 1$; choose $N = \lfloor \frac{1}{x-1} \rfloor + 1$, which gives $N \geq 1$ and $N \geq \frac{1}{x-1}$, hence $x \geq 1 + \frac{1}{N}$, contradicting $x \in [0, 1 + \frac{1}{N}]$. **The invention is always the same: build an index N from x that breaks the inequality.**

NOW YOU 2.2.20 · solutions in Appendix S

2.2.20 For $n \in \mathbb{Z}$ let $A_n = \{x \in \mathbb{R} \mid x \neq n\}$. Describe $\bigcap_{n \in \mathbb{Z}} A_n$ in plain terms.

Union

Definitions 2.2.21, 2.2.22, Notations 2.2.29, 2.2.33

The **union** of a collection of sets is the set of objects that are elements of at least one of them. $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$, so $x \in A \cup B \Leftrightarrow x \in A \vee x \in B$. $A_0 \cup \dots \cup A_{n-1} = \{x \mid x \in A_i \text{ for some } i \in [n]\}$. $\bigcup_{i \in I} A_i = \{x \mid x \in A_i \text{ for some } i \in I\}$, so $x \in \bigcup_{i \in I} A_i \Leftrightarrow \exists i \in I, x \in A_i$. Example 2.2.23: $E \cup O = \mathbb{Z}$ and $E \cap O = \emptyset$ for the even and odd integers.

Exercise 2.2.24**essential**

Write down the elements of $\{0, 1, 4, 7\} \cup \{1, 2, 3, 4, 5\}$.

Everything in either list, each element once: $\{0, 1, 2, 3, 4, 5, 7\}$. □

Exercise 2.2.25**recommended**

Express $[-2, 5) \cup [4, 7)$ as a single interval.

x lies in the union iff $-2 \leq x < 5$ or $4 \leq x < 7$; since the two ranges overlap on $[4, 5)$, this is iff $-2 \leq x < 7$. So $[-2, 5) \cup [4, 7) = [-2, 7)$. □

Exercise 2.2.26**essential**

Let X and Y be sets. Prove that $X \subseteq Y$ if and only if $X \cup Y = Y$.

($\Rightarrow$) Suppose $X \subseteq Y$; we prove $X \cup Y = Y$ by double containment. ($\subseteq$) Let $a \in X \cup Y$. Then $a \in X$ or $a \in Y$. If $a \in X$ then $a \in Y$ since $X \subseteq Y$; if $a \in Y$ then $a \in Y$. In both cases $a \in Y$. ($\supseteq$) Let $a \in Y$. Then $a \in X \cup Y$ by definition of union.

($\Leftarrow$) Suppose $X \cup Y = Y$. Let $a \in X$. Then $a \in X \cup Y$ by definition of union, so $a \in Y$ since $X \cup Y = Y$. Hence $X \subseteq Y$. □

Example 2.2.27 · distributivity of $\cap$ over $\cup$

$X \cap (Y \cup Z) = (X \cap Y) \cup (X \cap Z)$, by double containment; the $\subseteq$ direction splits into cases on $x \in Y \cup Z$.

Exercise 2.2.28**essential**

Let X, Y, Z be sets. Prove that $X \cup (Y \cap Z) = (X \cup Y) \cap (X \cup Z)$.

($\subseteq$) Let $x \in X \cup (Y \cap Z)$. Then $x \in X$ or $x \in Y \cap Z$. If $x \in X$, then $x \in X \cup Y$ and $x \in X \cup Z$, so $x \in (X \cup Y) \cap (X \cup Z)$. If $x \in Y \cap Z$, then $x \in Y$ and $x \in Z$, so $x \in X \cup Y$ and $x \in X \cup Z$, and again $x \in (X \cup Y) \cap (X \cup Z)$. ($\supseteq$) Let $x \in (X \cup Y) \cap (X \cup Z)$, so $x \in X \cup Y$ and $x \in X \cup Z$. By the law of excluded middle, $x \in X$ or $x \notin X$. If $x \in X$, then $x \in X \cup (Y \cap Z)$. If $x \notin X$, then from $x \in X \cup Y$ we get $x \in Y$, and from $x \in X \cup Z$ we get $x \in Z$; so $x \in Y \cap Z$, and $x \in X \cup (Y \cap Z)$. □

NOW YOU 2.2.31, 2.2.32 · solutions in Appendix S

2.2.31 Write $X = \{x \in \mathbb{R} \mid x^4 - 5x^2 + 4 > 0\}$ as a union of intervals. **2.2.32** State and prove the distributivity laws for finitary unions and intersections.

Exercise 2.2.35**essential**

Express $\bigcup_{n \geq 1} [0, 1 - \frac{1}{n}]$ as an interval.

We claim $\bigcup_{n \geq 1} [0, 1 - \frac{1}{n}] = [0, 1)$. ($\subseteq$) Let x be in the union; take $n \geq 1$ with $x \in [0, 1 - \frac{1}{n}]$. Then $0 \leq x \leq 1 - \frac{1}{n} < 1$, so $x \in [0, 1)$. ($\supseteq$) Let $x \in [0, 1)$, so $0 \leq x < 1$ and $1 - x > 0$. Define $N = \lceil \frac{1}{1-x} \rceil$; since $\frac{1}{1-x} \geq 1$ we have $N \geq 1$, and $N \geq \frac{1}{1-x}$ gives $\frac{1}{N} \leq 1 - x$, i.e. $x \leq 1 - \frac{1}{N}$. So $x \in [0, 1 - \frac{1}{N}]$ and hence x is in the union. $\square$

Exercise 2.2.36**essential**

Prove that $\bigcap_{n \in \mathbb{N}} [n] = \emptyset$ and $\bigcup_{n \in \mathbb{N}} [n] = \mathbb{N}$.

Intersection. Suppose $x \in \bigcap_{n \in \mathbb{N}} [n]$. Then $x \in [n]$ for all $n \in \mathbb{N}$, in particular $x \in [0]$. But $[0] = \{k \in \mathbb{N} \mid k \leq 0\}$ is empty (Example 2.1.22): contradiction. So the intersection is empty.

Union. ($\subseteq$) If $x \in [n]$ for some n , then $x \in \mathbb{N}$ by definition of $[n]$. ($\supseteq$) Let $k \in \mathbb{N}$. Then $k < k + 1$ and $k + 1 \in \mathbb{N}$, so $k \in [k + 1]$, hence $k \in \bigcup_{n \in \mathbb{N}} [n]$. $\square$

Exercise 2.2.37**essential**

Find a family $\{X_n \mid n \in \mathbb{N}\}$ with (i) $\bigcup_{n \in \mathbb{N}} X_n = \mathbb{N}$; (ii) $\bigcap_{n \in \mathbb{N}} X_n = \emptyset$; (iii) $X_i \cap X_j \neq \emptyset$ for all $i, j \in \mathbb{N}$.

Take $X_n = \{k \in \mathbb{N} \mid k \neq n\}$ (“ $\mathbb{N}$ with n removed”). (i) Each $X_n \subseteq \mathbb{N}$; and any $k \in \mathbb{N}$ lies in X_{k+1} since $k \neq k + 1$. (ii) If k were in every X_n then in particular $k \in X_k$, i.e. $k \neq k$: contradiction. (iii) Given i, j , the number $i + j + 1$ differs from both i and j , so $i + j + 1 \in X_i \cap X_j$. $\square$

Relative complement**Definition 2.2.38**

$X \setminus A = \{x \in X \mid x \notin A\}$, the **relative complement** of A in X (“ X minus A ”); $x \in X \setminus A \Leftrightarrow x \in X \wedge x \notin A$. The book uses no absolute complement A^c . Example 2.2.39: $\mathbb{Z} \setminus E$ is the set of odd integers.

Exercise 2.2.40**essential**

Write down the elements of $\{0, 1, 4, 7\} \setminus \{1, 2, 3, 4, 5\}$.

Keep the elements of the first set that are not in the second: $\{0, 7\}$. $\square$

Exercise 2.2.41**recommended**

Express $[-2, 5) \setminus [4, 7)$ and $[4, 7) \setminus [-2, 5)$ as intervals.

$x \in [-2, 5) \setminus [4, 7)$ iff $-2 \leq x < 5$ and not $4 \leq x < 7$, iff $-2 \leq x < 4$: the set is $[-2, 4)$.

$x \in [4, 7) \setminus [-2, 5)$ iff $4 \leq x < 7$ and not $-2 \leq x < 5$, iff $5 \leq x < 7$: the set is $[5, 7)$. $\square$

NOW YOU 2.2.42, 2.2.43 · solutions in Appendix S

2.2.42 A = the reals that are not integers: (a) write A as a relative complement; (b) as a union of intervals.

2.2.43 Prove $X \setminus (X \setminus A) = X \cap A$, and deduce that $A \subseteq X$ iff $X \setminus (X \setminus A) = A$.

Sets versus logic

The dictionary (page 92)

With $p: a \in X$, $q: a \in Y$, $r(i): a \in X_i$:

$$\begin{array}{lll} a \notin X & \neg p & a \in \bigcap_{i \in I} X_i \quad \forall i \in I, r(i) \\ a \in X \cap Y & p \wedge q & a \in \bigcup_{i \in I} X_i \quad \exists i \in I, r(i) \\ a \in X \cup Y & p \vee q & a \in X \setminus Y \quad p \wedge (\neg q) \end{array}$$

Every set identity in this section is a logical law in disguise; the proof unfolds membership, applies the law, folds back.

Theorem 2.2.44 · de Morgan's laws for sets

(a) $X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B)$; (b) $X \setminus (A \cap B) = (X \setminus A) \cup (X \setminus B)$; (c) $X \setminus \bigcup_{i \in I} A_i = \bigcap_{i \in I} (X \setminus A_i)$; (d) $X \setminus \bigcap_{i \in I} A_i = \bigcup_{i \in I} (X \setminus A_i)$. The book proves (a) by double containment, using Theorem 1.3.24 to turn $x \notin A \cup B$ into $x \notin A$ and $x \notin B$.

Exercise 2.2.45**recommended**

Complete the proof of de Morgan's laws for sets: parts (b), (c), (d).

(b) ($\subseteq$) Let $x \in X \setminus (A \cap B)$: $x \in X$ and $x \notin A \cap B$, i.e. $\neg(x \in A \wedge x \in B)$. By de Morgan (Theorem 1.3.24(a)), $x \notin A$ or $x \notin B$. If $x \notin A$ then $x \in X \setminus A$; if $x \notin B$ then $x \in X \setminus B$. Either way $x \in (X \setminus A) \cup (X \setminus B)$. ($\supseteq$) Let $x \in (X \setminus A) \cup (X \setminus B)$. If $x \in X \setminus A$: $x \in X$ and $x \notin A$, so $x \notin A \cap B$ (an element of $A \cap B$ would be in A), hence $x \in X \setminus (A \cap B)$. The case $x \in X \setminus B$ is the same.

(c) For any x : $x \in X \setminus \bigcup_i A_i \Leftrightarrow x \in X \wedge \neg(\exists i \in I, x \in A_i) \Leftrightarrow x \in X \wedge \forall i \in I, x \notin A_i$ {Theorem 1.3.29(b)}
 $\Leftrightarrow \forall i \in I, (x \in X \wedge x \notin A_i)$ { $x \in X$ does not depend on i ; see note} $\Leftrightarrow \forall i \in I, x \in X \setminus A_i \Leftrightarrow x \in \bigcap_i (X \setminus A_i)$.

Note: the third step needs I non-empty for the direction $\Leftarrow$; for $I = \emptyset$ one reads the conventions of the book for empty intersections.

(d) For any x : $x \in X \setminus \bigcap_i A_i \Leftrightarrow x \in X \wedge \neg(\forall i \in I, x \in A_i) \Leftrightarrow x \in X \wedge \exists i \in I, x \notin A_i$ {Theorem 1.3.29(a)}
 $\Leftrightarrow \exists i \in I, (x \in X \wedge x \notin A_i) \Leftrightarrow \exists i \in I, x \in X \setminus A_i \Leftrightarrow x \in \bigcup_i (X \setminus A_i)$. $\square$

Watch out

- From $x \in A \cup B$ you get a *disjunction*: do cases. From $x \in A \cap B$ you get two facts. From $x \notin A \cup B$ you get $x \notin A$ and $x \notin B$ (de Morgan), not just one of them.
- $X \setminus A = X \setminus B$ does not give $A = B$ (aside on page 91): the complement operation does not cancel like subtraction.
- An “express as an interval” question still deserves the one-line justification “ x is in the set iff ...”.

C TUPLES AND CARTESIAN PRODUCTS**Definition 2.2.46 · ordered lists (tuples)**

A (finite) **ordered list** is an expression $(x_0, x_1, \dots, x_{k-1})$ with $k \in \mathbb{N}$; k is its **length** and $i \in [k]$ is the **position** of x_i . Two lists are equal exactly when they have the same length and the same objects in all positions:

$$(x_0, \dots, x_{k-1}) = (y_0, \dots, y_{\ell-1}) \Leftrightarrow k = \ell \text{ and } x_i = y_i \text{ for all } i \in [k].$$

A list of length k is a **k -tuple**; $()$ is the empty list; the 1-tuple (x) is identified with x ; 2-, 3-, 4-tuples are **ordered pairs, triples, quadruples**. Example 2.2.47: $(1, 2, 3) \neq (1, 2, 3, 3, 3)$ (different lengths) and $(1, 2, 3) \neq (3, 2, 1)$ (different objects in position 0), although $\{1, 2, 3\} = \{3, 2, 1\} = \{1, 2, 3, 3, 3\}$.

Exercise 2.2.48

from the book

Write down all ordered lists that have all of the elements of $\{1, 2, 3\}$ as terms and have no repeated terms.

Such a list has length 3 (three distinct terms, each of 1, 2, 3 appearing). Choosing the term in position 0, then position 1, leaves the last term forced:

$$(1, 2, 3), (1, 3, 2), (2, 1, 3), (2, 3, 1), (3, 1, 2), (3, 2, 1).$$

These six lists are pairwise different by the equality rule of Definition 2.2.46 (they differ in some position).

□

Definition 2.2.49 and Notation 2.2.54 · cartesian product, A^k

For $k \in \mathbb{N}$ and sets $A_0, \dots, A_{k-1}$,

$$\prod_{i \in [k]} A_i = A_0 \times A_1 \times \dots \times A_{k-1} = \{(x_0, x_1, \dots, x_{k-1}) \mid x_i \in A_i \text{ for all } i \in [k]\}.$$

So $(x, y) \in X \times Y$ means exactly $x \in X$ and $y \in Y$. The **k -fold product** is $A^k = A \times \dots \times A$ (k times), the set of all k -tuples of elements of A ; $A^0 = \{()\}$ and $A^1 = A$. Example 2.2.50: $\mathbb{R} \times \mathbb{R} = \{(x, y) \mid x, y \in \mathbb{R}\}$ is the plane, $\mathbb{R} \times \{0\}$ is the x -axis, and the graph of $y = x^2$ is $\{(x, x^2) \mid x \in \mathbb{R}\} \subseteq \mathbb{R} \times \mathbb{R}$.

Exercise 2.2.51

essential

Write down the elements of the set $\{1, 2\} \times \{3, 4, 5\}$.

By Definition 2.2.49 the elements are the ordered pairs (x, y) with $x \in \{1, 2\}$ and $y \in \{3, 4, 5\}$:

$$\{1, 2\} \times \{3, 4, 5\} = \{(1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5)\}.$$

(Two choices for the first term, three for the second: $2 \cdot 3 = 6$ pairs.)

□

Exercise 2.2.52

from the book

Let X be a set. Prove that $X \times \emptyset = \emptyset$.

By Strategy 2.1.20 we prove that $X \times \emptyset$ has no elements. Suppose $t \in X \times \emptyset$. Then $t = (x, y)$ with $x \in X$ and $y \in \emptyset$ {Definition 2.2.49}. But $\emptyset$ has no elements, so $y \in \emptyset$ is a contradiction. Hence $X \times \emptyset$ is empty, and by Theorem 2.1.25 it equals $\emptyset$.

□

Exercise 2.2.53

recommended

Let X and Y be sets. Under what conditions is it true that $X \times Y = Y \times X$?

Claim. $X \times Y = Y \times X$ if and only if $X = Y$ or $X = \emptyset$ or $Y = \emptyset$.

($\Leftarrow$) If $X = Y$ the two sets are literally the same. If $X = \emptyset$ then $X \times Y = \emptyset \times Y = \emptyset$ (as in Exercise 2.2.52: a pair in it would have a first term in $\emptyset$) and $Y \times X = Y \times \emptyset = \emptyset$ by Exercise 2.2.52, so both sides equal $\emptyset$. The case $Y = \emptyset$ is the same.

($\Rightarrow$) Suppose $X \times Y = Y \times X$, and suppose moreover that $X \neq \emptyset$ and $Y \neq \emptyset$; we prove $X = Y$ by double containment. Let $x \in X$. Since Y is non-empty, fix some $y \in Y$. Then $(x, y) \in X \times Y = Y \times X$, so $x \in Y$ {first term of an element of $Y \times X$ }. Hence $X \subseteq Y$. Symmetrically, let $y \in Y$ and fix $x \in X$; then $(y, x) \in Y \times X = X \times Y$, so $y \in X$. Hence $Y \subseteq X$, and $X = Y$. □

NOW YOU 2.2.55 · solutions in Appendix S

2.2.55 Are $\mathbb{R}^3$, $\mathbb{R}^2 \times \mathbb{R}$, $\mathbb{R} \times \mathbb{R}^2$, $\mathbb{R} \times \mathbb{R} \times \mathbb{R}$, $(\mathbb{R} \times \mathbb{R}) \times \mathbb{R}$, $\mathbb{R} \times (\mathbb{R} \times \mathbb{R})$ all equal? Which pairs are equal, and in what other sense are they all equivalent? [Hint: write down a typical element of each: is it a triple, or a pair one of whose terms is a pair?]

D CHAPTER 2 EXERCISES**Sets, subsets and power sets (2.E.1–2.E.7)****What the chapter exercises test**

Everything in 2.E is done with four moves from §2.1–§2.2: **(1)** unfold membership (“ $a \in X \setminus V$ means $a \in X$ and $a \notin V$ ”); **(2)** prove $\subseteq$ by “let $a \in \dots$ ” (Strategy 2.1.7) and $=$ by double containment (Strategy 2.1.15); **(3)** prove a set is empty by assuming it has an element and deriving a contradiction (Strategy 2.1.20); **(4)** to refute a general claim, give a concrete counterexample, usually built from $\emptyset$, $\{0\}$, $\{1\}$, $\{0, 1\}$. Remember $\mathcal{P}(X) = \{U \mid U \subseteq X\}$, so $U \in \mathcal{P}(X) \Leftrightarrow U \subseteq X$ (Definition 2.1.29), and $\emptyset \subseteq X$ always (Exercise 2.1.27).

Exercise 2.E.4**recommended**

Let X be a set and let $U, V \subseteq X$. Prove that U and V are disjoint if and only if $U \subseteq X \setminus V$.

($\Rightarrow$) Suppose $U \cap V = \emptyset$. Let $a \in U$. Then $a \in X$ since $U \subseteq X$. If $a \in V$ we would have $a \in U \cap V = \emptyset$, a contradiction; so $a \notin V$. Hence $a \in X \setminus V$ {Definition 2.2.38}, and $U \subseteq X \setminus V$.

($\Leftarrow$) Suppose $U \subseteq X \setminus V$. Suppose towards a contradiction that $a \in U \cap V$. Then $a \in U$, so $a \in X \setminus V$, so $a \notin V$; but also $a \in V$. Contradiction. So $U \cap V$ is empty: U and V are disjoint. $\square$

Exercise 2.E.5**recommended**

For each statement, determine whether it is true for all sets A and X , and prove your claim.

(a) If $X \setminus A = \emptyset$ then $X = A$. (b) If $X \setminus A = X$ then $A = \emptyset$. (c) If $X \setminus A = A$ then $A = \emptyset$. (d) $X \setminus (X \setminus A) = A$.

(a) *False*. Take $X = \emptyset$ and $A = \{0\}$. Then $X \setminus A = \emptyset$ but $X \neq A$. (In general $X \setminus A = \emptyset$ says only that $X \subseteq A$.)

(b) *False*. Take $X = \emptyset$ and $A = \{0\}$. Then $X \setminus A = \emptyset = X$ but $A \neq \emptyset$. (In general $X \setminus A = X$ says only that X and A are disjoint.)

(c) *True*. Suppose $X \setminus A = A$ and suppose towards a contradiction that $a \in A$. Then $a \in X \setminus A$, so $a \notin A$. Contradiction. Hence A is empty.

(d) *False*. Take $X = \emptyset$ and $A = \{0\}$: $X \setminus (X \setminus A) = \emptyset \setminus \emptyset = \emptyset \neq A$. (By Exercise 2.2.43, $X \setminus (X \setminus A) = X \cap A$, which equals A exactly when $A \subseteq X$.) $\square$

Exercise 2.E.6**essential**

For each statement, determine whether it is true for all sets X, Y , false for all sets X, Y , or true for some and false for others.

(a) $\mathcal{P}(X \cup Y) = \mathcal{P}(X) \cup \mathcal{P}(Y)$ (b) $\mathcal{P}(X \cap Y) = \mathcal{P}(X) \cap \mathcal{P}(Y)$ (c) $\mathcal{P}(X \times Y) = \mathcal{P}(X) \times \mathcal{P}(Y)$
(d) $\mathcal{P}(X \setminus Y) = \mathcal{P}(X) \setminus \mathcal{P}(Y)$

(a) *Sometimes*. True for $X = Y = \emptyset$: both sides are $\mathcal{P}(\emptyset) = \{\emptyset\}$. False for $X = \{0\}$, $Y = \{1\}$:

$\{0, 1\} \in \mathcal{P}(X \cup Y)$, but $\{0, 1\} \not\subseteq X$ and $\{0, 1\} \not\subseteq Y$, so $\{0, 1\} \notin \mathcal{P}(X) \cup \mathcal{P}(Y)$. (The inclusion $\supseteq$ always holds: a subset of X or of Y is a subset of $X \cup Y$.)

(b) *Always.* For any U : $U \in \mathcal{P}(X \cap Y) \Leftrightarrow U \subseteq X \cap Y \Leftrightarrow (U \subseteq X \wedge U \subseteq Y) \Leftrightarrow U \in \mathcal{P}(X) \wedge U \in \mathcal{P}(Y) \Leftrightarrow U \in \mathcal{P}(X) \cap \mathcal{P}(Y)$. The middle step: if $U \subseteq X \cap Y$ then each $a \in U$ lies in $X \cap Y$, hence in X and in Y ; conversely if $U \subseteq X$ and $U \subseteq Y$ then each $a \in U$ lies in X and in Y , hence in $X \cap Y$.

(c) *Never.* The elements of $\mathcal{P}(X \times Y)$ are *sets* (subsets of $X \times Y$), whereas the elements of $\mathcal{P}(X) \times \mathcal{P}(Y)$ are *ordered pairs* (U, V) . In particular $\emptyset \in \mathcal{P}(X \times Y)$ (Exercise 2.1.27), but $\emptyset$ is not an ordered pair, so $\emptyset \notin \mathcal{P}(X) \times \mathcal{P}(Y)$. Hence the two sets are never equal.

(d) *Never.* $\emptyset \in \mathcal{P}(X \setminus Y)$ always. But $\emptyset \in \mathcal{P}(Y)$, so $\emptyset \notin \mathcal{P}(X) \setminus \mathcal{P}(Y)$. Hence the sets are never equal. $\square$

NOW YOU 2.E.1, 2.E.2, 2.E.3, 2.E.7 • solutions in Appendix S

2.E.1 Express: (a) $\{n \in \mathbb{Z} \mid n^2 < 20\}$ in list notation; (b) $\{4k+3 \mid k \in \mathbb{N}\}$ in implied list notation; (c) the odd multiples of six in set-builder notation; (d) $\{1, 2, 5, 10, 17, \dots, n^2+1, \dots\}$ in set-builder notation. **2.E.2** Find sets X_n ($n \in \mathbb{N}$) with $X_{n+1} \subsetneq X_n$ for all n . Can any X_n be empty? **2.E.3** Express $\mathcal{P}(\{\emptyset, \{\emptyset, \{\emptyset\}\})$ in list notation. **2.E.7** F is a set of sets with $\forall A \in F, \forall x \in A, x \in F$. Prove $F \subseteq \mathcal{P}(F)$.

Symmetric difference (2.E.8–2.E.14)

Definition 2.E.8 • symmetric difference

$X \triangle Y = \{a \mid a \in X \text{ or } a \in Y \text{ but not both}\}$. In the dictionary of page 92: $a \in X \triangle Y$ is $(p \vee q) \wedge \neg(p \wedge q)$ where p : $a \in X$, q : $a \in Y$. The book's hint for 2.E.13–14: rather than long chains of equations, use double containment with cases, or “a cunning use of truth tables” (a truth table in p, q, r decides any identity built from $\cap, \cup, \setminus, \triangle$).

Exercise 2.E.9

from the book

Prove that $X \triangle Y = (X \setminus Y) \cup (Y \setminus X) = (X \cup Y) \setminus (X \cap Y)$ for all sets X and Y .

Let a be arbitrary and write p for $a \in X$ and q for $a \in Y$. Then $a \in X \triangle Y$ is $(p \vee q) \wedge \neg(p \wedge q)$; $a \in (X \setminus Y) \cup (Y \setminus X)$ is $(p \wedge \neg q) \vee (q \wedge \neg p)$; and $a \in (X \cup Y) \setminus (X \cap Y)$ is $(p \vee q) \wedge \neg(p \wedge q)$, literally the same formula as the first. For the middle one:

p	q	$(p \vee q) \wedge \neg(p \wedge q)$	$(p \wedge \neg q) \vee (q \wedge \neg p)$
✓	✓	×	×
✓	×	✓	✓
×	✓	✓	✓
×	×	×	×

The columns agree, so the three membership conditions are equivalent for every a ; by set extensionality (Axiom 2.1.14) the three sets are equal. $\square$

NOW YOU 2.E.10–2.E.14 • solutions in Appendix S

2.E.10 $X \triangle X = \emptyset$ and $X \triangle \emptyset = X$. **2.E.11** $X = Y \Leftrightarrow X \triangle Y = \emptyset$. **2.E.12** X, Y disjoint $\Leftrightarrow X \triangle Y = X \cup Y$. **2.E.13** $X \triangle (Y \triangle Z) = (X \triangle Y) \triangle Z$. **2.E.14** $X \cap (Y \triangle Z) = (X \cap Y) \triangle (X \cap Z)$.

Open subsets of $\mathbb{R}$ (2.E.15–2.E.20)

Definition 2.E.15 · open set

$U \subseteq \mathbb{R}$ is **open** if for all $a \in U$ there exists $\delta > 0$ such that $(a - \delta, a + \delta) \subseteq U$. **To prove U open:** “Let $a \in U$. Put $\delta = \dots$ {in terms of a }; then $\delta > 0$, and if $a - \delta < x < a + \delta$ then $\dots$ so $x \in U$.” **To prove U not open:** negate $(\exists a \in U, \forall \delta > 0, (a - \delta, a + \delta) \subseteq U)$: “Take $a = \dots \in U$. Let $\delta > 0$. Then $x = \dots \in (a - \delta, a + \delta)$ but $x \notin U$.”

Exercise 2.E.16

recommended

For each subset of $\mathbb{R}$, determine (with proof) whether it is open: (a) $\emptyset$; (b) $(0, 1)$; (c) $(0, 1]$; (d) $\mathbb{Z}$; (e) $\mathbb{R} \setminus \mathbb{Z}$; (f) $\mathbb{Q}$.

(a) *Open.* The condition “for all $a \in \emptyset, \dots$ ” is vacuously true (Exercise 2.1.24).

(b) *Open.* Let $a \in (0, 1)$ and put $\delta = \min(a, 1 - a)$; then $\delta > 0$ since $0 < a < 1$. If $x \in (a - \delta, a + \delta)$ then $x > a - \delta \geq a - a = 0$ and $x < a + \delta \leq a + (1 - a) = 1$, so $x \in (0, 1)$. Hence $(a - \delta, a + \delta) \subseteq (0, 1)$.

(c) *Not open.* Take $a = 1 \in (0, 1]$ and let $\delta > 0$. Then $x = 1 + \frac{\delta}{2}$ satisfies $1 - \delta < x < 1 + \delta$, but $x > 1$, so $x \notin (0, 1]$. Hence no δ works for $a = 1$.

(d) *Not open.* Take $a = 0 \in \mathbb{Z}$ and let $\delta > 0$. Put $x = \frac{1}{2} \min(\delta, 1)$. Then $0 < x < \delta$, so $x \in (-\delta, \delta)$, and $0 < x \leq \frac{1}{2}$, so x is not an integer.

(e) *Open.* Let $a \in \mathbb{R} \setminus \mathbb{Z}$ and let $n = \lfloor a \rfloor$, the greatest integer $\leq a$; since $a \notin \mathbb{Z}$ we have $n < a < n + 1$. Put $\delta = \min(a - n, n + 1 - a) > 0$. If $x \in (a - \delta, a + \delta)$ then $n \leq a - \delta < x < a + \delta \leq n + 1$, so x lies strictly between the consecutive integers n and $n + 1$ and is not an integer: $x \in \mathbb{R} \setminus \mathbb{Z}$.

(f) *Not open.* Take $a = 0 \in \mathbb{Q}$ and let $\delta > 0$. Choose $n \in \mathbb{N}$ with $n > \sqrt{2}/\delta$ (the natural numbers are unbounded in $\mathbb{R}$) and put $x = \sqrt{2}/n$. Then $0 < x < \delta$, so $x \in (-\delta, \delta)$; but $x \notin \mathbb{Q}$, for if $\sqrt{2}/n = r \in \mathbb{Q}$ then $\sqrt{2} = nr \in \mathbb{Q}$, contradicting Assumption 0.49. $\square$

Exercise 2.E.18

bonus

(a) Let $n \geq 1$ and let $U_1, \dots, U_n$ be open subsets of $\mathbb{R}$. Prove that $U_1 \cap \dots \cap U_n$ is open. (b) Prove that $(0, 1 + \frac{1}{n})$ is open for all $n \geq 1$, but that $\bigcap_{n \geq 1} (0, 1 + \frac{1}{n})$ is not open.

(a) Let $a \in U_1 \cap \dots \cap U_n$. For each i we have $a \in U_i$, and U_i is open, so there is $\delta_i > 0$ with $(a - \delta_i, a + \delta_i) \subseteq U_i$. Put $\delta = \min(\delta_1, \dots, \delta_n)$; then $\delta > 0$ (a minimum of finitely many positive numbers). If $x \in (a - \delta, a + \delta)$ then, for each i , $a - \delta_i \leq a - \delta < x < a + \delta \leq a + \delta_i$, so $x \in U_i$. Hence $x \in U_1 \cap \dots \cap U_n$, and $(a - \delta, a + \delta) \subseteq U_1 \cap \dots \cap U_n$.

(b) Fix $n \geq 1$. Let $a \in (0, 1 + \frac{1}{n})$ and put $\delta = \min(a, 1 + \frac{1}{n} - a) > 0$. If $a - \delta < x < a + \delta$ then $x > 0$ and $x < 1 + \frac{1}{n}$, so $(a - \delta, a + \delta) \subseteq (0, 1 + \frac{1}{n})$: the interval is open.

Now $\bigcap_{n \geq 1} (0, 1 + \frac{1}{n}) = (0, 1]$. ($\supseteq$) If $0 < x \leq 1$ then $0 < x < 1 + \frac{1}{n}$ for every $n \geq 1$. ($\subseteq$) Let x be in every $(0, 1 + \frac{1}{n})$; then $x > 0$. If $x > 1$, choose $n \geq 1$ with $n > \frac{1}{x-1}$; then $\frac{1}{n} < x - 1$, i.e. $1 + \frac{1}{n} < x$, so $x \notin (0, 1 + \frac{1}{n})$, a contradiction. Hence $x \leq 1$. Finally $(0, 1]$ is not open by Exercise 2.E.16(c). So an intersection of infinitely many open sets need not be open: the proof of (a) breaks down because infinitely many δ_i need not have a positive minimum. $\square$

NOW YOU 2.E.17, 2.E.19, 2.E.20 · solutions in Appendix S

2.E.17 U is open iff for all $a \in U$ there are $u < a < v$ with $(u, v) \subseteq U$. **2.E.19** U is open iff $U = \bigcup_{i \in I} (a_i, b_i)$ for some family of open intervals. **2.E.20** Families $\{A_i\}$, $\{B_j\}$ with (i) $\forall i \exists j \geq i, B_j \subseteq A_i$ and (ii) $\forall j \exists i \geq j, A_i \subseteq B_j$: prove $\bigcap_i A_i = \bigcap_j B_j$.

True–false and always–sometimes–never (2.E.21–2.E.42)

How to answer these

True/false: decide, then prove the statement or exhibit a counterexample. **Always:** prove the conclusion from the hypotheses for arbitrary $X, Y, \dots$. **Never:** prove the *negation* of the conclusion from the hypotheses. **Sometimes:** give one concrete example satisfying the hypotheses where the conclusion holds and one where it fails. Tiny sets $(\emptyset, \{0\}, \{1\}, \{0, 1\}, \{\emptyset\})$ settle almost every “sometimes”.

Exercise 2.E.28**recommended**

Let a, b, c be arbitrary objects. Then $\{a, b\} = \{a, c\}$.

Sometimes. If $b = c$ (for instance $a = b = c = 0$) the two sets are the same set, so the conclusion holds. If $a = 0, b = 1, c = 2$ then $1 \in \{a, b\}$ but $1 \notin \{a, c\} = \{0, 2\}$, so $\{a, b\} \neq \{a, c\}$ by Axiom 2.1.14. $\square$

Exercise 2.E.32**recommended**

Let X and Y be sets with $X \cap Y = \emptyset$. Then $\forall a, a \in X \Rightarrow a \notin Y$.

Always. Let a be arbitrary and suppose $a \in X$. If $a \in Y$, then $a \in X \cap Y = \emptyset$, which is impossible. Hence $a \notin Y$. (This is just the unfolding of “ $X \cap Y$ is empty”.) $\square$

Exercise 2.E.33**essential**

Let X and Y be sets with $X \neq Y$. Then $\exists a, a \in X \wedge a \notin Y$.

Sometimes. Take $X = \{0\}$ and $Y = \emptyset$: $X \neq Y$, and $a = 0$ satisfies $a \in X$ and $a \notin Y$. Now take $X = \emptyset$ and $Y = \{0\}$: $X \neq Y$, but there is no a with $a \in X$ at all, so the conclusion is false.

Why: $X \neq Y$ means, by Axiom 2.1.14, that $\neg \forall a, (a \in X \Leftrightarrow a \in Y)$, i.e. some a lies in exactly one of X, Y : either $a \in X \setminus Y$ or $a \in Y \setminus X$. The hypothesis does not tell you which, so the one-sided conclusion can fail. $\square$

Exercise 2.E.38**essential**

Let X and Y be sets such that $\mathcal{P}(X) = \mathcal{P}(Y)$. Then $X = Y$.

Always. Since $X \subseteq X$ we have $X \in \mathcal{P}(X) = \mathcal{P}(Y)$, so $X \subseteq Y$. Likewise $Y \in \mathcal{P}(Y) = \mathcal{P}(X)$, so $Y \subseteq X$. Hence $X = Y$ by double containment. $\square$

Exercise 2.E.41**recommended**

Let X and Y be sets. Then $X \cup Y = X \cap Y$.

Sometimes. For $X = Y$ it holds: $X \cup X = X = X \cap X$ (Exercise 2.2.26 with $X \subseteq X$, and Exercise 2.2.9). For $X = \{0\}, Y = \emptyset$ it fails: $X \cup Y = \{0\} \neq \emptyset = X \cap Y$.

In fact $X \cup Y = X \cap Y$ if and only if $X = Y$: if $X \cup Y = X \cap Y$ and $a \in X$ then $a \in X \cup Y = X \cap Y$, so $a \in Y$; symmetrically $Y \subseteq X$. $\square$

NOW YOU 2.E.21–2.E.27, 2.E.29–2.E.31, 2.E.34–2.E.37, 2.E.39, 2.E.40, 2.E.42 · solutions in Appendix S

True/false: **2.E.21** $\neg \forall x, x \in E$ implies $E = \emptyset$. **2.E.22** $\forall x, \neg x \in E$ implies $E = \emptyset$. **2.E.23** $\{1, 2, 3\} = \{1, 2, 1, 3, 2, 1\}$. **2.E.24** $\emptyset \in \emptyset$. **2.E.25** $\emptyset \in \{\emptyset\}$. **2.E.26** $\emptyset \in \{\{\emptyset\}\}$.

Always/sometimes/never: **2.E.27** $\{a, b\} = \{b, a\}$. **2.E.29** some a with $a \in X \Leftrightarrow a \in Y$; then $X = Y$. **2.E.30**

some $a \in X$ with $a \notin Y$; then $X = Y$. **2.E.31** $X \neq Y$; then $\forall a, a \in X \Rightarrow a \notin Y$. **2.E.34** X, Y sets of sets, $X \in Y$; then $X \subseteq Y$. **2.E.35** $\emptyset \in \mathcal{P}(X)$. **2.E.36** $\{\emptyset\} \in \mathcal{P}(X)$. **2.E.37** $\{\emptyset, X\} \subseteq \mathcal{P}(X)$. **2.E.39** $X \setminus Y = \emptyset$; then $X = Y$. **2.E.40** $X \setminus Y = Z$ and $Y \subseteq Z$; then $X = Y \cup Z$. **2.E.42** $A \subseteq X$; then $A \in X$.

Chapter 3

Functions

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- B Injections and surjections (§3.2)
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Planning

- ★ 3.1: 11, 15abc, 36
 - ★ 3.2: 5, 8, 12, 14, 20, 40, 47
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A FUNCTIONS

Definition 3.1.1 · function, domain, codomain

A **function** f from a set X to a set Y is a specification of elements $f(x) \in Y$ for $x \in X$ such that

$$\forall x \in X, \exists! y \in Y, y = f(x).$$

$f(x)$ is the **value** of f at x ; X is the **domain**, Y the **codomain**; $f : X \rightarrow Y$ asserts that f is a function with this domain and codomain. Any true proposition $\forall x \in X, \exists! y \in Y, p(x, y)$ defines a function $X \rightarrow Y$: send x to the unique y with $p(x, y)$ (Example 3.1.2: $r : (0, \infty) \rightarrow (0, \infty)$, $r(x) = \sqrt{x}$).

Well-definedness (page 101) · the three checks

A specification of $f : X \rightarrow Y$ is **well-defined** if it satisfies: **totality** (a value $f(x)$ is specified for *each* $x \in X$), **existence** (the specified value exists and lies in Y), **uniqueness** (the specification refers to only one element of Y). Usually only uniqueness needs an argument, but check all three. Ways to specify: **lists** (finite X), **formulae** ($f(x) = x^2 + 3$), **by cases** (the conditions must be *exhaustive* and *compatible*: where two cases overlap they must give the same value, like $|x|$ at $x = 0$), **algorithms** (must be deterministic).

Exercise 3.1.15

essential (a)–(c)

Which of the following specifications of functions are well-defined?

- (a) $g : \mathbb{Q} \rightarrow \mathbb{Q}$ defined by $(x+1)g(x) = 1$ for all $x \in \mathbb{Q}$; (b) $h : \mathbb{N} \rightarrow \mathbb{Q}$ defined by $(x+1)h(x) = 1$ for all $x \in \mathbb{N}$;
 (c) $k : \mathbb{N} \rightarrow \mathbb{N}$ defined by $(x+1)k(x) = 1$ for all $x \in \mathbb{N}$; (d) $\ell : \mathbb{N} \rightarrow \mathbb{N}$ defined by $\ell(x) = \ell(x)$ for all $x \in \mathbb{N}$.

(a) *Not well-defined.* For $x = -1 \in \mathbb{Q}$ the condition reads $0 \cdot g(-1) = 1$, which no rational number $g(-1)$ satisfies: existence fails at $x = -1$.

(b) *Well-defined.* For $x \in \mathbb{N}$ we have $x+1 \geq 1$, so $x+1 \neq 0$ and the equation has exactly one solution $h(x) = \frac{1}{x+1}$, which is a rational number. Totality, existence and uniqueness all hold.

(c) *Not well-defined.* As in (b) the only candidate is $k(x) = \frac{1}{x+1}$, but for $x = 1$ this is $\frac{1}{2} \notin \mathbb{N}$: existence (of

a value in the codomain) fails.

(d) *Not well-defined.* The equation $\ell(x) = \ell(x)$ is satisfied by every natural number, so it does not single out one value: uniqueness fails. $\square$

Exercise 3.1.16

recommended

Find a condition on sets X and Y such that the specification of $i : X \cup Y \rightarrow \{0, 1\}$ given by $i(z) = 0$ if $z \in X$ and $i(z) = 1$ if $z \in Y$ is well-defined.

The cases are exhaustive: every $z \in X \cup Y$ is in X or in Y . They are compatible exactly when no z falls under both with different values. If $z \in X \cap Y$ the two cases prescribe $i(z) = 0$ and $i(z) = 1$, which differ, so uniqueness fails. Hence the specification is well-defined if and only if $X \cap Y = \emptyset$ (X and Y are disjoint). (Then $i = \chi_Y$ in the notation of Definition 3.1.23.) $\square$

NOW YOU 3.1.3 · solutions in Appendix S

3.1.3 Which of the statements (a), (b), (c) of Exercise 1.2.37 has the form $\forall x \in X, \exists! y \in Y, p(x, y)$? For each such statement give the domain, codomain and an expression for the function.

Function equality and graphs

Axiom 3.1.4 and Strategy 3.1.5 · function extensionality

$f : X \rightarrow Y$ and $g : A \rightarrow B$ are equal iff (i) $X = A$ and $Y = B$, and (ii) $f(x) = g(x)$ for all $x \in X$. **To prove** $f = g$ for $f, g : X \rightarrow Y$: “Let $x \in X$. Then $f(x) = \dots = g(x)$.” (Two functions given by different procedures with the same values are equal.)

Definition 3.1.6 and Theorem 3.1.9 · graphs

$\text{Gr}(f) = \{(x, f(x)) \mid x \in X\} = \{(x, y) \in X \times Y \mid y = f(x)\} \subseteq X \times Y$. A subset $G \subseteq X \times Y$ is the graph of a function $X \rightarrow Y$ **iff** $\forall x \in X, \exists! y \in Y, (x, y) \in G$: every $x \in X$ occurs as a first coordinate (totality/existence) exactly once (uniqueness). Example 3.1.10: $\{(1, \text{red}), (2, \text{red}), (3, \text{green})\}$ is the graph of a function $\{1, 2, 3\} \rightarrow \{\text{red}, \text{green}, \text{blue}\}$ but not of one with domain $\{1, 2, 3, 4\}$; adding $(2, \text{blue})$ destroys uniqueness.

Exercise 3.1.8

recommended

Find a function $f : \mathbb{Z} \rightarrow \mathbb{Z}$ whose graph is $\{\dots, (-2, -5), (-1, -2), (0, 1), (1, 4), (2, 7), (3, 10), \dots\}$.

The second coordinates increase by 3 when the first increases by 1, and $(0, 1)$ is in the graph, so try $f(n) = 3n + 1$: $f(-2) = -5$, $f(-1) = -2$, $f(0) = 1$, $f(1) = 4$, $f(2) = 7$, $f(3) = 10$. So $f : \mathbb{Z} \rightarrow \mathbb{Z}$, $f(n) = 3n + 1$ for all $n \in \mathbb{Z}$, has $\text{Gr}(f) = \{(n, 3n + 1) \mid n \in \mathbb{Z}\}$, the given set. $\square$

Exercise 3.1.11

essential

For each of the following, determine whether G is the graph of a function from X to Y .

- (a) $X = Y = \mathbb{R}$, $G = \{(a, a^2) \mid a \in \mathbb{R}\}$; (b) $X = Y = \mathbb{R}$, $G = \{(a^2, a) \mid a \in \mathbb{R}\}$; (c) $X = Y = [0, \infty)$, $G = \{(a^2, a) \mid a \in [0, \infty)\}$;
(d) $X = Y = \mathbb{Q}$, $G = \{(x, y) \in \mathbb{Q} \times \mathbb{Q} \mid xy = 1\}$; (e) $X = Y = \mathbb{Q}$, $G = \{(a, a) \mid a \in \mathbb{Z}\}$.

Use Theorem 3.1.9: for every $x \in X$ there must be exactly one $y \in Y$ with $(x, y) \in G$.

(a) *Yes.* For $x \in \mathbb{R}$, $(x, y) \in G$ iff $y = x^2$; this y exists and is unique. (G is the graph of $x \mapsto x^2$.)

- (b) *No*. For $x = -1$ there is no $a \in \mathbb{R}$ with $a^2 = -1$, so no $(x, y) \in G$ (existence fails). Also for $x = 1$ both $(1, 1)$ and $(1, -1)$ lie in G (uniqueness fails). Either failure suffices.
- (c) *Yes*. For $x \in [0, \infty)$, $(x, y) \in G$ iff $y \in [0, \infty)$ and $y^2 = x$; the unique such y is $\sqrt{x}$ (Example 1.2.36 for $x > 0$; $y = 0$ for $x = 0$). So G is the graph of $x \mapsto \sqrt{x}$.
- (d) *No*. For $x = 0$ there is no y with $0 \cdot y = 1$: existence fails.
- (e) *No*. For $x = \frac{1}{2} \in \mathbb{Q}$ there is no y with $(\frac{1}{2}, y) \in G$, since first coordinates of elements of G are integers: totality fails. $\square$

Definitions 3.1.12, 3.1.13 and Example 3.1.14 · identity, empty function, a non-function

$\text{id}_X : X \rightarrow X$, $\text{id}_X(x) = x$. The **empty function** $\emptyset \rightarrow X$ is the unique function with domain $\emptyset$; it is well-defined because there is nothing to check. Example 3.1.14: “ $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x)^2 = x$ ” is *not* well-defined (no value for $x < 0$; two values for $x > 0$), while the same equation does define $r : (0, \infty) \rightarrow (0, \infty)$. **Domain and codomain are part of the function.**

Composition

Definition 3.1.17 · composite

For $f : X \rightarrow Y$ and $g : Y \rightarrow Z$, the **composite** $g \circ f : X \rightarrow Z$ (“ g after f ”) is $(g \circ f)(x) = g(f(x))$. **Common error:** the function applied *first* is written *last*. The codomain of f must equal the domain of g . Example 3.1.18: the machine “multiply by 2, add 5, square, divide by 6” is $M = ((k \circ h) \circ g) \circ f$ with $f(x) = 2x$, $g(x) = x + 5$, $h(x) = x^2$, $k(x) = \frac{x}{6}$. Example 3.1.20: $\text{id}_Y \circ f = f = f \circ \text{id}_X$.

Exercise 3.1.19
recommended

With $f, g, h, k : \mathbb{Q} \rightarrow \mathbb{Q}$ as in Example 3.1.18, compute equations defining (a) $f \circ g$; (b) $g \circ f$; (c) $((f \circ g) \circ h) \circ k$; (d) $f \circ (g \circ (h \circ k))$; (e) $(g \circ g) \circ (g \circ g)$.

- (a) $(f \circ g)(x) = f(g(x)) = f(x + 5) = 2(x + 5) = 2x + 10$. (b) $(g \circ f)(x) = g(2x) = 2x + 5$. (So $f \circ g \neq g \circ f$.)
- (c) Apply k , then h , then g , then f : $x \mapsto \frac{x}{6} \mapsto \frac{x^2}{36} \mapsto \frac{x^2}{36} + 5 \mapsto \frac{x^2}{18} + 10$. So $((f \circ g) \circ h) \circ k(x) = \frac{x^2}{18} + 10$.
- (d) $(f \circ (g \circ (h \circ k)))(x) = f(g(h(k(x))))$, the same chain of operations as in (c): $\frac{x^2}{18} + 10$ (associativity, Exercise 3.1.21).
- (e) $(g \circ g)(x) = x + 10$, so $((g \circ g) \circ (g \circ g))(x) = (x + 10) + 10 = x + 20$. $\square$

Exercise 3.1.21
bonus

Prove that composition is associative: if $f : X \rightarrow Y$, $g : Y \rightarrow Z$, $h : Z \rightarrow W$ then $h \circ (g \circ f) = (h \circ g) \circ f : X \rightarrow W$.

Both composites have domain X and codomain W , so by Strategy 3.1.5 it suffices to check values. Let $x \in X$. Then

$$(h \circ (g \circ f))(x) = h((g \circ f)(x)) = h(g(f(x))) = (h \circ g)(f(x)) = ((h \circ g) \circ f)(x),$$

each step by Definition 3.1.17. Hence the functions are equal, and we may write $h \circ g \circ f$. $\square$

NOW YOU 3.1.22 · solutions in Appendix S

3.1.22 $f : X \rightarrow Y$, $g : Z \rightarrow W$ with $Y \subseteq Z$; $h(x) = g(f(x))$ defines $h : X \rightarrow W$. Write h as a composite involving f and g . [Hint: look closely at Definition 3.1.17: why is $g \circ f$ itself not allowed?]

Characteristic functions**Definition 3.1.23, Theorems 3.1.25 and 3.1.27, Strategy 3.1.26**

For $A \subseteq X$, $\chi_A : X \rightarrow [2] = \{0, 1\}$ is $\chi_A(x) = 1$ if $x \in A$ and 0 if $x \notin A$. **Theorem 3.1.25:** $A = B \Leftrightarrow \chi_A = \chi_B$, so to prove $A = B$ it suffices to prove $\chi_A = \chi_B$. **Theorem 3.1.27:** for all $x \in X$,

$$\chi_{A \cap B}(x) = \chi_A(x)\chi_B(x), \quad \chi_{A \cup B}(x) = \chi_A(x) + \chi_B(x) - \chi_A(x)\chi_B(x), \quad \chi_{X \setminus A}(x) = 1 - \chi_A(x).$$

Example 3.1.29 proves $X \cap (Y \cup Z) = (X \cap Y) \cup (X \cap Z)$ by computing $\chi_{X \cap (Y \cup Z)} = \chi_X(\chi_Y + \chi_Z - \chi_Y\chi_Z) = \chi_X\chi_Y + \chi_X\chi_Z - \chi_X\chi_Y\chi_Z$, then using $\chi_X^2 = \chi_X$ to rewrite the last term as $\chi_{X \cap Y}\chi_{X \cap Z}$, giving $\chi_{(X \cap Y) \cup (X \cap Z)}$.

NOW YOU 3.1.28, 3.1.30 · solutions in Appendix S

3.1.28 Prove parts (b) and (c) of Theorem 3.1.27. [Hint: the only possible values are 0 and 1: check the cases $x \in A / x \notin A$, $x \in B / x \notin B$.] **3.1.30** Use characteristic functions to prove de Morgan's laws for pairwise unions and intersections (Theorem 2.2.44(a),(b)).

Images and preimages**Definitions 3.1.31 and 3.1.37**

For $f : X \rightarrow Y$, $U \subseteq X$, $V \subseteq Y$:

$$f[U] = \{f(x) \mid x \in U\} = \{y \in Y \mid \exists x \in U, y = f(x)\} \subseteq Y, \quad f^{-1}[V] = \{x \in X \mid f(x) \in V\} \subseteq X.$$

$f[U]$ is the **image** of U (also $f_*(U)$, $f(U)$); $f[X]$ is the *image of f* . $f^{-1}[V]$ is the **preimage** of V (also $f^*(V)$, $f^{-1}(V)$); it is defined for every function, whether or not f has an inverse. Example 3.1.38 ($f : \mathbb{Z} \rightarrow \mathbb{Z}$, $f(x) = x^2$): $f^{-1}[\{1, 4, 9\}] = \{-3, -2, -1, 1, 2, 3\} = f^{-1}[\{1, \dots, 9\}]$, $f^{-1}[\mathbb{N}] = \mathbb{Z}$.

Working with images and preimages

Using $y \in f[U]$: “fix $x \in U$ with $y = f(x)$ ”. **Proving $y \in f[U]$:** exhibit $x \in U$ with $f(x) = y$. $x \in f^{-1}[V]$ means exactly $x \in X$ and $f(x) \in V$: no choice is involved. Example 3.1.32: $f[\mathbb{R}] = [0, \infty)$ for $f(x) = x^2$, by double containment ($y = x^2 \geq 0$; conversely $y = (\sqrt{y})^2$). Example 3.1.35: $f[U \cap V] \subseteq f[U] \cap f[V]$. Example 3.1.39: $f[U] \subseteq V \Leftrightarrow U \subseteq f^{-1}[V]$.

Exercise 3.1.33**recommended**

Describe the image $f[U]$: (a) $f : \mathbb{Z} \rightarrow \mathbb{Z}$, $f(n) = 3n$, $U = \mathbb{N}$; (b) $f : X \rightarrow X \times X$, $f(x) = (x, x)$, $U = X$; (c) $f : \{a, b, c\} \rightarrow \{1, 2, 3\}$, $f(a) = 1$, $f(b) = 3$, $f(c) = 1$, $U = \{a, b, c\}$.

(a) $f[\mathbb{N}] = \{3n \mid n \in \mathbb{N}\} = \{0, 3, 6, 9, \dots\}$, the natural multiples of 3. (b) $f[X] = \{(x, x) \mid x \in X\}$, the “diagonal” of $X \times X$. (c) $f[\{a, b, c\}] = \{f(a), f(b), f(c)\} = \{1, 3, 1\} = \{1, 3\}$. $\square$

Exercise 3.1.36

essential

Example 3.1.35 showed $f[U \cap V] \subseteq f[U] \cap f[V]$. Determine which of the following hold for all sets U, V, X, Y and functions $f : X \rightarrow Y$, with proof of truth or falsity: (a) $f[U] \cap f[V] \subseteq f[U \cap V]$; (b) $f[U \cup V] \subseteq f[U] \cup f[V]$; (c) $f[U] \cup f[V] \subseteq f[U \cup V]$.

- (a) *False.* Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$, $U = \{1\}$, $V = \{-1\}$. Then $f[U] = \{1\} = f[V]$, so $f[U] \cap f[V] = \{1\}$, while $U \cap V = \emptyset$ and $f[\emptyset] = \emptyset$ (Exercise 3.1.34). So $1 \in f[U] \cap f[V]$ but $1 \notin f[U \cap V]$.
 (b) *True.* Let $y \in f[U \cup V]$; fix $x \in U \cup V$ with $y = f(x)$. If $x \in U$ then $y = f(x) \in f[U]$; if $x \in V$ then $y \in f[V]$. Either way $y \in f[U] \cup f[V]$.
 (c) *True.* Let $y \in f[U] \cup f[V]$. If $y \in f[U]$, fix $x \in U$ with $y = f(x)$; then $x \in U \cup V$, so $y \in f[U \cup V]$. The case $y \in f[V]$ is identical. (So $f[U \cup V] = f[U] \cup f[V]$, but for intersections only one inclusion holds.) $\square$

Exercise 3.1.40

recommended

Let $f : X \rightarrow Y$ and $A, B \subseteq Y$. Prove that $f^{-1}[A \cap B] = f^{-1}[A] \cap f^{-1}[B]$, $f^{-1}[A \cup B] = f^{-1}[A] \cup f^{-1}[B]$ and $f^{-1}[Y \setminus A] = X \setminus f^{-1}[A]$.

For any $x \in X$, unfolding Definition 3.1.37:

$$\begin{aligned} x \in f^{-1}[A \cap B] &\Leftrightarrow f(x) \in A \cap B \Leftrightarrow f(x) \in A \wedge f(x) \in B \\ &\Leftrightarrow x \in f^{-1}[A] \wedge x \in f^{-1}[B] \Leftrightarrow x \in f^{-1}[A] \cap f^{-1}[B]; \\ x \in f^{-1}[A \cup B] &\Leftrightarrow f(x) \in A \vee f(x) \in B \Leftrightarrow x \in f^{-1}[A] \vee x \in f^{-1}[B] \Leftrightarrow x \in f^{-1}[A] \cup f^{-1}[B]; \\ x \in f^{-1}[Y \setminus A] &\Leftrightarrow f(x) \in Y \wedge f(x) \notin A \Leftrightarrow x \notin f^{-1}[A] \Leftrightarrow x \in X \setminus f^{-1}[A] \end{aligned}$$

(in the last line $f(x) \in Y$ is automatic and $x \in X$ throughout). Each identity follows by set extensionality. $\square$

Exercise 3.1.41

recommended

Let $f : X \rightarrow Y$. Prove that $f^{-1}[\emptyset] = \emptyset$ and $f^{-1}[Y] = X$.

If $x \in f^{-1}[\emptyset]$ then $f(x) \in \emptyset$, impossible; so $f^{-1}[\emptyset]$ is empty. For the second: $f^{-1}[Y] \subseteq X$ by definition, and if $x \in X$ then $f(x) \in Y$ (the codomain), so $x \in f^{-1}[Y]$. Hence $f^{-1}[Y] = X$. $\square$

NOW YOU 3.1.34, 3.1.42 · solutions in Appendix S

3.1.34 Prove $f[\emptyset] = \emptyset$ for every function f . **3.1.42** Prove that every $f : X \rightarrow \{0, 1\}$ is the characteristic function of $f^{-1}[\{1\}] \subseteq X$.

Watch out

- A function is *not* a formula: $x \mapsto x^2$ is a different function on $\mathbb{R} \rightarrow \mathbb{R}$, on $\mathbb{R} \rightarrow [0, \infty)$ and on $[0, \infty) \rightarrow [0, \infty)$ (Example 3.1.14, 3.E.25). Always state domain and codomain, and check well-definedness when a value is specified by an equation or by cases.
- $f^{-1}[V]$ is a set and exists for every f ; it does not say f is invertible. $f^{-1}[\{y\}]$ can be empty or have many elements (Example 3.1.38).
- From $y \in f[U]$ you get *some* $x \in U$ with $f(x) = y$, not a specific one: this is why $f[U] \cap f[V] \subseteq f[U \cap V]$ fails (3.1.36(a)).

B INJECTIONS AND SURJECTIONS**Definition 3.2.1 and Strategy 3.2.2 · injective**

$f : X \rightarrow Y$ is **injective** (one-to-one) if $\forall a, b \in X, f(a) = f(b) \Rightarrow a = b$. **Proof:** “Let $a, b \in X$ and suppose $f(a) = f(b)$ So $a = b$.” Contrapositive form: $a \neq b \Rightarrow f(a) \neq f(b)$ (distinct inputs, distinct outputs). Example 3.2.3: $f(n) = 2n + 1$ on $\mathbb{Z}$ is injective ($2m + 1 = 2n + 1 \Rightarrow m = n$). Proposition 3.2.4: composites of injections are injective ($g(f(a)) = g(f(b)) \Rightarrow f(a) = f(b) \Rightarrow a = b$).

Exercise 3.2.5**essential**

Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$. Prove that if $g \circ f$ is injective then f is injective.

Suppose $g \circ f$ is injective. Let $a, b \in X$ and suppose $f(a) = f(b)$. Applying g to both sides, $g(f(a)) = g(f(b))$, i.e. $(g \circ f)(a) = (g \circ f)(b)$. Since $g \circ f$ is injective, $a = b$. Hence f is injective. $\square$

Exercise 3.2.6**recommended**

Write out what it means for $f : X \rightarrow Y$ to be *not* injective, say how you would prove it, give an example and prove it is not injective.

Negating Definition 3.2.1 (de Morgan for quantifiers, Theorem 1.3.29, and $\neg(p \Rightarrow q) \equiv p \wedge \neg q$):

$$\exists a, b \in X, f(a) = f(b) \wedge a \neq b.$$

To prove f not injective: exhibit two specific elements $a \neq b$ of X and check $f(a) = f(b)$. Example: $g : \mathbb{Z} \rightarrow \mathbb{Z}, g(n) = n^2$. Take $a = 1, b = -1$: $a \neq b$ but $g(1) = 1 = g(-1)$. So g is not injective. $\square$

Exercise 3.2.8**essential**

Let $a, b \in \mathbb{R}$ with $b \neq 0$ and define $f : \mathbb{R} \rightarrow \mathbb{R}$ by $f(t) = a + bt$. Prove that f is injective.

Let $s, t \in \mathbb{R}$ and suppose $f(s) = f(t)$, i.e. $a + bs = a + bt$. Subtracting a gives $bs = bt$; since $b \neq 0$ we may divide by b to get $s = t$. Hence f is injective. $\square$

NOW YOU 3.2.7 · solutions in Appendix S

3.2.7 Injective or not? $f : \mathbb{N} \rightarrow \mathbb{Z}, f(n) = n^2$; $g : \mathbb{Z} \rightarrow \mathbb{N}, g(n) = n^2$; $h : \mathbb{N} \times \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}, h(x, y, z) = 2^x 3^y 5^z$.

Definition 3.2.9 and Strategy 3.2.10 · surjective

$f : X \rightarrow Y$ is **surjective** (onto) if $\forall y \in Y, \exists x \in X, f(x) = y$. **Proof:** “Let $y \in Y$. Put $x = \dots$ {solve $f(x) = y$ for x on scrap paper}. Then $x \in X$ {check!} and $f(x) = \dots = y$.” The verification that $f(x) = y$ and that x really lies in X is part of the proof. Example 3.2.11: $r : \mathbb{Z} \rightarrow [n], r(a) = \text{remainder of } a \text{ on division by } n$, is surjective since $r(k) = k$ for $k \in [n]$.

Exercise 3.2.12**essential**

For each pair (X, Y) determine whether $f : X \rightarrow Y, f(x) = 2x + 1$, is surjective: (a) $X = \mathbb{Z}, Y = \{x \in \mathbb{Z} \mid$

x odd}; (b) $X = Y = \mathbb{Z}$; (c) $X = Y = \mathbb{Q}$; (d) $X = Y = \mathbb{R}$.

(a) *Surjective.* Let $y \in Y$, so y is odd. By Proposition 1.1.58, $y = 2k + 1$ for some $k \in \mathbb{Z}$. Then $k \in X$ and $f(k) = 2k + 1 = y$.

(b) *Not surjective.* $0 \in \mathbb{Z}$, but $f(x) = 0$ means $2x = -1$, which has no solution in $\mathbb{Z}$ (-1 is odd). So no $x \in \mathbb{Z}$ has $f(x) = 0$.

(c) *Surjective.* Let $y \in \mathbb{Q}$. Put $x = \frac{y-1}{2}$; then $x \in \mathbb{Q}$ and $f(x) = 2 \cdot \frac{y-1}{2} + 1 = y$.

(d) *Surjective.* Same as (c) with $x = \frac{y-1}{2} \in \mathbb{R}$. □

Exercise 3.2.14

essential

Let $f : X \rightarrow Y$. Prove that f is surjective if and only if $Y = f[X]$.

Note first that $f[X] \subseteq Y$ always (its elements are values $f(x) \in Y$). So $Y = f[X]$ iff $Y \subseteq f[X]$.

($\Rightarrow$) Suppose f is surjective and let $y \in Y$. Then there is $x \in X$ with $f(x) = y$, so $y \in f[X]$ by Definition 3.1.31. Hence $Y \subseteq f[X]$, so $Y = f[X]$.

($\Leftarrow$) Suppose $Y = f[X]$ and let $y \in Y$. Then $y \in f[X]$, so $y = f(x)$ for some $x \in X$. This is exactly surjectivity. □

Exercise 3.2.16

recommended

Let $f : X \rightarrow \mathcal{P}(X)$. By considering $B = \{x \in X \mid x \notin f(x)\}$, prove that f is not surjective.

$B \subseteq X$, so $B \in \mathcal{P}(X)$. We show there is no $a \in X$ with $f(a) = B$. Suppose, towards a contradiction, that $a \in X$ and $f(a) = B$. Either $a \in B$ or $a \notin B$.

If $a \in B$, then by definition of B , $a \notin f(a) = B$: contradiction. If $a \notin B$, then $a \in f(a)$ (as $f(a) = B$), so a satisfies the defining condition of B and $a \in B$: contradiction.

So B is an element of the codomain that is not a value of f , and f is not surjective. (Cantor's argument: no set surjects onto its power set.) □

NOW YOU 3.2.13, 3.2.15 · solutions in Appendix S

3.2.13 Find $V \subseteq Y$ and a surjection $g : X \rightarrow V$ with $g(x) = f(x)$ for all x . [Hint: Definition 3.1.31.] **3.2.15**

Find a set Z and functions $p : X \rightarrow Z$ (surjective), $i : Z \rightarrow Y$ (injective) with $f = i \circ p$. [Hint: if $Z \subseteq Y$, $i(z) = z$ is an injection.]

Definition 3.2.17 · bijective

f is **bijective** if it is injective and surjective. **Proof tip:** prove both, in two labelled parts. Example 3.2.18: on the dyadic rationals $D = \{\frac{a}{2^n} \mid a \in \mathbb{Z}, n \in \mathbb{N}\}$, $f(x) = \frac{x}{2^k}$ is a bijection $D \rightarrow D$ (injective: $\frac{x}{2^k} = \frac{y}{2^k} \Rightarrow x = y$; surjective: $f(2^k y) = y$, after checking $2^k y \in D$ by the cases $k \leq n, k > n$).

Exercise 3.2.19

recommended

Let X be a set. Prove that $\text{id}_X : X \rightarrow X$ is a bijection.

Injective: let $a, b \in X$ with $\text{id}_X(a) = \text{id}_X(b)$. Then $a = b$ by definition of id_X .

Surjective: let $x \in X$. Then $x = \text{id}_X(x)$, so x is a value of id_X at the element $x \in X$.

Hence id_X is bijective. □

Exercise 3.2.20**essential**

Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be bijections. Prove that $g \circ f$ is a bijection.

Injective: f and g are injective, so $g \circ f$ is injective by Proposition 3.2.4.

Surjective: let $z \in Z$. Since g is surjective there is $y \in Y$ with $g(y) = z$; since f is surjective there is $x \in X$ with $f(x) = y$. Then $(g \circ f)(x) = g(f(x)) = g(y) = z$. Hence $g \circ f$ is surjective, and so bijective. $\square$

Inverses**Definitions 3.2.23, 3.2.29, 3.2.37 · left, right and two-sided inverses**

For $f : X \rightarrow Y$, a function $g : Y \rightarrow X$ is a **left inverse** if $g \circ f = \text{id}_X$ (i.e. $g(f(x)) = x$ for all $x \in X$); a **right inverse** if $f \circ g = \text{id}_Y$ (i.e. $f(g(y)) = y$ for all $y \in Y$); an **inverse** if both. Example 3.2.24: $e(m, n) = 2^m 3^n$ has left inverse $d(k) = (m, n)$ if $k = 2^m 3^n$, $(0, 0)$ otherwise. Example 3.2.30: $f(x) = x^2 : \mathbb{R} \rightarrow [0, \infty)$ has infinitely many right inverses ($\sqrt{y}$, $-\sqrt{y}$, and mixtures), none of them left inverses since f is not injective.

The dictionary (3.2.25–3.2.44)

- f has a left inverse $\Rightarrow f$ injective (Exercise 3.2.25); conversely injective and $X \neq \emptyset \Rightarrow$ left inverse (Proposition 3.2.27). **Strategy 3.2.26:** to prove injective, find g with $g(f(x)) = x$.
- f has a right inverse $\Rightarrow f$ surjective (Exercise 3.2.31); conversely every surjection has a right inverse (Proposition 3.2.36, needs the **axiom of choice** 3.2.33: from $\forall y \exists x f(x) = y$ one may *choose* such an x for each y , Strategy 3.2.35). **Strategy 3.2.32:** to prove surjective, find g with $f(g(y)) = y$.
- f bijective $\Leftrightarrow f$ has an inverse (Exercise 3.2.40). **Strategy 3.2.41:** to prove bijective, find g and check both $g(f(x)) = x$ and $f(g(y)) = y$.
- A left inverse and a right inverse are equal (Proposition 3.2.42: $\ell(y) = \ell(f(r(y))) = r(y)$), so the inverse is unique: $f^{-1} : Y \rightarrow X$ (Notation 3.2.43). For a bijection, left inverse $\Leftrightarrow$ right inverse (Proposition 3.2.44).

Exercise 3.2.25**recommended**

Let $f : X \rightarrow Y$. Prove that if f has a left inverse then f is injective.

Let $g : Y \rightarrow X$ with $g \circ f = \text{id}_X$. Let $a, b \in X$ and suppose $f(a) = f(b)$. Applying g : $g(f(a)) = g(f(b))$, i.e. $a = b$ since $g(f(x)) = x$ for all x . So f is injective. $\square$

Exercise 3.2.39**recommended (a), (b)**

Find the inverse and prove it is an inverse: (a) $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \frac{2x+1}{3}$; (b) $g : \mathcal{P}(\mathbb{N}) \rightarrow \mathcal{P}(\mathbb{N})$, $g(X) = \mathbb{N} \setminus X$.

(a) Solving $y = \frac{2x+1}{3}$ gives $x = \frac{3y-1}{2}$, so define $f^{-1} : \mathbb{R} \rightarrow \mathbb{R}$ by $f^{-1}(y) = \frac{3y-1}{2}$. Left inverse: for $x \in \mathbb{R}$, $f^{-1}(f(x)) = \frac{3 \cdot \frac{2x+1}{3} - 1}{2} = \frac{2x+1-1}{2} = x$. Right inverse: for $y \in \mathbb{R}$, $f(f^{-1}(y)) = \frac{2 \cdot \frac{3y-1}{2} + 1}{3} = \frac{3y-1+1}{3} = y$. So f^{-1} is a two-sided inverse.

(b) g is its own inverse: for $X \in \mathcal{P}(\mathbb{N})$, $g(g(X)) = \mathbb{N} \setminus (\mathbb{N} \setminus X) = \mathbb{N} \cap X = X$ by Exercise 2.2.43 (and $X \subseteq \mathbb{N}$). This single computation shows $g \circ g = \text{id}_{\mathcal{P}(\mathbb{N})}$, which is both the left- and the right-inverse condition. $\square$

Exercise 3.2.40**essential**

Let $f : X \rightarrow Y$. Prove that f is bijective if and only if f has an inverse.

($\Rightarrow$) Suppose f is bijective. For each $y \in Y$ there is a unique $x \in X$ with $f(x) = y$: existence by surjectivity, uniqueness by injectivity (if $f(x) = y = f(x')$ then $x = x'$). So by Definition 3.1.1 we may define $g : Y \rightarrow X$ by letting $g(y)$ be this unique x ; thus $f(g(y)) = y$ for all $y \in Y$. Also for $x \in X$, the unique element mapping to $f(x)$ is x itself, so $g(f(x)) = x$. Hence g is an inverse for f .

($\Leftarrow$) Suppose $g : Y \rightarrow X$ is an inverse. Since g is a left inverse, f is injective (Exercise 3.2.25). Since g is a right inverse, f is surjective: for $y \in Y$, $x = g(y) \in X$ satisfies $f(x) = f(g(y)) = y$. Hence f is bijective. $\square$

Exercise 3.2.45**recommended**

Let $f : X \rightarrow Y$ be a bijection. Prove that $f^{-1} : Y \rightarrow X$ is a bijection.

By Exercise 3.2.40, f has an inverse f^{-1} , with $f^{-1} \circ f = \text{id}_X$ and $f \circ f^{-1} = \text{id}_Y$. Read these two equations from the point of view of f^{-1} : $f \circ f^{-1} = \text{id}_Y$ says f is a left inverse for f^{-1} , and $f^{-1} \circ f = \text{id}_X$ says f is a right inverse for f^{-1} . So f is an inverse for f^{-1} , and by Exercise 3.2.40 (direction $\Leftarrow$) f^{-1} is a bijection. (Moreover $(f^{-1})^{-1} = f$.) $\square$

Exercise 3.2.46**recommended**

Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be bijections. Prove that $g \circ f$ is a bijection and express its inverse in terms of f^{-1} and g^{-1} .

Claim: $(g \circ f)^{-1} = f^{-1} \circ g^{-1} : Z \rightarrow X$. Using associativity (3.1.21):

$$(f^{-1} \circ g^{-1}) \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ \text{id}_Y \circ f = f^{-1} \circ f = \text{id}_X,$$

$$(g \circ f) \circ (f^{-1} \circ g^{-1}) = g \circ (f \circ f^{-1}) \circ g^{-1} = g \circ \text{id}_Y \circ g^{-1} = g \circ g^{-1} = \text{id}_Z.$$

So $f^{-1} \circ g^{-1}$ is a two-sided inverse for $g \circ f$; by Strategy 3.2.41, $g \circ f$ is a bijection with this inverse (“socks and shoes”: undo the last step first). $\square$

Exercise 3.2.47**essential**

Let $f : X \rightarrow A$ and $g : Y \rightarrow B$ be bijections. Prove that there is a bijection $X \times Y \rightarrow A \times B$ and describe its inverse.

Define $h : X \times Y \rightarrow A \times B$ by $h(x, y) = (f(x), g(y))$ (well-defined: $f(x) \in A$, $g(y) \in B$). Define $k : A \times B \rightarrow X \times Y$ by $k(a, b) = (f^{-1}(a), g^{-1}(b))$. Then for $(x, y) \in X \times Y$,

$$k(h(x, y)) = k(f(x), g(y)) = (f^{-1}(f(x)), g^{-1}(g(y))) = (x, y),$$

and for $(a, b) \in A \times B$, $h(k(a, b)) = h(f^{-1}(a), g^{-1}(b)) = (f(f^{-1}(a)), g(g^{-1}(b))) = (a, b)$. So k is an inverse for h , and h is a bijection by Strategy 3.2.41, with $h^{-1} = k$. $\square$

NOW YOU 3.2.21, 3.2.22, 3.2.28, 3.2.31, 3.2.39(c) · solutions in Appendix S

3.2.21 f injective iff $\forall y \in f[X], \exists! x \in X, y = f(x)$. **3.2.22** $e(m, n) = 2^m 3^n$ is injective; decode 1, 24, 7776, 59049, 396718580736. **3.2.28** A left inverse g of f is surjective. [Hint: one sentence.] **3.2.31** If f has a right inverse then f is surjective. **3.2.39(c)** Inverse of $h : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$, $h(m, n) = 2^m(2n + 1) - 1$. [Hint: every positive integer is uniquely $2^m \cdot (\text{odd})$.]

Watch out

- If you only prove $g(f(x)) = x$ you have only proved f injective (text after Strategy 3.2.41). A bijection proof via an inverse needs both equations.
- In a surjectivity proof, after solving for x check that x lies in the *domain* (3.2.12(b): $x = -\frac{1}{2} \notin \mathbb{Z}$) and that $f(x) = y$ really holds.
- “Not injective” needs two *specific* unequal inputs with equal outputs; “not surjective” needs one *specific* element of the codomain and a proof that nothing maps to it.

C CHAPTER 3 EXERCISES**Functions, graphs, even and odd functions (3.E.1–3.E.11)****Definition 3.E.4 · even and odd functions**

$f : \mathbb{R} \rightarrow \mathbb{R}$ is **even** if $f(-x) = f(x)$ for all $x \in \mathbb{R}$, and **odd** if $f(-x) = -f(x)$ for all $x \in \mathbb{R}$.

Exercise 3.E.2**recommended**

Let X be a set. Prove that $\forall a \in X, \exists! U \in \mathcal{P}(X), (a \in U \wedge \exists! x \in X, x \in U)$, and describe explicitly the function $X \rightarrow \mathcal{P}(X)$ this suggests.

Let $a \in X$. *Existence:* take $U = \{a\}$. Then $U \subseteq X$, so $U \in \mathcal{P}(X)$; $a \in U$; and a is the unique element of U (any $x \in U$ equals a), so $\exists! x \in X, x \in U$ holds. *Uniqueness:* let $U \in \mathcal{P}(X)$ with $a \in U$ and with exactly one element. Let $u \in U$; since $a \in U$ and U has exactly one element, $u = a$. So $U \subseteq \{a\}$, and $a \in U$ gives $\{a\} \subseteq U$; hence $U = \{a\}$.

The function is $X \rightarrow \mathcal{P}(X), a \mapsto \{a\}$ (“singleton”), well-defined by what we just proved. $\square$

NOW YOU 3.E.1, 3.E.3, 3.E.5–3.E.11 · solutions in Appendix S

3.E.1 For which equations is there $f : \mathbb{R} \rightarrow \mathbb{R}$ with “equation holds iff $y = f(x)$ ” for all x, y : (a) $x + y = 1$; (b) $x^2 + y^2 = 1$; (c) $x = 0$; (d) $y = 0$; (e) $(1 + x^2)y = 1$; (f) $(1 - x^2)y = 0$? **3.E.3** There is only one function with empty codomain; what is its domain? **3.E.5** x^n is even iff n even, odd iff n odd. **3.E.6** Unique f both even and odd. **3.E.7** $\chi_U : \mathbb{R} \rightarrow \{-1, +1\}$: even iff $\{-u \mid u \in U\} = U$; odd iff $\{-u \mid u \in U\} = \mathbb{R} \setminus U$. **3.E.8** Every f is uniquely $g + h$ with g even, h odd. **3.E.9** If $f \circ \theta_m = \theta_n \circ f$ for all $f : [m] \rightarrow [n]$ then $\theta_n = \text{id}_{[n]}$. **3.E.10** $\chi_{U \Delta V}$ in terms of χ_U, χ_V . **3.E.11** $\int \frac{1}{x} dx = \log|x| + a\chi_U(x) + b\chi_V(x)$: find U, V .

Images and preimages (3.E.12–3.E.23)**Exercise 3.E.13****essential**

$f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}, f(a, b) = a + 2b; U = \{1\} \times \mathbb{Z}$. Find $f[U]$.

$f[U] = \{f(1, b) \mid b \in \mathbb{Z}\} = \{1 + 2b \mid b \in \mathbb{Z}\}$, the set of odd integers (Proposition 1.1.58). Explicitly: $(\subseteq)$ $1 + 2b$ is odd. $(\supseteq)$ If n is odd then $n = 2b + 1$ for some $b \in \mathbb{Z}$, and $n = f(1, b)$ with $(1, b) \in U$. $\square$

Exercise 3.E.17**essential**

$f: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$, $f(a, b) = a + 2b$; $V = \{n \in \mathbb{Z} \mid n \text{ is odd}\}$. Find $f^{-1}[V]$.

$f^{-1}[V] = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} \mid a + 2b \text{ is odd}\}$. Now $a + 2b$ is odd iff a is odd: if $a = 2k + 1$ then $a + 2b = 2(k + b) + 1$ is odd; if $a = 2k$ then $a + 2b = 2(k + b)$ is even, hence not odd (Definition 0.41). So

$$f^{-1}[V] = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} \mid a \text{ is odd}\} = \{n \in \mathbb{Z} \mid n \text{ odd}\} \times \mathbb{Z}.$$

□

Exercise 3.E.19**essential**

Let $f: X \rightarrow Y$. Prove or find a counterexample: (a) $U \subseteq f^{-1}[f[U]]$ for all $U \subseteq X$; (b) $f^{-1}[f[U]] \subseteq U$; (c) $V \subseteq f[f^{-1}[V]]$ for all $V \subseteq Y$; (d) $f[f^{-1}[V]] \subseteq V$.

(a) *True*. Let $x \in U$. Then $f(x) \in f[U]$ (witness x), so $x \in f^{-1}[f[U]]$ by definition of preimage.

(b) *False*. $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$, $U = \{1\}$: $f[U] = \{1\}$ and $f^{-1}[\{1\}] = \{-1, 1\} \not\subseteq \{1\}$.

(c) *False*. Same f , $V = \{-1\}$: $f^{-1}[V] = \emptyset$ (no real square is -1), so $f[f^{-1}[V]] = f[\emptyset] = \emptyset \not\subseteq \{-1\}$.

(d) *True*. Let $y \in f[f^{-1}[V]]$; fix $x \in f^{-1}[V]$ with $y = f(x)$. Then $f(x) \in V$ by definition of preimage, i.e. $y \in V$.

(So in general: $U \subseteq f^{-1}[f[U]]$ with equality when f is injective, and $f[f^{-1}[V]] \subseteq V$ with equality when f is surjective.) □

NOW YOU 3.E.12, 3.E.14–3.E.16, 3.E.18, 3.E.20–3.E.23 • solutions in Appendix S

3.E.12 $f(x) = \sqrt{1+x^2}$, $U = \mathbb{R}$: find $f[U]$. **3.E.14** $f: \mathbb{N} \rightarrow \mathcal{P}(\mathbb{N})$, $f(0) = \emptyset$, $f(n+1) = f(n) \cup \{n\}$: find $f[\mathbb{N}]$. **3.E.15** $f: \mathbb{R}^{\mathbb{R}} \rightarrow \mathbb{R}^{\mathbb{R}}$, $f(h)(x) = h(|x|)$: find $f[\mathbb{R}^{\mathbb{R}}]$. [Hint: each $f(h)$ is even.] **3.E.16** $f(x) = \sqrt{1+x^2}$: find $f^{-1}[(-5, 5)]$. **3.E.18** $f(A, B) = A \cap B$ on $\mathcal{P}(\mathbb{N}) \times \mathcal{P}(\mathbb{N})$: find $f^{-1}[\{\emptyset\}]$. **3.E.20** The universal property of $f[X]$ (fiddly; see Appendix S). **3.E.21** = Exercise 3.1.40. **3.E.22** $(g \circ f)[U] = g[f[U]]$ and $(g \circ f)^{-1}[W] = f^{-1}[g^{-1}[W]]$. **3.E.23** $p[S] = \text{Gr}(g \circ f)$ for $S = \{(x, y, z) \mid y = f(x), z = g(y)\}$, $p(x, y, z) = (x, z)$.

Injections, surjections and bijections (3.E.24–3.E.28)**Exercise 3.E.24****essential**

(a) Prove that for all $f: X \rightarrow Y$, $g: Y \rightarrow Z$: if $g \circ f$ is bijective then f is injective and g is surjective. (b) Find f, g with $g \circ f$ bijective, f not surjective and g not injective.

(a) f is injective by Exercise 3.2.5 (since $g \circ f$ is injective). For g : let $z \in Z$. As $g \circ f$ is surjective, there is $x \in X$ with $g(f(x)) = z$; put $y = f(x) \in Y$, then $g(y) = z$. So g is surjective.

(b) $X = \{0\}$, $Y = \{0, 1\}$, $Z = \{0\}$; $f(0) = 0$; $g(0) = g(1) = 0$. Then $g \circ f: \{0\} \rightarrow \{0\}$ sends $0 \mapsto 0$, so $g \circ f = \text{id}_{\{0\}}$ is a bijection (3.2.19). But f is not surjective ($1 \notin f[X]$) and g is not injective ($g(0) = g(1)$, $0 \neq 1$). □

Exercise 3.E.25**essential; (e) recommended later**

For each pair (U, V) of subsets of $\mathbb{R}$, determine whether “ $f(x) = x^2 - 4x + 7$ for all $x \in U$ ” defines a function $f: U \rightarrow V$, and if so whether f is injective and whether it is surjective.

(a) $U = V = \mathbb{R}$; (b) $U = (1, 4)$, $V = [3, 7)$; (c) $U = [3, 4)$, $V = [4, 7)$; (d) $U = (3, 4]$, $V = [4, 7)$; (e)

$U = [2, \infty)$, $V = [3, \infty)$; (f) $U = [2, \infty)$, $V = \mathbb{R}$.

Complete the square: $f(x) = (x-2)^2 + 3$. So $f(x) \geq 3$ with equality only at $x = 2$, f is symmetric about $x = 2$ ($f(2+t) = f(2-t)$), and on $x \geq 2$ the equation $f(x) = y$ with $y \geq 3$ has exactly the solution $x = 2 + \sqrt{y-3}$.

(a) Defined (every real x gives a real value). Not injective: $f(0) = 7 = f(4)$. Not surjective: $0 \in V$ but $f(x) \geq 3 > 0$ for all x .

(b) For $1 < x < 4$: $-1 < x-2 < 2$, so $0 \leq (x-2)^2 < 4$ and $3 \leq f(x) < 7$: defined. Not injective: $f(1.5) = f(2.5) = 3.25$. Surjective: let $y \in [3, 7)$; put $x = 2 + \sqrt{y-3}$. Then $0 \leq \sqrt{y-3} < 2$, so $x \in [2, 4) \subseteq (1, 4)$ and $f(x) = (\sqrt{y-3})^2 + 3 = y$.

(c) For $3 \leq x < 4$: $1 \leq x-2 < 2$, so $1 \leq (x-2)^2 < 4$, $4 \leq f(x) < 7$: defined. Injective: if $f(a) = f(b)$ with $a, b \in [3, 4)$ then $(a-2)^2 = (b-2)^2$ with $a-2, b-2 > 0$, so $a-2 = b-2$ (positive square roots are unique), $a = b$. Surjective: for $y \in [4, 7)$, $x = 2 + \sqrt{y-3}$ has $1 \leq \sqrt{y-3} < 2$, so $x \in [3, 4)$, and $f(x) = y$. Hence bijective.

(d) Not a function: $4 \in U$ but $f(4) = 7 \notin [4, 7)$ (existence of a value in the codomain fails).

(e) For $x \geq 2$: $f(x) \geq 3$: defined. Injective as in (c) (now $a-2, b-2 \geq 0$). Surjective: for $y \geq 3$, $x = 2 + \sqrt{y-3} \geq 2$ and $f(x) = y$. Bijective.

(f) Defined and injective as in (e). Not surjective: $0 \in \mathbb{R}$ is not a value since $f(x) \geq 3$. $\square$

Exercise 3.E.26(b)

recommended

Find (with proof) a bijection $\mathbb{R} \rightarrow (-1, 1)$.

Define $f: \mathbb{R} \rightarrow (-1, 1)$ by $f(x) = \frac{x}{1+|x|}$. Well-defined: $1+|x| > 0$, and $|f(x)| = \frac{|x|}{1+|x|} < 1$, so $f(x) \in (-1, 1)$. Define $g: (-1, 1) \rightarrow \mathbb{R}$ by $g(y) = \frac{y}{1-|y|}$ (well-defined as $1-|y| > 0$). We check g is an inverse (Strategy 3.2.41).

For $x \in \mathbb{R}$: $|f(x)| = \frac{|x|}{1+|x|}$, so $1-|f(x)| = \frac{1}{1+|x|}$, and $g(f(x)) = \frac{x}{1+|x|} \cdot (1+|x|) = x$.

For $y \in (-1, 1)$: $|g(y)| = \frac{|y|}{1-|y|}$, so $1+|g(y)| = \frac{1}{1-|y|}$, and $f(g(y)) = \frac{y}{1-|y|} \cdot (1-|y|) = y$.

Hence f is a bijection with inverse g . $\square$

NOW YOU 3.E.26(a),(c),(d), 3.E.27, 3.E.28 (26cd bonus) · solutions in Appendix S

3.E.26 Bijections (a) $\mathbb{Z} \rightarrow \mathbb{N}$; (c) $[0, 1] \rightarrow (0, 1)$; (d) $[a, b] \rightarrow (c, d)$. [Hint: for (c) move $0, 1, \frac{1}{2}, \frac{1}{3}, \dots$ along and fix everything else.] **3.E.27** $f(a, b) = \binom{a+b+1}{2} + b$ is a bijection $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$. [Hint: for each n there is a unique k with $\binom{k+1}{2} \leq n < \binom{k+2}{2}$.] **3.E.28** If $e \circ e = e$, find Y , $f: X \rightarrow Y$, $g: Y \rightarrow X$ with $g \circ f = e$, $f \circ g = \text{id}_Y$. [Hint: fixed points of e .]

True–false (3.E.29–3.E.36)

Exercise 3.E.34

essential

Given $f: X \rightarrow Y$, the assignment $V \mapsto f^{-1}[V]$ defines a function $\mathcal{P}(Y) \rightarrow \mathcal{P}(X)$.

True. Let $V \in \mathcal{P}(Y)$, i.e. $V \subseteq Y$. Then $f^{-1}[V] = \{x \in X \mid f(x) \in V\}$ is a uniquely specified subset of X (Definition 3.1.37; it is given by set-builder notation, so it exists and is unique), hence an element of $\mathcal{P}(X)$. Totality, existence and uniqueness hold, so $V \mapsto f^{-1}[V]$ is a well-defined function $\mathcal{P}(Y) \rightarrow \mathcal{P}(X)$ (the book's f^*). $\square$

Exercise 3.E.35**recommended**

Let f, g, h be composable functions. If $h \circ g \circ f$ is injective, then g is injective.

False. Let $X = \{0\}$, $Y = \{0, 1\}$, $Z = \{0\}$, $W = \{0\}$; $f : X \rightarrow Y$, $f(0) = 0$; $g : Y \rightarrow Z$, $g(0) = g(1) = 0$; $h = \text{id}_Z$. Then $h \circ g \circ f : \{0\} \rightarrow \{0\}$ is injective (a function on a one-element set is trivially injective: if $a, b \in \{0\}$ then $a = b$), but $g(0) = g(1)$ with $0 \neq 1$, so g is not injective. (Exercise 3.2.5 only gives that f , the *first* function applied, is injective.) $\square$

NOW YOU 3.E.29–3.E.33, 3.E.36 · solutions in Appendix S

3.E.29 $\{(x, y) \in \mathbb{R} \times \mathbb{R} \mid x^2 = y^2\}$ is a graph of a function. **3.E.30** $\{(x, y) \in \mathbb{Z} \times \mathbb{N} \mid x^2 = y^2\}$ is a graph of a function. **3.E.31** Every function with empty domain has empty codomain. **3.E.32** Every function with empty codomain has empty domain. **3.E.33** The image of a function is a subset of its domain. **3.E.36** Every left inverse is surjective and every right inverse is injective.

Always–sometimes–never (3.E.37–3.E.48)**Exercise 3.E.37****recommended**

Let $f : X \rightarrow Y$ and $U \subseteq X$. Then there is a function $g : U \rightarrow Y$ defined by $g(x) = f(x)$ for all $x \in U$.

Always. Let $x \in U$. Then $x \in X$, so $f(x)$ is a uniquely specified element of Y . Thus the specification satisfies totality, existence and uniqueness, and g (the *restriction* of f to U) is well-defined. $\square$

Exercise 3.E.38**recommended**

Let $f : X \rightarrow Y$ and $V \subseteq Y$. Then there is a function $g : X \rightarrow V$ defined by $g(x) = f(x)$ for all $x \in X$.

Sometimes. The specification is well-defined exactly when $f(x) \in V$ for every $x \in X$, i.e. $f[X] \subseteq V$ (existence of a value in the codomain). It holds for $V = Y$ (then $g = f$). It fails for $f = \text{id}_{\mathbb{R}}$ and $V = \{0\}$: $f(1) = 1 \notin V$. $\square$

Exercise 3.E.41**essential**

Let X, Y be sets and $G \subseteq X \times Y$. Then G is the graph of a function $f : X \rightarrow Y$.

Sometimes. By Theorem 3.1.9, G is a graph iff $\forall x \in X, \exists! y \in Y, (x, y) \in G$. Holds: $X = Y = \{0\}$, $G = \{(0, 0)\}$ (the graph of $\text{id}_{\{0\}}$). Fails: $X = Y = \{0\}$, $G = \emptyset$ (no pair with first coordinate 0). Also fails for $X = \{0\}$, $Y = \{0, 1\}$, $G = X \times Y$ (two pairs with first coordinate 0). $\square$

Exercise 3.E.47**recommended**

Let $a, b, c, d \in \mathbb{R}$ and define $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $f(x, y) = (ax + by, cx + dy)$. Then f is injective if and only if f is surjective.

Always. We prove: f injective $\Leftrightarrow ad - bc \neq 0 \Leftrightarrow f$ surjective.

If $ad - bc \neq 0$, f is bijective. Define $g(u, v) = \left(\frac{du - bv}{ad - bc}, \frac{av - cu}{ad - bc}\right)$. Then $f(g(u, v)) = \left(\frac{a(du - bv) + b(av - cu)}{ad - bc}, \frac{c(du - bv) + d(av - cu)}{ad - bc}\right) = \left(\frac{(ad - bc)u}{ad - bc}, \frac{(ad - bc)v}{ad - bc}\right) = (u, v)$, and similarly $g(f(x, y)) = (x, y)$ (substitute $u = ax + by$, $v = cx + dy$: $du - bv = (ad - bc)x$ and $av - cu = (ad - bc)y$). By Strategy 3.2.41, f is bijective, so both injective and surjective.

If $ad - bc = 0$, f is neither. **Not injective:** we find $(x, y) \neq (0, 0)$ with $f(x, y) = (0, 0) = f(0, 0)$. If

$(a, b) \neq (0, 0)$ take $(x, y) = (b, -a)$: $ax + by = ab - ba = 0$ and $cx + dy = cb - da = -(ad - bc) = 0$. If $(a, b) = (0, 0)$ take $(x, y) = (d, -c)$ if $(c, d) \neq (0, 0)$ (then $cx + dy = cd - dc = 0$), and $(1, 0)$ if $(c, d) = (0, 0)$ too. **Not surjective:** if $(a, c) \neq (0, 0)$, every value satisfies $c(ax + by) - a(cx + dy) = (cb - ad)y = 0$, but $(u, v) = (c, -a)$ has $cu - av = c^2 + a^2 \neq 0$, so it is not a value. If $(a, c) = (0, 0)$ but $(b, d) \neq (0, 0)$, every value (by, dy) satisfies $du - bv = 0$, but $(d, -b)$ has $du - bv = d^2 + b^2 \neq 0$. If all four are 0, f is constantly $(0, 0)$ and $(1, 0)$ is not a value. $\square$

Exercise 3.E.48**essential**

Let $U, V \subseteq \mathbb{R}$ with $x^2 \in V$ for all $x \in U$. The function $f : U \rightarrow V$ defined by $f(x) = x^2$ is injective.

Sometimes. (The hypothesis guarantees f is well-defined.) For $U = [0, \infty)$, $V = \mathbb{R}$: if $a, b \geq 0$ and $a^2 = b^2$ then $a = b$ (uniqueness of nonnegative square roots), so f is injective. For $U = V = \mathbb{R}$: $f(-1) = 1 = f(1)$ with $-1 \neq 1$, so f is not injective. $\square$

NOW YOU 3.E.39, 3.E.40, 3.E.42–3.E.46 · solutions in Appendix S

3.E.39 There is a unique function $X \rightarrow \{0\}$. **3.E.40** There is a unique function $X \rightarrow \emptyset$. **3.E.42** $f : \{1, 2, 3\} \rightarrow \{1, 2\}$, $G' = \{(y, x) \mid (x, y) \in \text{Gr}(f)\}$ is the graph of a function $\{1, 2\} \rightarrow \{1, 2, 3\}$. **3.E.43** $U \neq \emptyset \Rightarrow f[U] \neq \emptyset$. **3.E.44** $V \neq \emptyset \Rightarrow f^{-1}[V] \neq \emptyset$. **3.E.45** $f : \{1, 2\} \rightarrow \{1, 2, 3\}$ is injective. **3.E.46** $f : \{1, 2\} \rightarrow \{1, 2, 3\}$ is surjective.

Chapter 4

Mathematical induction

Contents

- A** Peano's axioms and recursion (§4.1)
- B** Weak induction (§4.2)
- C** Strong induction and well-ordering (§4.3)
- D** Chapter 4 exercises (4.E)

Planning

§4.1 is not in the planning (covered briefly).

★ 4.2: 5, 11 ★ 4.2: 8, 17, 20

★ 4.3: 8, 9, 14, 15 ★ 4.3: 10, 18

★ 4.E: 10, 11, 13, 16–20, 25–27, 29–32

★ 4.E: 12, 14, 15, 21, 24, 28, 33–35

Tested in the **exam** (4 Nov).

A PEANO'S AXIOMS AND RECURSION

Definition 4.1.1 and Theorem 4.1.2 · what $\mathbb{N}$ is

A **notion of natural numbers** is a set $\mathbb{N}$ with a zero $z \in \mathbb{N}$ and a successor function $s : \mathbb{N} \rightarrow \mathbb{N}$ such that (i) $z \notin s[\mathbb{N}]$; (ii) s is injective; (iii) $\mathbb{N}$ is generated by z and s : for every set X , if $z \in X$ and $(n \in X \Rightarrow s(n) \in X)$ for all $n \in \mathbb{N}$, then $\mathbb{N} \subseteq X$ (**Peano's axioms**). Write $0 = z$, $1 = s(0)$, $2 = s(1)$, $\dots$ **Recursion theorem**: for any set X , $a \in X$ and $h : \mathbb{N} \times X \rightarrow X$ there is a unique $f : \mathbb{N} \rightarrow X$ with $f(0) = a$ and $f(s(n)) = h(n, f(n))$. **Strategy 4.1.3 (definition by recursion)**: define $f(0)$, then define $f(s(n))$ in terms of n and $f(n)$.

Examples 4.1.4, 4.1.7 and Definitions 4.1.10–4.1.15 · the recursive definitions

$m + 0 = m$, $m + s(n) = s(m + n)$; $m \cdot 0 = 0$, $m \cdot (n + 1) = (m \cdot n) + m$; $s(n) = n + 1$ (Prop. 4.1.6).

$$\sum_{k=1}^0 a_k = 0, \quad \sum_{k=1}^{n+1} a_k = \left(\sum_{k=1}^n a_k \right) + a_{n+1}; \quad \prod_{k=1}^0 a_k = 1, \quad \prod_{k=1}^{n+1} a_k = \left(\prod_{k=1}^n a_k \right) \cdot a_{n+1}.$$

$$0! = 1, \quad (n+1)! = (n+1) \cdot n!; \quad \binom{n}{0} = 1, \quad \binom{0}{k+1} = 0, \quad \binom{n+1}{k+1} = \binom{n}{k} + \binom{n}{k+1}.$$

(The last three rules generate Pascal's triangle: each entry is the sum of the two above it.) Model (Proposition 4.1.5, $2 + 2 = 4$): $2 + 2 = 2 + s(1) = s(2 + 1) = s(2 + s(0)) = s(s(2 + 0)) = s(s(2)) = s(3) = 4$, one definition per step.

Exercise 4.1.13

from the book

Let $x_1, x_2 \in \mathbb{R}$. Working strictly from the definitions of indexed sum and product, prove that $\sum_{i=1}^2 x_i = x_1 + x_2$ and $\prod_{i=1}^2 x_i = x_1 \cdot x_2$.

By Definition 4.1.10 (three times): $\sum_{i=1}^2 x_i = (\sum_{i=1}^1 x_i) + x_2 = ((\sum_{i=1}^0 x_i) + x_1) + x_2 = (0 + x_1) + x_2 = x_1 + x_2$. By Definition 4.1.11: $\prod_{i=1}^2 x_i = (\prod_{i=1}^1 x_i) \cdot x_2 = ((\prod_{i=1}^0 x_i) \cdot x_1) \cdot x_2 = (1 \cdot x_1) \cdot x_2 = x_1 \cdot x_2$. $\square$

NOW YOU 4.1.9, 4.1.16 · solutions in Appendix S

4.1.9 Define m^n by recursion on n and prove $2^2 = 4$ from the recursive definitions. **4.1.16** Write down the next two rows of Pascal's triangle (after 1 5 10 10 5 1).

B WEAK INDUCTION**Theorem 4.2.1 and Strategy 4.2.2 · proof by (weak) induction**

To prove $\forall n \geq n_0, p(n)$ it suffices to prove: **(base case)** $p(n_0)$; **(induction step)** for all $n \geq n_0$, if $p(n)$ then $p(n+1)$. In the step, $p(n)$ is the **induction hypothesis** and $p(n+1)$ the **induction goal**. (Proof: $X = \{n \in \mathbb{N} \mid p(n_0 + n)\}$ contains 0 and is closed under $n \mapsto n+1$, so $\mathbb{N} \subseteq X$ by Peano axiom (iii).)

How the book writes an induction proof (page 136)

- 1 Say “We proceed by induction on $n \geq n_0$ ” and **state** $p(n)$ **explicitly**.
- 2 **(Base case)** “We need to prove $p(n_0)$: ...”.
- 3 **(Induction step)** “Let $n \geq n_0$ and suppose $p(n)$. We need to prove $p(n+1)$: ...”. Write out the goal before proving it; mark the line where the induction hypothesis is used.
- 4 “The result follows by induction.”

Models: Proposition 4.2.3 ($\sum_{k=1}^n k = \frac{n(n+1)}{2}$: $\sum_{k=1}^{n+1} k = \frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2}$); Proposition 4.2.4 ($3 \mid n^3 - n$: $(n+1)^3 - (n+1) = (n^3 - n) + 3n^2 + 3n = 3(k + n^2 + n)$); Proposition 4.2.6 ($3n < 2^n$ for $n \geq 4$: $3(n+1) = 3n + 3 < 2^n + 3 < 2^n + 2^n = 2^{n+1}$ since $3 < 2^4 \leq 2^n$). The proof for $n \geq 4$ says nothing about $n < 4$: check those separately ($p(0)$ true, $p(1), p(2), p(3)$ false).

Exercise 4.2.5**essential**

Prove by induction that $\sum_{k=0}^n 2^k = 2^{n+1} - 1$ for all $n \in \mathbb{N}$.

We proceed by induction on $n \geq 0$, with $p(n)$: $\sum_{k=0}^n 2^k = 2^{n+1} - 1$.

(Base case) $\sum_{k=0}^0 2^k = 2^0 = 1 = 2^1 - 1$. So $p(0)$ holds.

(Induction step) Let $n \geq 0$ and suppose $\sum_{k=0}^n 2^k = 2^{n+1} - 1$. We need to prove $\sum_{k=0}^{n+1} 2^k = 2^{n+2} - 1$. Indeed

$$\begin{aligned} \sum_{k=0}^{n+1} 2^k &= \left(\sum_{k=0}^n 2^k \right) + 2^{n+1} \quad \{\text{Definition 4.1.10}\} \\ &= (2^{n+1} - 1) + 2^{n+1} \quad \{\text{induction hypothesis}\} \\ &= 2 \cdot 2^{n+1} - 1 = 2^{n+2} - 1. \end{aligned}$$

The result follows by induction. □

Exercise 4.2.8**recommended**

Write down the statement $p(n)$ that Example 4.2.7 (“all horses have the same colour”) tried to prove for $n \geq 1$; write down, with quantifiers, exactly what the induction step must prove; explain why the argument fails.

$p(n)$: “every set of n horses has the same colour” (all horses in the set share one colour). Base case $p(1)$: a set with one horse trivially has one colour: correct. The induction step must prove $\forall n \geq 1, (p(n) \Rightarrow p(n+1))$: for every $n \geq 1$, if every set of n horses is monochrome then every set of $n+1$ horses is. The

argument removes the first horse and the last horse from a set X of $n + 1$ horses and uses that the two resulting n -sets overlap, so that the colour of the overlap forces all $n + 1$ to agree. For $n = 1$ the two n -sets $\{h_2\}$ and $\{h_1\}$ do *not* overlap, so nothing links the colour of h_1 to that of h_2 : the implication $p(1) \Rightarrow p(2)$ was never proved. The step works for $n \geq 2$ only, and the chain from the base case is broken at its first link. $\square$

Exercise 4.2.11**essential**

Let $a, r \in \mathbb{R}$ with $r \neq 1$. Prove that $\sum_{k=0}^n ar^k = \frac{a(1-r^{n+1})}{1-r}$ for all $n \in \mathbb{N}$.

We proceed by induction on $n \geq 0$.

(Base case) $\sum_{k=0}^0 ar^k = ar^0 = a$, and $\frac{a(1-r^{1})}{1-r} = a$ since $r \neq 1$. So the formula holds for $n = 0$.

(Induction step) Let $n \geq 0$ and suppose $\sum_{k=0}^n ar^k = \frac{a(1-r^{n+1})}{1-r}$. We need to prove $\sum_{k=0}^{n+1} ar^k = \frac{a(1-r^{n+2})}{1-r}$. Now

$$\begin{aligned} \sum_{k=0}^{n+1} ar^k &= \sum_{k=0}^n ar^k + ar^{n+1} \quad \{\text{Definition 4.1.10}\} \\ &= \frac{a(1-r^{n+1})}{1-r} + ar^{n+1} \quad \{\text{induction hypothesis}\} \\ &= \frac{a(1-r^{n+1}) + ar^{n+1}(1-r)}{1-r} = \frac{a - ar^{n+1} + ar^{n+1} - ar^{n+2}}{1-r} = \frac{a(1-r^{n+2})}{1-r}. \end{aligned}$$

The result follows by induction. (Exercise 4.2.5 is the case $a = 1, r = 2$.) $\square$

NOW YOU 4.2.12 · solutions in Appendix S

4.2.12 For sets X_k ($k \in [n+1]$), prove by induction on n that there is a bijection $\prod_{k \in [n+1]} X_k \rightarrow (\prod_{k \in [n]} X_k) \times X_n$. [Hint: elements of the left are $(a_0, \dots, a_n)$; of the right, $((a_0, \dots, a_{n-1}), a_n)$.]

Binomials and the binomial theorem**Theorem 4.2.15, Example 4.2.16, Theorem 4.2.18**

$\binom{n}{k} = \frac{n!}{k!(n-k)!}$ if $k \leq n$, and $\binom{n}{k} = 0$ if $k > n$ (proved from the recursion by induction on n , with four cases in the step). Consequences: $\binom{n}{0} = \binom{n}{n} = 1$; $\binom{n}{k} \binom{k}{\ell} = \binom{n}{\ell} \binom{n-\ell}{k-\ell}$ for $\ell \leq k \leq n$. **Binomial theorem:** $(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$ for $n \in \mathbb{N}, x, y \in \mathbb{R}$ (proved for $(1+x)^n$ by induction, reindexing and using the Pascal recursion, then scaling by y^n). Example 4.2.13/4.2.19: $\sum_{k=0}^n \binom{n}{k} = 2^n$, by induction or at once from $(1+1)^n$.

Exercise 4.2.17**recommended**

Prove that $\binom{n}{k} = \binom{n}{n-k}$ for all $n, k \in \mathbb{N}$ with $k \leq n$.

Since $k \leq n$ we have $n-k \in \mathbb{N}$ and $n-k \leq n$, so Theorem 4.2.15 applies to both sides:

$$\binom{n}{n-k} = \frac{n!}{(n-k)!(n-(n-k))!} = \frac{n!}{(n-k)!k!} = \frac{n!}{k!(n-k)!} = \binom{n}{k}.$$

 $\square$

Exercise 4.2.20**recommended**

Give a proof of $\sum_{i=0}^n (-1)^i \binom{n}{i} = 0$ using the binomial theorem.

For $n \geq 1$, apply Theorem 4.2.18 with $x = -1$, $y = 1$: $\sum_{i=0}^n \binom{n}{i} (-1)^i 1^{n-i} = (-1 + 1)^n = 0^n = 0$ (the hint: $(-1)^i \binom{n}{i} = \binom{n}{i} (-1)^i 1^{n-i}$). Note that the identity needs $n \geq 1$: for $n = 0$ the sum is $(-1)^0 \binom{0}{0} = 1$, matching $0^0 = 1$. $\square$

NOW YOU 4.2.14 · solutions in Appendix S

4.2.14 Prove $\sum_{i=0}^n (-1)^i \binom{n}{i} = 0$ (for $n \geq 1$) by induction, in the style of Example 4.2.13.

C STRONG INDUCTION AND WELL-ORDERING**Theorems 4.3.2, 4.3.5 and Strategies 4.3.3, 4.3.6 · strong induction**

To prove $\forall n \geq n_0, p(n)$ it suffices to prove $p(n_0)$ and: for all $n \geq n_0$, if $p(k)$ holds for all $n_0 \leq k \leq n$, then $p(n+1)$. **Multiple base cases:** prove $p(n_0), \dots, p(n_1)$ and: for all $n \geq n_1$, if $p(k)$ for all $n_0 \leq k \leq n$ then $p(n+1)$. Strong induction is no stronger than weak induction (it is proved from it, applied to $q(n)$: “ $p(k)$ for all $n_0 \leq k \leq n$ ”); use it when the goal $p(n+1)$ depends on earlier terms than $p(n)$ (Example 4.3.1: $b_{n+1} = 1 + \sum_{k \leq n} b_k$ needs all b_k). **Get the quantifiers right:** “Fix $n \geq n_1$ and assume $p(k)$ for all $n_0 \leq k \leq n$.”

Model: Example 4.3.7

$a_0 = 0, a_1 = 1, a_n = 3a_{n-1} - 2a_{n-2}$ ($n \geq 2$); claim $a_n = 2^n - 1$. **Base cases:** $a_0 = 0 = 2^0 - 1, a_1 = 1 = 2^1 - 1$. **Step:** fix $n \geq 1$ and assume $a_k = 2^k - 1$ for all $0 \leq k \leq n$. Since $n+1 \geq 2$ the recursion applies: $a_{n+1} = 3a_n - 2a_{n-1} = 3(2^n - 1) - 2(2^{n-1} - 1) = 3 \cdot 2^n - 3 - 2^n + 2 = 2 \cdot 2^n - 1 = 2^{n+1} - 1$. The step uses a_n and a_{n-1} , which is why there are two base cases and why $n \geq 1$ (not $n \geq 0$) is fixed in the step.

Exercise 4.3.8**essential**

Define $a_0 = 4, a_1 = 9$ and $a_n = 5a_{n-1} - 6a_{n-2}$ for $n \geq 2$. Prove that $a_n = 3 \cdot 2^n + 3^n$ for all $n \in \mathbb{N}$.

We proceed by strong induction on $n \geq 0$ with two base cases.

(Base cases) $a_0 = 4 = 3 \cdot 2^0 + 3^0$ and $a_1 = 9 = 3 \cdot 2^1 + 3^1 = 6 + 3$.

(Induction step) Fix $n \geq 1$ and assume $a_k = 3 \cdot 2^k + 3^k$ for all $0 \leq k \leq n$. We need to prove $a_{n+1} = 3 \cdot 2^{n+1} + 3^{n+1}$. Since $n+1 \geq 2$,

$$\begin{aligned} a_{n+1} &= 5a_n - 6a_{n-1} \quad \{\text{definition}\} \\ &= 5(3 \cdot 2^n + 3^n) - 6(3 \cdot 2^{n-1} + 3^{n-1}) \quad \{\text{induction hypothesis for } k = n, n-1\} \\ &= 15 \cdot 2^n + 5 \cdot 3^n - 9 \cdot 2^{n-1} - 2 \cdot 3^n \quad \{6 \cdot 2^{n-1} = 3 \cdot 2^n, 6 \cdot 3^{n-1} = 2 \cdot 3^n\} \\ &= 6 \cdot 2^n + 3 \cdot 3^n = 3 \cdot 2^{n+1} + 3^{n+1}. \end{aligned}$$

The result follows by induction. $\square$

Exercise 4.3.9**essential**

The Tribonacci sequence is $t_0 = 0, t_1 = 0, t_2 = 1, t_n = t_{n-1} + t_{n-2} + t_{n-3}$ for $n \geq 3$. Prove that $t_n \leq 2^{n-3}$ for all $n \geq 3$.

We proceed by strong induction on $n \geq 3$ with base cases $n = 3, 4, 5$ (the step uses three previous terms).
(Base cases) $t_3 = 1 + 0 + 0 = 1 \leq 2^0$; $t_4 = 1 + 1 + 0 = 2 \leq 2^1$; $t_5 = 2 + 1 + 1 = 4 \leq 2^2$.

(Induction step) Fix $n \geq 5$ and assume $t_k \leq 2^{k-3}$ for all $3 \leq k \leq n$. We need $t_{n+1} \leq 2^{n-2}$. Since $n + 1 \geq 6$, $t_{n+1} = t_n + t_{n-1} + t_{n-2}$, and $n, n-1, n-2 \geq 3$, so the hypothesis applies to all three:

$$t_{n+1} \leq 2^{n-3} + 2^{n-4} + 2^{n-5} = 2^{n-5}(4 + 2 + 1) = 7 \cdot 2^{n-5} < 8 \cdot 2^{n-5} = 2^{n-2}.$$

The result follows by induction. □

Exercise 4.3.10**recommended**

Prove that for all $n \geq 8$, a value of n dubloons can be obtained using only 3- and 5-dubloon coins.

Let $p(n)$: “ $n = 3a + 5b$ for some $a, b \in \mathbb{N}$ ”. We proceed by strong induction on $n \geq 8$ with base cases 8, 9, 10 (hint: if $p(n)$ then $p(n+3)$, by adding one 3-coin).

(Base cases) $8 = 3 + 5$; $9 = 3 + 3 + 3$; $10 = 5 + 5$.

(Induction step) Fix $n \geq 10$ and assume $p(k)$ for all $8 \leq k \leq n$. We need $p(n+1)$. Since $n \geq 10$, $n-2 \geq 8$, so $p(n-2)$ holds: $n-2 = 3a + 5b$. Then $n+1 = 3(a+1) + 5b$, so $p(n+1)$ holds.

The result follows by induction. (For $n = 7$: $7 = 3a + 5b$ would need $b \in \{0, 1\}$, giving $3a = 7$ or $3a = 2$, impossible.) □

The well-ordering principle**Theorem 4.3.11, Lemma 4.3.13, Proposition 4.3.12**

Well-ordering principle: every non-empty set of natural numbers has a least element (proved by strong induction on an element n of the set). **Lemma 4.3.13:** every positive rational q is $\frac{a}{b}$ with $a, b \in \mathbb{N}$ non-zero and 1 the only natural number dividing both (take the least a with $q = \frac{a}{b}$; a common divisor $d \geq 2$ gives $q = \frac{a'}{b'}$ with $a' = \frac{a}{d} < a$). **$\sqrt{2}$ is irrational:** if $\sqrt{2} = \frac{a}{b}$ in lowest terms then $a^2 = 2b^2$, so a is even by Proposition 1.1.63 (“ a^2 even $\Rightarrow a$ even”), $a = 2c$, $2c^2 = b^2$, so b is even too; then 2 divides both a and b , contradiction. *Writing tip:* prove a repeated step once as a lemma and quote it.

Exercise 4.3.14**essential**

What goes wrong in the proof of Proposition 4.3.12 if we try to prove that $\sqrt{4}$ is irrational?

Suppose $\sqrt{4} = \frac{a}{b}$ in lowest terms. Squaring gives $a^2 = 4b^2$, so a^2 is even and hence a is even, $a = 2c$. Substituting: $4c^2 = 4b^2$, so $c^2 = b^2$. At this point the argument for $\sqrt{2}$ deduced “ b^2 is even”; here we only get $c^2 = b^2$, which says nothing about the parity of b . No contradiction arises, and indeed $\sqrt{4} = \frac{2}{1}$ with $a = 2, b = 1$ satisfies everything. The step that fails is the one that depended on the prime 2 dividing the right-hand side an odd number of times. □

Exercise 4.3.15**essential**

Prove that $\sqrt{3}$ is irrational.

Lemma. If $a \in \mathbb{Z}$ and 3 divides a^2 , then 3 divides a . *Proof.* By the division theorem $a = 3q + r$ with

$r \in \{0, 1, 2\}$. If $r = 1$ then $a^2 = 9q^2 + 6q + 1 = 3(3q^2 + 2q) + 1$; if $r = 2$ then $a^2 = 9q^2 + 12q + 4 = 3(3q^2 + 4q + 1) + 1$. In both cases a^2 leaves remainder 1, so $3 \nmid a^2$ (remainders are unique, Theorem 0.42). Hence $3 \mid a^2$ forces $r = 0$, i.e. $3 \mid a$.

Proof of the claim. Suppose $\sqrt{3}$ is rational. Since $\sqrt{3} > 0$, by Lemma 4.3.13 we can write $\sqrt{3} = \frac{a}{b}$ with a, b positive natural numbers whose only common natural divisor is 1. Squaring: $a^2 = 3b^2$, so $3 \mid a^2$ and by the Lemma $3 \mid a$: $a = 3c$. Then $9c^2 = 3b^2$, so $b^2 = 3c^2$, $3 \mid b^2$, and $3 \mid b$ by the Lemma. So 3 divides both a and b , contradicting lowest terms. Hence $\sqrt{3}$ is irrational. $\square$

Infinite descent

Definition 4.3.16 and Strategy 4.3.19

$X \subseteq \mathbb{R}$ has **infinite descent** if for all $x \in X$ there is $y \in X$ with $y < x$ (Example 4.3.17: $\mathbb{Z}$). The well-ordering principle says no *non-empty subset of $\mathbb{N}$* has infinite descent. **Proof by infinite descent:** to derive a contradiction from p , use p to build a non-empty $X \subseteq \mathbb{N}$ with infinite descent. Typical use: “ X = the set of counterexamples” (Example 4.3.21, existence of prime factorisation: a smallest counterexample $n = ab$ yields a smaller one) or “ X = the set of candidate remainders” (Example 4.3.20: if the least r with $a = qb + r$ had $r \geq b$, then $r - b$ would be a smaller candidate).

Exercise 4.3.18

recommended

Prove that the interval $(0, 1)$ has infinite descent.

Let $x \in (0, 1)$. Put $y = \frac{x}{2}$. Then $0 < y$ (as $x > 0$), $y < x$ (as $x > 0$), and $y < x < 1$, so $y \in (0, 1)$ and $y < x$. Hence $(0, 1)$ has infinite descent. (No contradiction follows: $(0, 1) \not\subseteq \mathbb{N}$.) $\square$

NOW YOU 4.3.22 · solutions in Appendix S

4.3.22 Prove by infinite descent: if $\sqrt{n}$ ($n \in \mathbb{N}$) is not an integer then it is irrational. [Hint: if $\sqrt{n} = \frac{a}{b}$, then $a(\sqrt{n} - \lfloor \sqrt{n} \rfloor)$ and $b(\sqrt{n} - \lfloor \sqrt{n} \rfloor)$ are integers; consider $\sqrt{n} \cdot \frac{\sqrt{n} - \lfloor \sqrt{n} \rfloor}{\sqrt{n} - \lfloor \sqrt{n} \rfloor}$.]

Watch out

- The induction step must work for *every* $n \geq n_0$, including $n = n_0$ (the horses, 4.2.8). When the step uses $p(n-1)$ or $p(n-2)$ you need extra base cases and must fix $n \geq n_1$.
- Write the induction goal before proving it and never assume it; mark where the induction hypothesis is used.
- Divisibility statements (“ $3 \mid n^3 - n$ ”, 4.E.16, 4.E.17): unfold “ $= 3k$ for some $k \in \mathbb{Z}$ ” and exhibit the new quotient explicitly.
- “ $3n < 2^n$ for $n \geq 4$ ” says nothing about $n < 4$: check small cases one by one (4.E.10).

D CHAPTER 4 EXERCISES

Recursive definitions (4.E.1–4.E.9)

NOW YOU 4.E.1–4.E.9 · solutions in Appendix S

4.E.1–4.E.5 From the recursive definitions: $1 + 3 = 4$; $0 + 5 = 5$; $2 \cdot 3 = 6$; $0 \cdot 5 = 0$; $2^3 = 8$. **4.E.6** Define $\exists^n$ by recursion on n . **4.E.7** From Definition 4.1.15, $\binom{4}{2} = 6$. **4.E.8** Trailing zeros of 41! in

decimal and in binary. [Hint: the number of trailing zeros in base b is the largest r with $b^r \mid n$.] **4.E.9** (N, z, s) is a notion of natural numbers iff for every X , $a \in X$, $f : X \rightarrow X$ there is a unique $h : N \rightarrow X$ with $h(z) = a$ and $f \circ h = h \circ s$.

Proofs by induction (4.E.10–4.E.17)

Exercise 4.E.10

essential

Find all natural numbers n such that $n^5 < 5^n$.

Small cases: $0 < 1$ (true); $1 < 5$ (true); $32 < 25$ (false); $243 < 125$ (false); $1024 < 625$ (false); $3125 < 3125$ (false); $7776 < 15625$ (true). **Claim:** $n^5 < 5^n$ for all $n \geq 6$. By induction on $n \geq 6$. *Base case:* $6^5 = 7776 < 15625 = 5^6$. *Step:* let $n \geq 6$ and suppose $n^5 < 5^n$. Then $\left(\frac{n+1}{n}\right)^5 = \left(1 + \frac{1}{n}\right)^5 \leq \left(\frac{7}{6}\right)^5 = \frac{16807}{7776} < 5$, so $(n+1)^5 < 5n^5 < 5 \cdot 5^n = 5^{n+1}$. The result follows by induction.

Answer: $n^5 < 5^n$ exactly for $n \in \{0, 1\} \cup \{n \in \mathbb{N} \mid n \geq 6\}$. □

Exercise 4.E.11

essential

Prove that $(1+x)^{123456789} \geq 1 + 123456789x$ for all real $x \geq -1$.

Nothing is special about 123456789 (the hint): we prove $(1+x)^n \geq 1 + nx$ for all $n \in \mathbb{N}$ and all real $x \geq -1$, by induction on n (not on x : induction only reaches natural numbers). Fix $x \geq -1$. *Base case:* $(1+x)^0 = 1 \geq 1 + 0 \cdot x$. *Step:* let $n \geq 0$ and suppose $(1+x)^n \geq 1 + nx$. Since $1+x \geq 0$ we may multiply the hypothesis by $1+x$ without changing the inequality:

$$(1+x)^{n+1} = (1+x)(1+x)^n \geq (1+x)(1+nx) = 1 + (n+1)x + nx^2 \geq 1 + (n+1)x,$$

as $nx^2 \geq 0$. By induction the inequality holds for all n ; take $n = 123456789$. □

Exercise 4.E.12

recommended

Let $a \in \mathbb{N}$ end in the digit 6. Prove that a^n ends in 6 for all $n \geq 1$.

“Ends in 6” means $a = 10q + 6$ for some $q \in \mathbb{N}$ (the last decimal digit is the remainder on division by 10). Induction on $n \geq 1$. *Base case:* $a^1 = a$ ends in 6. *Step:* let $n \geq 1$ and suppose $a^n = 10m + 6$. Then $a^{n+1} = a^n \cdot a = (10m + 6)(10q + 6) = 100mq + 60m + 60q + 36 = 10(10mq + 6m + 6q + 3) + 6$, which ends in 6. The result follows by induction. □

Exercise 4.E.13

essential

Let $a, b \in \mathbb{R}$ and let $a_0 = a$, $a_1 = b$ and $a_n = \frac{a_{n-1} + a_{n+1}}{2}$ for all $n \geq 1$. Prove that $a_n = a + (b-a)n$ for all $n \in \mathbb{N}$.

Rewrite the recursion as $a_{n+1} = 2a_n - a_{n-1}$ for $n \geq 1$. We proceed by strong induction with base cases $n = 0, 1$. *Base cases:* $a_0 = a = a + (b-a) \cdot 0$ and $a_1 = b = a + (b-a) \cdot 1$. *Step:* fix $n \geq 1$ and assume $a_k = a + (b-a)k$ for all $0 \leq k \leq n$. Then

$$a_{n+1} = 2a_n - a_{n-1} = 2(a + (b-a)n) - (a + (b-a)(n-1)) = a + (b-a)(2n - n + 1) = a + (b-a)(n+1).$$

The result follows by induction. □

Exercise 4.E.14**recommended**

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $f(x+y) = f(x) + f(y)$ for all x, y . (a) Prove by induction that there is $a \in \mathbb{R}$ with $f(n) = an$ for all $n \in \mathbb{N}$. (b) Deduce $f(n) = an$ for all $n \in \mathbb{Z}$. (c) Deduce $f(x) = ax$ for all $x \in \mathbb{Q}$.

(a) Put $a = f(1)$. First $f(0) = f(0+0) = 2f(0)$, so $f(0) = 0 = a \cdot 0$: base case. Step: if $f(n) = an$ then $f(n+1) = f(n) + f(1) = an + a = a(n+1)$. By induction $f(n) = an$ for all $n \in \mathbb{N}$.

(b) For $n \in \mathbb{N}$: $0 = f(0) = f(n + (-n)) = f(n) + f(-n) = an + f(-n)$, so $f(-n) = -an = a(-n)$. Every integer is n or $-n$ for some $n \in \mathbb{N}$.

(c) Let $x \in \mathbb{Q}$ and write $x = \frac{m}{b}$ with $m \in \mathbb{Z}$ and $b \geq 1$. First, $f(bx) = bf(x)$ for all $b \in \mathbb{N}$, by induction on b : $f(0 \cdot x) = f(0) = 0$, and $f((b+1)x) = f(bx+x) = f(bx) + f(x) = bf(x) + f(x)$. Hence $bf(x) = f(bx) = f(m) = am$ by (b), so $f(x) = \frac{am}{b} = ax$. $\square$

Exercise 4.E.15**recommended**

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(0) > 0$ and $f(x+y) = f(x)f(y)$ for all x, y . Prove there is a positive real a with $f(x) = a^x$ for all $x \in \mathbb{Q}$.

$f(0) = f(0)f(0)$ with $f(0) > 0$ gives $f(0) = 1$. For any x , $f(x)f(-x) = f(0) = 1$, so $f(x) \neq 0$, and $f(x) = f(\frac{x}{2})^2 > 0$. Put $a = f(1) > 0$. *Naturals*: $f(0) = 1 = a^0$; if $f(n) = a^n$ then $f(n+1) = f(n)f(1) = a^{n+1}$; by induction $f(n) = a^n$ for $n \in \mathbb{N}$. *Integers*: $f(-n) = 1/f(n) = a^{-n}$. *Rationals*: for $x = \frac{m}{b}$ with $m \in \mathbb{Z}$, $b \geq 1$: by induction on b , $f(bx) = f(x)^b$; so $f(x)^b = f(m) = a^m$, and since $f(x) > 0$, $f(x)$ is the unique positive b -th root of a^m , namely $a^{m/b} = a^x$. $\square$

Exercise 4.E.16**essential**

Let $a, b \in \mathbb{Z}$. Prove that $b - a$ divides $b^n - a^n$ for all $n \in \mathbb{N}$.

Induction on $n \geq 0$; $p(n)$: $b^n - a^n = (b-a)k$ for some $k \in \mathbb{Z}$. *Base case*: $b^0 - a^0 = 0 = (b-a) \cdot 0$. *Step*: let $n \geq 0$ and suppose $b^n - a^n = (b-a)k$. Then

$$b^{n+1} - a^{n+1} = b \cdot b^n - a \cdot a^n = b(b^n - a^n) + (b-a)a^n = (b-a)(bk + a^n),$$

and $bk + a^n \in \mathbb{Z}$. So $b-a \mid b^{n+1} - a^{n+1}$. The result follows by induction. $\square$

Exercise 4.E.17**essential**

Prove by induction that $7^n - 2 \cdot 4^n + 1$ is divisible by 18 for all $n \in \mathbb{N}$.

The book's hint: $7^{n+1} - 2 \cdot 4^{n+1} + 1 = 4(7^n - 2 \cdot 4^n + 1) + 3(7^n - 1)$ for all n (check: the right side is $7 \cdot 7^n - 8 \cdot 4^n + 4 - 3 = 7^{n+1} - 2 \cdot 4^{n+1} + 1$). So we need a separate fact about $7^n - 1$.

Lemma: $6 \mid 7^n - 1$ for all $n \in \mathbb{N}$. By induction: $7^0 - 1 = 0 = 6 \cdot 0$; and if $7^n - 1 = 6m$ then $7^{n+1} - 1 = 7(7^n - 1) + 6 = 6(7m + 1)$.

Main proof, by induction on $n \geq 0$. *Base case*: $7^0 - 2 \cdot 4^0 + 1 = 1 - 2 + 1 = 0 = 18 \cdot 0$. *Step*: let $n \geq 0$ and suppose $7^n - 2 \cdot 4^n + 1 = 18k$ with $k \in \mathbb{Z}$. By the Lemma, $7^n - 1 = 6m$ for some $m \in \mathbb{Z}$. Then

$$7^{n+1} - 2 \cdot 4^{n+1} + 1 = 4 \cdot 18k + 3 \cdot 6m = 18(4k + m),$$

so 18 divides $7^{n+1} - 2 \cdot 4^{n+1} + 1$. The result follows by induction. $\square$

Some examples from calculus (4.E.18–4.E.23)

Exercise 4.E.18

essential

Prove that $\frac{d^n}{dx^n}(xe^x) = (n+x)e^x$ for all $n \in \mathbb{N}$.

Induction on n . *Base case:* the 0-th derivative is the function itself: $xe^x = (0+x)e^x$. *Step:* suppose $\frac{d^n}{dx^n}(xe^x) = (n+x)e^x$. Differentiating once more (product rule): $\frac{d^{n+1}}{dx^{n+1}}(xe^x) = \frac{d}{dx}((n+x)e^x) = 1 \cdot e^x + (n+x)e^x = (n+1+x)e^x$. The result follows by induction. $\square$

Exercise 4.E.19

essential

Find and prove an expression for $\frac{d^n}{dx^n}(x^2e^x)$ for all $n \in \mathbb{N}$.

Compute: $n = 0$: x^2e^x ; $n = 1$: $(x^2 + 2x)e^x$; $n = 2$: $(x^2 + 4x + 2)e^x$; $n = 3$: $(x^2 + 6x + 6)e^x$. **Claim:** $\frac{d^n}{dx^n}(x^2e^x) = (x^2 + 2nx + n(n-1))e^x$. *Base case* $n = 0$: $(x^2 + 0 + 0)e^x$. *Step:* if the claim holds for n , then

$$\begin{aligned} \frac{d}{dx}((x^2 + 2nx + n(n-1))e^x) &= (2x + 2n)e^x + (x^2 + 2nx + n(n-1))e^x \\ &= (x^2 + 2(n+1)x + n(n-1) + 2n)e^x, \end{aligned}$$

and $n(n-1) + 2n = n(n+1) = (n+1)n$, which is the claim for $n+1$. By induction the formula holds for all n . $\square$

Exercise 4.E.20

essential

Let f, g be differentiable (as often as needed). Prove that $\frac{d^n}{dx^n}(f(x)g(x)) = \sum_{k=0}^n \binom{n}{k} f^{(k)}(x)g^{(n-k)}(x)$ for all $n \in \mathbb{N}$.

Induction on n . *Base case:* $n = 0$: $\binom{0}{0}f^{(0)}g^{(0)} = fg$. *Step:* suppose the formula for n . Differentiate term by term with the product rule:

$$\frac{d^{n+1}}{dx^{n+1}}(fg) = \sum_{k=0}^n \binom{n}{k} (f^{(k+1)}g^{(n-k)} + f^{(k)}g^{(n+1-k)}) = \sum_{k=1}^{n+1} \binom{n}{k-1} f^{(k)}g^{(n+1-k)} + \sum_{k=0}^n \binom{n}{k} f^{(k)}g^{(n+1-k)}$$

(reindexing the first sum with $k \rightarrow k-1$). Split off $k = n+1$ from the first sum and $k = 0$ from the second, and combine the rest with Pascal's rule $\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}$ (Definition 4.1.15):

$$= \binom{n}{n} f^{(n+1)}g^{(0)} + \sum_{k=1}^n \binom{n+1}{k} f^{(k)}g^{(n+1-k)} + \binom{n}{0} f^{(0)}g^{(n+1)} = \sum_{k=0}^{n+1} \binom{n+1}{k} f^{(k)}g^{(n+1-k)},$$

since $\binom{n}{n} = 1 = \binom{n+1}{n+1}$ and $\binom{n}{0} = 1 = \binom{n+1}{0}$. This is exactly the induction in the proof of the binomial theorem (Theorem 4.2.18), with derivatives in place of powers. $\square$

Exercise 4.E.21

recommended

Find and prove an expression for the n -th derivative of $\log x$ ($x > 0$) for all $n \in \mathbb{N}$.

$n = 0$: $\log x$; $n = 1$: x^{-1} ; $n = 2$: $-x^{-2}$; $n = 3$: $2x^{-3}$; $n = 4$: $-6x^{-4}$. **Claim:** for $n \geq 1$, $\frac{d^n}{dx^n} \log x = (-1)^{n-1}(n-1)!x^{-n}$. *Base case* $n = 1$: $(-1)^0 0!x^{-1} = \frac{1}{x}$. *Step:* if the claim holds for $n \geq 1$ then $\frac{d}{dx}((-1)^{n-1}(n-1)!x^{-n}) = (-1)^{n-1}(n-1)! \cdot (-n)x^{-n-1} = (-1)^n n!x^{-(n+1)}$, the claim for $n+1$. By induction the formula holds for all $n \geq 1$ (and for $n = 0$ the derivative is $\log x$ itself). $\square$

NOW YOU 4.E.22, 4.E.23 · solutions in Appendix S

4.E.22 $(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$ for all $n \in \mathbb{N}$. [Hint: angle addition formulae.] **4.E.23** (a) $\int_0^{\pi/2} \sin^{n+2} x dx = \frac{n+1}{n+2} \int_0^{\pi/2} \sin^n x dx$ (integration by parts); (b) $\int_0^{\pi/2} \sin^{2n} x dx = \frac{\pi}{2^{2n+1}} \binom{2n}{n}$ by induction; (c) find $\int_0^{\pi/2} \sin^{2n+1} x dx$.

True–false (4.E.24–4.E.32)**Exercise 4.E.24****recommended**

Every factorial is positive.

True. By induction on n : $0! = 1 > 0$; if $n! > 0$ then $(n+1)! = (n+1) \cdot n! > 0$ as a product of positive numbers. So $n! > 0$ for all $n \in \mathbb{N}$. ☐

Exercise 4.E.25**essential**

Every binomial coefficient is positive.

False. $\binom{0}{1} = 0$ by Definition 4.1.15 (the clause $\binom{0}{k+1} = 0$); more generally $\binom{n}{k} = 0$ whenever $k > n$ (Theorem 4.2.15). ☐

Exercise 4.E.26**essential**

Every proof by strong induction can be turned into a proof by weak induction by changing only the induction hypothesis.

False. A proof by strong induction may use $p(k)$ for $k < n$ in its step (Example 4.3.4 uses all of $b_0, \dots, b_n$; Example 4.3.7 uses a_{n-1}). Replacing the hypothesis by just $p(n)$ leaves those uses unjustified, so the step no longer proves $p(n+1)$: the rest of the proof must change too (e.g. by proving the stronger statement $q(n)$: “ $p(k)$ for all $k \leq n$ ”, as in the proof of Theorem 4.3.2). (Strong and weak induction prove the same statements, but not by the same proof.) ☐

Exercise 4.E.27**essential**

Every proof by weak induction can be turned into a proof by strong induction by changing only the induction hypothesis.

True. Replace the hypothesis “ $p(n)$ ” by “ $p(k)$ for all $n_0 \leq k \leq n$ ”. The new hypothesis contains the old one (take $k = n$), so every step of the original argument remains valid and still derives $p(n+1)$; the base case is unchanged. By Theorem 4.3.2 the conclusion follows. ☐

Exercise 4.E.28**recommended**

Let $f: \mathbb{N} \rightarrow X$ be a surjection and $p(x)$ a formula on X . If $p(f(0))$ and $\forall n \in \mathbb{N}, (p(f(n)) \Rightarrow p(f(n+1)))$, then $\forall x \in X, p(x)$.

True. Let $q(n)$ be $p(f(n))$, a formula with free variable $n \in \mathbb{N}$. The hypotheses say $q(0)$ and $q(n) \Rightarrow q(n+1)$ for all n , so $q(n)$ holds for all $n \in \mathbb{N}$ by weak induction. Now let $x \in X$; since f is surjective, $x = f(n)$ for some n , and $p(x) = p(f(n)) = q(n)$ is true. ☐

Exercise 4.E.29

essential

If for all $x \in \mathbb{N}$ there is some $y \leq x$ with $p(y)$ false, then $p(x)$ is false for all $x \in \mathbb{N}$.

False. Take $p(x)$: “ $x \geq 1$ ”. The hypothesis holds: for every $x \in \mathbb{N}$, $y = 0 \leq x$ and $p(0)$ is false. But the conclusion fails, since $p(1)$ is true. (The hypothesis only guarantees that $p(0)$ is false: it is not an induction principle.) $\square$

Exercise 4.E.30

essential

Every non-empty subset of $\mathbb{Q}_{\geq 0}$ has a least element.

False. $\{\frac{1}{n} \mid n \geq 1\} \subseteq \mathbb{Q}_{\geq 0}$ is non-empty and has no least element: for any $\frac{1}{n}$ in it, $\frac{1}{n+1} < \frac{1}{n}$ is also in it (infinite descent, Definition 4.3.16). $\square$

Exercise 4.E.31

essential

Every non-empty subset of $\mathbb{Z}$ has a least element.

False. $\mathbb{Z}$ itself is non-empty and has no least element: for $n \in \mathbb{Z}$, $n - 1 \in \mathbb{Z}$ and $n - 1 < n$ (Example 4.3.17). $\square$

Exercise 4.E.32

essential

Every non-empty subset of $\mathbb{Z}^-$ (the negative integers) has a greatest element.

True. Let $X \subseteq \mathbb{Z}^-$ be non-empty and put $Y = \{-x - 1 \mid x \in X\}$. For $x \leq -1$, $-x - 1 \geq 0$, so $Y \subseteq \mathbb{N}$, and Y is non-empty. By the well-ordering principle Y has a least element $-x_0 - 1$ with $x_0 \in X$. For any $x \in X$: $-x - 1 \geq -x_0 - 1$, so $x \leq x_0$. Hence x_0 is the greatest element of X . $\square$

Always–sometimes–never (4.E.33–4.E.35)
Exercise 4.E.33

recommended

Let $n, k, \ell \in \mathbb{N}$ (the book writes $\mathbb{Z}$, but binomial coefficients are defined for natural numbers). Then $\binom{n}{k+\ell} = \binom{n}{k} + \binom{n}{\ell}$.

Sometimes. True for $n = 0, k = \ell = 1$: $\binom{0}{2} = 0 = \binom{0}{1} + \binom{0}{1}$. False for $n = 1, k = 0, \ell = 1$: $\binom{1}{1} = 1$ but $\binom{1}{0} + \binom{1}{1} = 2$. $\square$

Exercise 4.E.34

recommended

Let $m, n \geq 2$. Then $(m+n)! = m! + n!$.

Never. First, $j! \leq \ell!$ whenever $j \leq \ell$ (each step multiplies by a number ≥ 1). Since $m+n-1 \geq m$ and $\geq n$, $(m+n-1)! \geq \max(m!, n!)$, so

$$(m+n)! = (m+n) \cdot (m+n-1)! \geq 4 \max(m!, n!) > 2 \max(m!, n!) \geq m! + n!,$$

using $m+n \geq 4$ and $\max(m!, n!) > 0$ (4.E.24). So $(m+n)! > m! + n!$ always. $\square$

Exercise 4.E.35**recommended**

Let $p(x)$ be a formula on $\mathbb{Z}$ with $p(0)$ true and, for all $n \in \mathbb{Z}$, $p(n) \Rightarrow p(n+1)$ and $p(n) \Rightarrow p(n-2)$. Then $\forall n \in \mathbb{Z}, p(n)$.

Always. By weak induction, $p(n)$ for all $n \in \mathbb{N}$ (base $p(0)$; step $p(n) \Rightarrow p(n+1)$). For negative integers prove $q(m)$: “ $p(-m)$ ” for all $m \in \mathbb{N}$ by strong induction with base cases $m = 0, 1$: $p(0)$ holds, and $p(-1)$ follows from $p(1)$ via $p(1) \Rightarrow p(1-2)$. Step: fix $m \geq 1$ and assume $p(-k)$ for all $0 \leq k \leq m$; then $p(-(m-1))$ holds, and $p(-(m-1)) \Rightarrow p(-(m-1)-2) = p(-(m+1))$. So $p(n)$ holds for every integer n . $\square$

Chapter 5

Relations

Contents

- A Relations (§5.1)
- B Equivalence relations and partitions (§5.2)
- C Chapter 5 exercises (5.E)

Planning

★ 5.1: 6, 13, 22, 23, 28, 39, 40, 41

★ 5.1: 10, 11, 18, 24, 29, 33, 34

★ 5.2: 24, 27, 33 ★ 5.2: 25, 28, 30, 34, 36

★ 5.E: 18, 22, 23

Tested in **Quiz 3** (21 Oct, relations and modular arithmetic) and the exam.

A RELATIONS

Definition 5.1.1, Axiom 5.1.4, Definition 5.1.7 · relations and their graphs

A **(binary) relation** from X to Y is a logical formula $R(x, y)$ with free variables $x \in X$ (the **domain**) and $y \in Y$ (the **codomain**); write $x R y$ when $R(x, y)$ is true. R is **homogeneous** (a relation *on* X) if $X = Y$. Examples 5.1.2: $|$ on $\mathbb{Z}$, $\leq$ on $\mathbb{R}$, $=$ on X , $\subseteq$ on $\mathcal{P}(X)$, $\in$ from X to $\mathcal{P}(X)$, and R_f from X to Y ($x R_f y \Leftrightarrow f(x) = y$) for any $f: X \rightarrow Y$.

Relation extensionality: $R = S$ iff they have the same domain and codomain and $\forall x \in X, \forall y \in Y, (x R y \Leftrightarrow x S y)$. **Graph:** $\text{Gr}(R) = \{(x, y) \in X \times Y \mid x R y\}$; it depends on the domain and codomain (Example 5.1.8: $\leq$ on $[3]$ versus from $[2]$ to $[4]$). **Theorem 5.1.14:** every $G \subseteq X \times Y$ is the graph of a unique relation from X to Y , so (**Strategy 5.1.15**) a relation is specified by domain, codomain and graph, and $R = S$ can be proved by showing equal domains, codomains and graphs. The **discrete** relation $D_{X,Y}$ relates everything; the **empty** relation $\emptyset_{X,Y}$ relates nothing. The **diagonal** is $\Delta_X = \{(x, x) \mid x \in X\}$.

Exercise 5.1.6

essential

Let R and S be the relations on $\mathbb{R}$ defined by $a R b \Leftrightarrow b - a \in \mathbb{Q}$ and $a S b \Leftrightarrow \exists n \in \mathbb{Z}, (n \neq 0) \wedge n(b - a) \in \mathbb{Z}$. Prove that $R = S$.

Both have domain and codomain $\mathbb{R}$, so by Axiom 5.1.4 it suffices to show $a R b \Leftrightarrow a S b$ for all $a, b \in \mathbb{R}$.
 $(\Rightarrow)$ Suppose $b - a \in \mathbb{Q}$, say $b - a = \frac{p}{q}$ with $p, q \in \mathbb{Z}, q \neq 0$. Take $n = q$: then $n \neq 0$ and $n(b - a) = p \in \mathbb{Z}$. So $a S b$.

$(\Leftarrow)$ Suppose $n \in \mathbb{Z}, n \neq 0$ and $n(b - a) = m \in \mathbb{Z}$. Then $b - a = \frac{m}{n} \in \mathbb{Q}$. So $a R b$. □

Exercise 5.1.10

recommended

Let R be the relation on $\mathbb{Z}$ with $x R y \Leftrightarrow x^2 = y^2$. Describe $\text{Gr}(R) \subseteq \mathbb{Z} \times \mathbb{Z}$.

$x^2 = y^2$ iff $(y - x)(y + x) = 0$ iff $y = x$ or $y = -x$. So $\text{Gr}(R) = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} \mid x^2 = y^2\} = \{(x, x) \mid x \in \mathbb{Z}\} \cup \{(x, -x) \mid x \in \mathbb{Z}\}$: the two diagonals of the integer grid, i.e. $\Delta_{\mathbb{Z}} \cup \{(x, -x) \mid x \in \mathbb{Z}\}$. □

Exercise 5.1.11**recommended**

Let $f : X \rightarrow Y$ and let R_f be the relation $x R_f y \Leftrightarrow f(x) = y$. Prove that $\text{Gr}(R_f) = \text{Gr}(f)$.

By Definition 5.1.7 and Definition 3.1.6, $\text{Gr}(R_f) = \{(x, y) \in X \times Y \mid x R_f y\} = \{(x, y) \in X \times Y \mid f(x) = y\} = \text{Gr}(f)$. (Both are the same set-builder expression, so no double containment is needed.) $\square$

Exercise 5.1.13**essential**

Let X, Y be sets. Describe $\text{Gr}(D_{X,Y})$ and $\text{Gr}(\emptyset_{X,Y})$.

$\text{Gr}(D_{X,Y}) = \{(x, y) \in X \times Y \mid \text{true}\} = X \times Y$, since every pair is related. $\text{Gr}(\emptyset_{X,Y}) = \{(x, y) \in X \times Y \mid \text{false}\} = \emptyset$, since no pair is related. $\square$

Exercise 5.1.18**recommended**

Let X be a set. Prove that $\Delta_X = \text{Gr}(=)$.

$\text{Gr}(=) = \{(x, y) \in X \times X \mid x = y\}$. ($\subseteq$) Let $(x, x) \in \Delta_X$; then $x = x$, so $(x, x) \in \text{Gr}(=)$. ($\supseteq$) Let $(x, y) \in \text{Gr}(=)$; then $x = y$, so $(x, y) = (x, x) \in \Delta_X$. $\square$

NOW YOU 5.1.3 · solutions in Appendix S

5.1.3 Give three more examples of relations, not all homogeneous.

Properties of homogeneous relations**Definitions 5.1.19, 5.1.25, 5.1.30, 5.1.36 · reflexive, symmetric, antisymmetric, transitive**

For a relation R on X :

- **reflexive**: $\forall a \in X, a R a$;
- **symmetric**: $\forall a, b \in X, a R b \Rightarrow b R a$;
- **antisymmetric**: $\forall a, b \in X, (a R b \wedge b R a) \Rightarrow a = b$ (not the same as “not symmetric”: $=$ is both; many relations are neither);
- **transitive**: $\forall a, b, c \in X, (a R b \wedge b R c) \Rightarrow a R c$.

Examples: $=$ has all four. $\leq$ on $\mathbb{R}$ is reflexive, antisymmetric, transitive. “ $b - a \in \mathbb{Q}$ ” on $\mathbb{R}$ is reflexive ($a - a = 0$), symmetric ($a - b = -(b - a)$), transitive ($c - a = (c - b) + (b - a)$), Examples 5.1.21, 5.1.27, 5.1.38). Proposition 5.1.42: R transitive iff $x_0 R x_1 R \cdots R x_n$ always implies $x_0 R x_n$ (induction on n). Reflexivity depends on the domain (5.1.24); symmetry and transitivity depend only on the graph (5.1.29, 5.1.41).

Proving and disproving the properties

Reflexive: “Let $a \in X$ so $a R a$.” **Symmetric**: “Let $a, b \in X$ and suppose $a R b$ so $b R a$.” **Antisymmetric**: “Let $a, b \in X$ with $a R b$ and $b R a$ so $a = b$.” **Transitive**: “Let $a, b, c \in X$ with $a R b$ and $b R c$ so $a R c$.” In each case unfold what $a R b$ means (e.g. “ $b = qa$ for some $q \in \mathbb{Z}$ ”) and name the witnesses. **To disprove**: one explicit counterexample (specific a , or a, b , or a, b, c).

Exercise 5.1.22**essential**

Let X be a set. Prove that $\subseteq$ is a reflexive relation on $\mathcal{P}(X)$, but $\subsetneq$ is not.

Let $U \in \mathcal{P}(X)$. Every $a \in U$ satisfies $a \in U$, so $U \subseteq U$ (Definition 2.1.6): $\subseteq$ is reflexive. On the other hand $U \subsetneq U$ would require $U \subseteq U$ and $U \neq U$, and $U \neq U$ is false; so no U satisfies $U \subsetneq U$. Since $\mathcal{P}(X)$ is non-empty ($\emptyset \in \mathcal{P}(X)$), there is an element, e.g. $\emptyset$, with $\emptyset \not\subsetneq \emptyset$: $\subsetneq$ is not reflexive. $\square$

Exercise 5.1.23**essential**

Prove that the relation “ x divides y ” on $\mathbb{Z}$ is reflexive.

Let $x \in \mathbb{Z}$. Then $x = 1 \cdot x$ with $1 \in \mathbb{Z}$, so $x \mid x$ by Definition 0.36. $\square$

Exercise 5.1.24**recommended**

Let $G = \{(0,0), (1,1), (2,2)\}$, R the relation on $[3]$ with graph G , S the relation on $[4]$ with graph G . Prove R is reflexive but S is not.

$[3] = \{0,1,2\}$, and $(a,a) \in G$ for each $a \in \{0,1,2\}$, so $a R a$ for all $a \in [3]$: R is reflexive. $[4] = \{0,1,2,3\}$ and $(3,3) \notin G$, so $3 S 3$ is false: S is not reflexive. (Same graph, different domain.) $\square$

Exercise 5.1.28**essential**

Find all subsets $U \subseteq \mathbb{Z}$ such that the relation “ x divides y ” on U is symmetric.

Lemma. For $x, y \in \mathbb{Z}$: $x \mid y$ and $y \mid x$ if and only if $y = x$ or $y = -x$. *Proof.* $(\Leftarrow)$ $y = \pm x$ gives $y = (\pm 1)x$ and $x = (\pm 1)y$. $(\Rightarrow)$ Write $y = qx$ and $x = ry$ with $q, r \in \mathbb{Z}$; then $x = rqx$. If $x \neq 0$ then $rq = 1$, so $q = r = \pm 1$ (the only integers whose product is 1) and $y = \pm x$. If $x = 0$ then $y = q \cdot 0 = 0 = x$.

Answer. Divisibility on U is symmetric iff $\forall x, y \in U, x \mid y \Rightarrow y \mid x$, iff (by the Lemma) **whenever** $x, y \in U$ **and** $x \mid y$, **then** $y = \pm x$. In words: no element of U divides an element of U of a different absolute value. Examples: $\emptyset, \{0\}, \{a, -a\}$ for any a , $\{2, 3\}$, the set of prime numbers (no prime divides another). Non-examples: $\{1, 2\}$ ($1 \mid 2, 2 \nmid 1$), $\{0, 5\}$ ($5 \mid 0, 0 \nmid 5$), $\mathbb{N}$. In particular, if $0 \in U$ then symmetry forces $U = \{0\}$, since $x \mid 0$ for every x but $0 \mid x$ only for $x = 0$. $\square$

Exercise 5.1.29**recommended**

Let R and S be relations with $\text{Gr}(R) = \text{Gr}(S)$ (possibly different domains). Prove that R is symmetric iff S is symmetric.

Let R have domain X . We claim: R is symmetric iff $\forall (a, b) \in \text{Gr}(R), (b, a) \in \text{Gr}(R)$. $(\Rightarrow)$ If $(a, b) \in \text{Gr}(R)$ then $a, b \in X$ and $a R b$, so $b R a$ by symmetry, i.e. $(b, a) \in \text{Gr}(R)$. $(\Leftarrow)$ If $a, b \in X$ and $a R b$ then $(a, b) \in \text{Gr}(R)$, so $(b, a) \in \text{Gr}(R)$, so $b R a$. The claim expresses symmetry purely in terms of the graph; since $\text{Gr}(R) = \text{Gr}(S)$, R is symmetric iff S is. $\square$

Exercise 5.1.33**recommended**

Prove that “ x divides y ” is antisymmetric on $\mathbb{N}$ but not on $\mathbb{Z}$.

On $\mathbb{N}$: let $x, y \in \mathbb{N}$ with $x \mid y$ and $y \mid x$. By the Lemma in 5.1.28, $y = x$ or $y = -x$. If $y = -x$ with $x, y \geq 0$ then $x = y = 0$. In both cases $x = y$. On $\mathbb{Z}$: $1 \mid -1$ (as $-1 = (-1) \cdot 1$) and $-1 \mid 1$ (as $1 = (-1)(-1)$), but

$$1 \neq -1.$$

**Exercise 5.1.34****recommended**

Let X be a set. Prove that $\subseteq$ is antisymmetric on $\mathcal{P}(X)$.

Let $U, V \in \mathcal{P}(X)$ with $U \subseteq V$ and $V \subseteq U$. Then $U = V$ by set extensionality (Axiom 2.1.14: double containment). ☐

Exercise 5.1.39**essential**

Let X be a set. Prove that $\subseteq$ is transitive on $\mathcal{P}(X)$.

Let $U, V, W \in \mathcal{P}(X)$ with $U \subseteq V$ and $V \subseteq W$. Let $a \in U$; then $a \in V$ since $U \subseteq V$, and then $a \in W$ since $V \subseteq W$. So $U \subseteq W$ (this is Proposition 2.1.12). ☐

Exercise 5.1.40**essential**

Prove that “ x divides y ” on $\mathbb{Z}$ is transitive.

Let $x, y, z \in \mathbb{Z}$ with $x \mid y$ and $y \mid z$. Then $y = qx$ and $z = ry$ for some $q, r \in \mathbb{Z}$ (Definition 0.36). Hence $z = r(qx) = (rq)x$ with $rq \in \mathbb{Z}$, so $x \mid z$. ☐

Exercise 5.1.41**essential**

Let R and S be relations with $\text{Gr}(R) = \text{Gr}(S)$. Prove that R is transitive iff S is transitive.

As in 5.1.29: R (with domain X) is transitive iff for all $(a, b), (b, c) \in \text{Gr}(R)$ we have $(a, c) \in \text{Gr}(R)$. Indeed, if R is transitive and $(a, b), (b, c) \in \text{Gr}(R)$ then $a, b, c \in X$ with $a R b, b R c$, so $a R c$ and $(a, c) \in \text{Gr}(R)$; conversely if the graph condition holds and $a, b, c \in X$ with $a R b, b R c$, then $(a, b), (b, c) \in \text{Gr}(R)$, so $(a, c) \in \text{Gr}(R)$, i.e. $a R c$. The condition depends only on the graph, which R and S share. ☐

NOW YOU 5.1.35 · solutions in Appendix S

5.1.35 R is symmetric and antisymmetric iff $\text{Gr}(R) \subseteq \Delta_X$; deduce that the only reflexive, symmetric, antisymmetric relation on X is $=$.

B EQUIVALENCE RELATIONS AND PARTITIONS**Definitions 5.2.1 and 5.2.6, Theorem 5.2.11 · equivalence relations, congruence**

An **equivalence relation** is a reflexive, symmetric and transitive relation on X (written $\sim, \equiv, \approx$). Examples: $=$; “ $b - a \in \mathbb{Q}$ ” on $\mathbb{R}$ (5.2.3); $a \sim_f b \Leftrightarrow f(a) = f(b)$ for any $f : X \rightarrow Y$ (5.2.4); $U \cong V \Leftrightarrow$ there is a bijection $U \rightarrow V$ (5.2.5).

Congruence: for $n \in \mathbb{Z}$, $a \equiv b \pmod{n}$ means $n \mid b - a$. Example 5.2.7: $a \equiv b \pmod{2}$ iff a, b have the same parity. Example 5.2.8: mod 10 on $\mathbb{N}$ means “same last decimal digit”. **Theorem 5.2.11:** congruence mod n is an equivalence relation: $a - a = 0 = 0 \cdot n$; if $b - a = kn$ then $a - b = (-k)n$; if $b - a = kn$ and $c - b = \ell n$ then $c - a = (k + \ell)n$.

Exercise 5.2.4

from the book

Given $f : X \rightarrow Y$, define $a \sim_f b \Leftrightarrow f(a) = f(b)$. Prove that $\sim_f$ is an equivalence relation on X .

Reflexive: for $a \in X$, $f(a) = f(a)$, so $a \sim_f a$. *Symmetric:* if $f(a) = f(b)$ then $f(b) = f(a)$. *Transitive:* if $f(a) = f(b)$ and $f(b) = f(c)$ then $f(a) = f(c)$. Each property of $\sim_f$ is inherited from the same property of $=$ on Y . $\square$

Exercise 5.2.9

from the book

Let $n \in \mathbb{Z}$, $n \neq 0$. Prove that $a \equiv b \pmod{n}$ iff a and b have the same remainder when divided by n .

By the division theorem (Theorem 0.42) write $a = q_1n + r_1$ and $b = q_2n + r_2$ with $0 \leq r_1, r_2 < |n|$.

($\Leftarrow$) If $r_1 = r_2$ then $b - a = (q_2 - q_1)n$, so $n \mid b - a$.

($\Rightarrow$) Suppose $b - a = kn$. Then $r_2 - r_1 = (b - q_2n) - (a - q_1n) = (k - q_2 + q_1)n$ is a multiple of n with $|r_2 - r_1| < |n|$. A non-zero multiple mn of n has $|mn| = |m||n| \geq |n|$, so $r_2 - r_1 = 0$: the remainders agree. $\square$

NOW YOU 5.2.5, 5.2.10 · solutions in Appendix S

5.2.5 $U \cong V$ (there is a bijection $U \rightarrow V$) is an equivalence relation on a set of sets. [Hint: 3.2.19, 3.2.45, 3.2.20.] **5.2.10** When is $a \equiv b \pmod{0}$? When is $a \equiv b \pmod{1}$?

Equivalence classes and quotients**Definitions 5.2.12, 5.2.18, Theorem 5.2.17**

For an equivalence relation $\sim$ on X and $a \in X$: the **equivalence class** $[a]_\sim = \{x \in X \mid a \sim x\}$ and the **quotient** $X/\sim = \{[a]_\sim \mid a \in X\}$ (a set of subsets of X). **Theorem 5.2.17:** $a \sim b \Leftrightarrow [a]_\sim = [b]_\sim$ (proof: if $a \sim b$, double containment using symmetry and transitivity; conversely $b \in [b]_\sim = [a]_\sim$ gives $a \sim b$). So $\sim$ on X becomes $=$ on $X/\sim$. Example 5.2.13: for $\equiv \pmod{2}$, $[a]$ is the set of integers of the same parity as a , and $\mathbb{Z}/\sim = \{[0], [1]\} = \{E, O\}$. Example 5.2.15: $[a]_{\sim_f} = f^{-1}[\{f(a)\}]$. **Congruence classes:** $[a]_n = \{x \in \mathbb{Z} \mid a \equiv x \pmod{n}\}$ and $\mathbb{Z}/n\mathbb{Z} = \{[a]_n \mid a \in \mathbb{Z}\}$; e.g. $\mathbb{Z}/2\mathbb{Z} = \{[0]_2, [1]_2\}$.

NOW YOU 5.2.14, 5.2.16, 5.2.20 · solutions in Appendix S

5.2.14 Describe the classes of congruence mod 10 on $\mathbb{N}$ and write $\mathbb{N}/\approx$ in list notation. **5.2.16** f is injective iff every $\sim_f$ -class has exactly one element. **5.2.20** $i : [|n|] \rightarrow \mathbb{Z}/n\mathbb{Z}$, $i(r) = [r]_n$, is a bijection (for $n \neq 0$). [Hint: 5.2.9 and Theorem 5.2.17.]

Partitions**Definition 5.2.21 and Theorem 5.2.26 · partitions**

A **partition** of X is a set $\mathcal{A} \subseteq \mathcal{P}(X)$ such that (a) every $A \in \mathcal{A}$ is non-empty; (b) $\mathcal{A}$ **covers** X : $\forall x \in X, \exists A \in \mathcal{A}, x \in A$ (equivalently $\bigcup \mathcal{A} = X$); (c) $\mathcal{A}$ is **pairwise disjoint**: for all $A, B \in \mathcal{A}$, if $A \cap B$ is non-empty then $A = B$ (the contrapositive of “distinct sets do not overlap”, chosen because it has no negations). Model proof: Example 5.2.23, $\{\{2n, 2n+1\} \mid n \in \mathbb{N}\}$ partitions $\mathbb{N}$ (non-empty: $2n \in A$; cover: every x is $2n$ or $2n+1$; pairwise disjoint: a common element forces $m = n$). **Theorem 5.2.26:** (a) $X/\sim$ is a partition of X ; (b) every partition $\mathcal{A}$ is $X/\sim$ for a unique $\sim$, namely $x \sim y \Leftrightarrow \exists A \in \mathcal{A}, (x \in A \wedge y \in A)$. So equivalence relations on X and partitions of X are “the same thing”.

Exercise 5.2.24**essential**

Let $f : X \rightarrow Y$ be a surjection and $\mathcal{F} = \{f^{-1}[\{b\}] \mid b \in Y\}$. Prove that $\mathcal{F}$ is a partition of X . Where is surjectivity used?

- (a) *Non-empty*: let $F \in \mathcal{F}$, say $F = f^{-1}[\{b\}]$ with $b \in Y$. Since f is **surjective** there is $x \in X$ with $f(x) = b$, so $x \in F$. (This is the only place surjectivity is used: without it, $f^{-1}[\{b\}] = \emptyset$ for $b \notin f[X]$.)
 (b) *Cover*: let $x \in X$. Then $f(x) \in Y$ and $x \in f^{-1}[\{f(x)\}] \in \mathcal{F}$.
 (c) *Pairwise disjoint*: let $F = f^{-1}[\{b\}]$, $F' = f^{-1}[\{b'\}]$ with $F \cap F'$ non-empty, say $x \in F \cap F'$. Then $f(x) = b$ and $f(x) = b'$, so $b = b'$ and $F = F'$. $\square$

Exercise 5.2.25**recommended**

Let $\mathcal{A}$ be a set of non-empty subsets of X . Prove that $\mathcal{A}$ is a partition of X iff for each $x \in X$ there is a unique $A \in \mathcal{A}$ with $x \in A$.

Condition (a) holds by assumption, so $\mathcal{A}$ is a partition iff (b) and (c) hold.

($\Rightarrow$) Let $x \in X$. Existence of $A \in \mathcal{A}$ with $x \in A$ is (b). Uniqueness: if $x \in A$ and $x \in B$ with $A, B \in \mathcal{A}$, then $A \cap B$ is non-empty, so $A = B$ by (c).

($\Leftarrow$) (b): for $x \in X$ there exists $A \in \mathcal{A}$ with $x \in A$. (c): let $A, B \in \mathcal{A}$ with $A \cap B$ non-empty; pick $x \in A \cap B$. Then A and B are both sets of $\mathcal{A}$ containing x , so $A = B$ by uniqueness. $\square$

Exercise 5.2.27**essential**

Prove Theorem 5.2.26(a): for every equivalence relation $\sim$ on X , the quotient $X/\sim$ is a partition of X .

$X/\sim = \{[a]_\sim \mid a \in X\}$ is a set of subsets of X .

(a) *Non-empty*: for $a \in X$, $a \sim a$ by reflexivity, so $a \in [a]_\sim$.

(b) *Cover*: for $x \in X$, $x \in [x]_\sim \in X/\sim$.

(c) *Pairwise disjoint*: let $[a]_\sim \cap [b]_\sim$ be non-empty, say x lies in both. Then $a \sim x$ and $b \sim x$; by symmetry $x \sim b$, and by transitivity $a \sim b$. Hence $[a]_\sim = [b]_\sim$ by Theorem 5.2.17. $\square$

Exercise 5.2.28**recommended**

For $i \in [0, 1)$ let $A_i = \{i + n \mid n \in \mathbb{Z}\}$ and $\mathcal{A} = \{A_i \mid i \in [0, 1)\}$. Prove that $\mathcal{A}$ is a partition of $\mathbb{R}$ and describe the equivalence relation $\sim$ with $\mathbb{R}/\sim = \mathcal{A}$.

Non-empty: $i = i + 0 \in A_i$. *Cover*: let $x \in \mathbb{R}$ and put $i = x - \lfloor x \rfloor$; then $0 \leq i < 1$ and $x = i + \lfloor x \rfloor \in A_i$.

Pairwise disjoint: suppose $x \in A_i \cap A_j$, say $x = i + n = j + m$ with $n, m \in \mathbb{Z}$. Then $i - j = m - n \in \mathbb{Z}$, and $|i - j| < 1$ since $i, j \in [0, 1)$; the only integer of absolute value less than 1 is 0, so $i = j$ and $A_i = A_j$.

By Theorem 5.2.26(b), $x \sim y$ iff x and y lie in a common A_i . We claim this holds iff $y - x \in \mathbb{Z}$. If $x = i + n$ and $y = i + m$ then $y - x = m - n \in \mathbb{Z}$. Conversely, if $y - x = k \in \mathbb{Z}$, let $i = x - \lfloor x \rfloor$; then $x \in A_i$ and $y = i + (\lfloor x \rfloor + k) \in A_i$. So $\sim$ is “ $y - x \in \mathbb{Z}$ ” (the equivalence classes are the sets A_i , and $[x]_\sim = A_{x - \lfloor x \rfloor}$). $\square$

Exercise 5.2.30**recommended**

Let $\sim, \approx$ be equivalence relations on X, Y , with a bijection $p : X/\sim \rightarrow Y/\approx$ and, for each class $E \in X/\sim$, a bijection $h_E : E \rightarrow p(E)$. Prove there is a bijection $X \rightarrow Y$.

By Theorem 5.2.26(a) (Exercise 5.2.27), $\mathcal{A} = X/\sim$ is a partition of X and $\mathcal{B} = Y/\approx$ is a partition of Y . The bijection $F = p : \mathcal{A} \rightarrow \mathcal{B}$ and the bijections $g_E = h_E : E \rightarrow F(E)$ for $E \in \mathcal{A}$ are exactly the data of

Lemma 5.2.29, which yields a bijection $h : X \rightarrow Y$ (namely $h(x) = h_{[x]_{\sim}}(x)$). $\square$

The quotient function

Definition 5.2.31 and Theorem 5.2.35

The **quotient function** $q_{\sim} : X \rightarrow X/\sim$ is $q_{\sim}(a) = [a]_{\sim}$. Example 5.2.32: $q_2 : \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ records the parity of an integer. **Theorem 5.2.35:** for every surjection $p : X \rightarrow Z$ there are a unique equivalence relation $\sim$ on X and a unique bijection $f : X/\sim \rightarrow Z$ with $f([x]_{\sim}) = p(x)$: take $x \sim y \Leftrightarrow p(x) = p(y)$ (so $[x]_{\sim} = p^{-1}[\{p(x)\}]$) and $f([x]) = p(x)$, well-defined by Theorem 5.2.17, injective since $p(x) = p(y) \Rightarrow x \sim y$, surjective since p is. So every surjection “is” a quotient function, and equivalence relations on X , partitions of X and surjections out of X are three views of one thing.

Exercise 5.2.33

essential

Let $n \in \mathbb{Z}, n \neq 0$. Describe $q_n : \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ in terms of remainders.

For $a \in \mathbb{Z}$ let r be the remainder of a on division by n ($a = qn + r, 0 \leq r < |n|$). Then $a \equiv r \pmod{n}$ (as $a - r = qn$), so $q_n(a) = [a]_n = [r]_n$ by Theorem 5.2.17. Thus q_n sends each integer to the class of its remainder: $q_n(a) = q_n(b)$ iff a and b have the same remainder (Exercise 5.2.9), and $\mathbb{Z}/n\mathbb{Z} = \{[0]_n, [1]_n, \dots, [|n| - 1]_n\}$ (Exercise 5.2.20). $\square$

Exercise 5.2.34

recommended

Prove that $q_{\sim} : X \rightarrow X/\sim$ is surjective.

Let $E \in X/\sim$. By Definition 5.2.12, $E = [a]_{\sim}$ for some $a \in X$, and then $q_{\sim}(a) = [a]_{\sim} = E$. $\square$

Exercise 5.2.36

recommended

Describe the correspondence between partitions of X and surjections with domain X .

Partition $\rightarrow$ surjection: given a partition $\mathcal{A}$ of X , define $q : X \rightarrow \mathcal{A}$ by $q(x) =$ the unique $A \in \mathcal{A}$ with $x \in A$ (well-defined by Exercise 5.2.25); q is surjective because each $A \in \mathcal{A}$ is non-empty, so $A = q(x)$ for any $x \in A$. (If $\mathcal{A} = X/\sim$ this is $q_{\sim}$.)

Surjection $\rightarrow$ partition: given a surjection $p : X \rightarrow A$, take $\mathcal{U}_p = \{p^{-1}[\{a\}] \mid a \in A\}$, a partition of X by Exercise 5.2.24, and the bijection $f : \mathcal{U}_p \rightarrow A$, $f(p^{-1}[\{a\}]) = a$ (well-defined and injective since $p^{-1}[\{a\}] = p^{-1}[\{a'\}]$ forces $a = a'$ by surjectivity, surjective by construction) satisfies $p(x) = f(U)$ for all $x \in U \in \mathcal{U}_p$. As in Theorem 5.2.35, $\mathcal{U}_p$ and f are unique with this property, and the two constructions are inverse to each other up to the bijection $f : \mathcal{U}_q = \mathcal{A}$ for the q above, and q built from $\mathcal{U}_p$ is $f^{-1} \circ p$. $\square$

Watch out

- “Antisymmetric” is not “not symmetric” (5.1.30). Check each property separately and give a counterexample for each failure.
- $a \equiv b \pmod{n}$ means $n \mid b - a$ (Definition 5.2.6), not “ a divided by b ”; equal remainders is a *theorem* (5.2.9), valid for $n \neq 0$.
- $[a]_{\sim}$ is a *set*; $X/\sim$ is a set of sets; $[a] = [b]$ iff $a \sim b$ (Theorem 5.2.17). To show a map $F([a]) = \dots$ is well-defined, show that $a \sim b$ gives the same value.
- Pairwise disjointness is proved by assuming $A \cap B \neq \emptyset$ and deriving $A = B$, not by contradiction (remark

after Definition 5.2.21).

C CHAPTER 5 EXERCISES

Definitions 5.E.4 and 5.E.6 · transport and generated equivalence relation

The **transport** of a relation R on X along $f : X \rightarrow Y$ is the relation S on Y with $\text{Gr}(S) = \{(f(a), f(b)) \mid a, b \in X, a R b\}$. The **equivalence relation generated by R** is $\sim_R : x \sim_R y$ iff there is a chain $x = a_0, a_1, \dots, a_k = y$ ($k \in \mathbb{N}$) with $a_i R a_{i+1}$ or $a_{i+1} R a_i$ for each $i \in [k]$. Exercises 5.E.10–5.E.13 show $\sim_R$ is the smallest equivalence relation containing R .

NOW YOU 5.E.1–5.E.3, 5.E.5, 5.E.7–5.E.14 · solutions in Appendix S

5.E.1 For each $P \subseteq \{\text{reflexive, symmetric, transitive}\}$, a relation with exactly the properties in P . **5.E.2** Symmetric and antisymmetric $\Rightarrow \text{Gr}(R) \subseteq \text{Gr}(=)$. **5.E.3** Left-total ($\forall x \exists y, x R y$), symmetric, transitive $\Rightarrow$ reflexive. **5.E.5** Transport of an equivalence relation: which properties survive, and when is it an equivalence relation? **5.E.7** $x R y \Leftrightarrow y = x + n$: then $\sim_R$ is congruence mod n . **5.E.8** $R \subseteq \subseteq$ on $\mathcal{P}(X)$: $U \sim_R V$ for all U, V . **5.E.9** $R = \{(a, b)\}$: $x \sim_R y$ iff $x = y$ or $\{x, y\} = \{a, b\}$. **5.E.10–5.E.13** $\sim_R$ is an equivalence relation; $R \subseteq \sim_R$; $\sim_R \subseteq \approx$ for every equivalence relation $\approx \supseteq R$; $\sim_R = R$ if R is an equivalence relation. **5.E.14** $F \sim G \Leftrightarrow F' = G'$ on $C^1(\mathbb{R})$.

True–false (5.E.15–5.E.21)

Exercise 5.E.18

essential

Every symmetric, antisymmetric relation is reflexive.

False. The empty relation $\emptyset_{\mathbb{N}, \mathbb{N}}$ on $\mathbb{N}$ is symmetric and antisymmetric: both definitions begin “if $a R b \dots$ ”, and $a R b$ never holds, so they are vacuously true. It is not reflexive, since $0 R 0$ is false. (By 5.1.35, symmetric and antisymmetric means $\text{Gr}(R) \subseteq \Delta_X$; reflexive would need $\text{Gr}(R) \supseteq \Delta_X$.) $\square$

NOW YOU 5.E.15–5.E.17, 5.E.19–5.E.21 · solutions in Appendix S

5.E.15 Every reflexive relation is symmetric. **5.E.16** There is a unique reflexive, symmetric, antisymmetric relation on $\mathbb{N}$. **5.E.17** Every relation that is not symmetric is antisymmetric. **5.E.19** $\emptyset$ is the graph of a relation on $\mathbb{N}$. **5.E.20** The graph of a function $X \rightarrow X$ is the graph of a relation on X . **5.E.21** The graph of a relation on X is the graph of a function $X \rightarrow X$.

Always–sometimes–never (5.E.22–5.E.23)

Exercise 5.E.22

essential

Let $\sim$ be an equivalence relation on X . Then the relation $\approx$ on $\mathcal{P}(X) \setminus \{\emptyset\}$ defined by $A \approx B \Leftrightarrow (x \sim y \text{ for all } x \in A \text{ and } y \in B)$ is an equivalence relation.

Sometimes. Symmetric, always: if $A \approx B$, let $y \in B, x \in A$; then $x \sim y$, so $y \sim x$ by symmetry of $\sim$; hence $B \approx A$. *Transitive, always:* let $A \approx B$ and $B \approx C$, and let $x \in A, z \in C$. Since $B \neq \emptyset$ pick $y \in B$; then $x \sim y$ and $y \sim z$, so $x \sim z$. Hence $A \approx C$. *Reflexive, not always:* $A \approx A$ says all elements of A are pairwise equivalent. Take $X = \{0, 1\}$ and $\sim$ the equality relation: $A = \{0, 1\}$ has $0 \not\sim 1$, so $A \not\approx A$ and $\approx$ is not an equivalence relation. But if $\sim$ is the discrete relation on X ($x \sim y$ for all x, y), then $A \approx B$ for all non-empty

A, B , and $\approx$ is an equivalence relation (reflexivity holds trivially). So the conclusion holds for some $\sim$ and fails for others. $\square$

Exercise 5.E.23**essential**

Let $\sim$ be an equivalence relation on X . Then the relation $\approx$ on $\mathcal{P}(X) \setminus \{\emptyset\}$ defined by $A \approx B \Leftrightarrow (x \sim y \text{ for some } x \in A \text{ and } y \in B)$ is an equivalence relation.

Sometimes. Reflexive, always: $A \neq \emptyset$, pick $x \in A$; $x \sim x$, so $A \approx A$. *Symmetric, always:* if $x \in A$, $y \in B$ with $x \sim y$ then $y \sim x$ with $y \in B$, $x \in A$, so $B \approx A$. *Transitive, not always:* take $X = \{0, 1\}$, $\sim$ the equality relation, $A = \{0\}$, $B = \{0, 1\}$, $C = \{1\}$. Then $A \approx B$ (via $0 \sim 0$) and $B \approx C$ (via $1 \sim 1$), but $A \approx C$ would need $0 \sim 1$, which is false. So $\approx$ is not an equivalence relation here. If instead $\sim$ is the discrete relation on X , then $A \approx B$ for all non-empty A, B and $\approx$ is an equivalence relation. Hence sometimes. $\square$

Appendix S

Solutions to the NOW YOU exercises

Contents

- A** Chapter 0
- B** Chapter 1
- C** Chapter 2
- D** Chapter 3
- E** Chapter 4
- F** Chapter 5

Planning

Every book exercise in Chapters 0–5 that is not worked in the main text is solved here, in the order of the book. Compare line by line: did you introduce every variable, unfold every definition, and name the strategy?

A CHAPTER 0

0.2 (a) $2 + 2 = 4$. (b) $2 + 2 = 5$. (c) “There are infinitely many pairs of primes that differ by 2” (the twin prime conjecture; nobody knows its truth value). Any proposition whose truth value is unknown is fine, e.g. $\pi^\pi \in \mathbb{Q}$ (Exercise 0.21(f)).

0.47 Repeated division by 17: $408735787 = 24043281 \cdot 17 + 10$, $24043281 = 1414310 \cdot 17 + 11$, $1414310 = 83194 \cdot 17 + 12$, $83194 = 4893 \cdot 17 + 13$, $4893 = 287 \cdot 17 + 14$, $287 = 16 \cdot 17 + 15$, $16 = 0 \cdot 17 + 16$. Reading the remainders upwards, with $A = 10, \dots, G = 16$: $408735787 = GFEDCBA_{(17)}$. Likewise $1442151747 = 40059770 \cdot 36 + 27$, $40059770 = 1112771 \cdot 36 + 14$, $1112771 = 30910 \cdot 36 + 11$, $30910 = 858 \cdot 36 + 22$, $858 = 23 \cdot 36 + 30$, $23 = 0 \cdot 36 + 23$, and $27 = R$, $14 = E$, $11 = B$, $22 = M$, $30 = U$, $23 = N$: $1442151747 = NUMBER_{(36)}$.

0.55 $s(w) = \sqrt{2}w^4 - 81\sqrt{2}$ has coefficients $\sqrt{2}, -81\sqrt{2} \notin \mathbb{Q}$ (so it is not a polynomial over $\mathbb{Q}$), is quartic, and $s(3) = \sqrt{2}(81 - 81) = 0 \in \mathbb{Q}$.

0.64 Since $i^2 = -1$, $i^3 = -i$ and $i^4 = 1$. For $n \in \mathbb{Z}$ write $n = 4q + r$ with $r \in \{0, 1, 2, 3\}$ (Theorem 0.42); then $i^n = (i^4)^q i^r = i^r$. So $i^n = 1$ if $r = 0$; $i^n = 0 + 1 \cdot i$ if $r = 1$; $i^n = -1$ if $r = 2$; $i^n = 0 + (-1)i$ if $r = 3$. (For negative n this uses $i^{-1} = -i$, which is consistent with the formula.)

0.65 Let $p(x) = ax^2 + bx + c$ with $a \neq 0$ and $\Delta < 0$. By Definition 0.62, $\sqrt{\Delta} = \sqrt{|\Delta|}i$, so $(\sqrt{\Delta})^2 = |\Delta|i^2 = -|\Delta| = \Delta$. The computation in the proof of Theorem 0.60 used only $(\sqrt{\Delta})^2 = \Delta$ and the completed square $p(x) = a(x + \frac{b}{2a})^2 - \frac{\Delta}{4a}$, so $p(\alpha) = p(\beta) = 0$ holds verbatim. For “no other roots”: if $p(\gamma) = 0$ then $(\gamma + \frac{b}{2a})^2 = \frac{\Delta}{4a^2} = (\frac{\sqrt{\Delta}}{2a})^2$, so $(\gamma + \frac{b}{2a} - \frac{\sqrt{\Delta}}{2a})(\gamma + \frac{b}{2a} + \frac{\sqrt{\Delta}}{2a}) = 0$, and a product of complex numbers is 0 only if a factor is 0; hence $\gamma = \alpha$ or $\gamma = \beta$. The cases $\Delta \geq 0$ are Theorem 0.60.

0.66 Let $\alpha = a + bi$ and put $p(x) = x^2 - 2ax + (a^2 + b^2)$, a quadratic over $\mathbb{R}$. Then $p(\alpha) = (a + bi)^2 - 2a(a + bi) + a^2 + b^2 = a^2 + 2abi - b^2 - 2a^2 - 2abi + a^2 + b^2 = 0$. The other root is $a - bi$: $p(a - bi) = a^2 - 2abi - b^2 - 2a^2 + 2abi + a^2 + b^2 = 0$. (Indeed $p(x) = (x - (a + bi))(x - (a - bi))$.)

0.67 Quadratic formula over $\mathbb{C}$. Let $p(x) = ax^2 + bx + c$ with $a, b, c \in \mathbb{C}$, $a \neq 0$, and let $\delta \in \mathbb{C}$ be any number with $\delta^2 = b^2 - 4ac$ (every complex number has a square root; we take this for granted). Then the roots of $p(x)$ are precisely $\frac{-b+\delta}{2a}$ and $\frac{-b-\delta}{2a}$. *Proof:* completing the square, $p(x) = a(x + \frac{b}{2a})^2 - \frac{\delta^2}{4a}$; substituting the two values gives $a \cdot \frac{\delta^2}{4a^2} - \frac{\delta^2}{4a} = 0$; and if $p(\gamma) = 0$ then $(\gamma + \frac{b}{2a})^2 = (\frac{\delta}{2a})^2$, so $\gamma + \frac{b}{2a} = \pm \frac{\delta}{2a}$ by factorising the difference of squares.

0.68 *Subtraction* $a - b$: translate a by b units to the *left* (it is the addition of $-b$): put the 0 of the lower line under a and read $a - b$ above the point $-b$ of the lower line. *Division* a/b ($b \neq 0$): it is the inverse of multiplication. Align the zeros, then stretch (and reflect if $b < 0$) the lower line until its point b lies beneath the point a of the upper line; then a/b is the point of the upper line above the point 1 of the lower line (because the scaling sends $1 \mapsto a/b$ and $b \mapsto a$).

0.E.1 With $A, \dots, Z \mapsto 0, \dots, 25$; $a, \dots, z \mapsto 26, \dots, 51$; $0, \dots, 9 \mapsto 52, \dots, 61$; $- \mapsto 62$; $_ \mapsto 63$, the digits of dQw4w9WgXcQ are

$$29, 16, 48, 56, 48, 61, 22, 32, 23, 28, 16, \quad \text{and} \quad \sum_k d_k 64^k = 33\,736\,714\,463\,647\,725\,328.$$

Repeated division of 7 159 047 702 620 056 984 by 64 gives the remainders 24, 28, 2, 29, 45, 18, 3, 44, 26, 13, 6; reading them last-first, 6, 13, 26, 44, 3, 18, 45, 29, 2, 28, 24 are the characters G, N, a, C, _, S, M, v, q, 2, Y, so the base-64 expansion is the string GNaC_SMvq2Y.

0.E.2 $(a + b\sqrt{2})(c + d\sqrt{2}) = (ac + 2bd) + (ad + bc)\sqrt{2}$. If $ad + bc = 0$ the product is $ac + 2bd \in \mathbb{Z}$. If $ad + bc \neq 0$ and the product were an integer m , then $\sqrt{2} = \frac{m - (ac + 2bd)}{ad + bc} \in \mathbb{Q}$, contradicting Assumption 0.49. So the product is an integer exactly when $ad + bc = 0$.

0.E.6–0.E.11 $a \odot b = a + b$: $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ all closed (Assumptions 0.7, 0.9, Theorem 0.16, Assumption 0.5). $a - b$: $\mathbb{N}$ not closed ($1 - 2 = -1 \notin \mathbb{N}$); $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ closed. $(a - b)(a + b)$: $\mathbb{N}$ not closed ($0 \odot 1 = -1$); $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$ closed (products and sums). $(a - 1)(b - 1) + 2(a + b) = ab + a + b + 1$: all four closed (sums and products only). $\frac{a}{b^2 + 1}$: $\mathbb{N}$ and $\mathbb{Z}$ not closed ($1 \odot 1 = \frac{1}{2}$); $\mathbb{Q}$ and $\mathbb{R}$ closed, as $b^2 + 1 \neq 0$. $\frac{a}{\sqrt{b^2 + 1}}$: $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ not closed, since $1 \odot 1 = \frac{1}{\sqrt{2}}$ is irrational (if $\frac{1}{\sqrt{2}} = r \in \mathbb{Q}$ then $r \neq 0$ and $\sqrt{2} = \frac{1}{r} \in \mathbb{Q}$); $\mathbb{R}$ closed, as $\sqrt{b^2 + 1}$ is a non-zero real.

0.E.13 Let $x \in \mathbb{Q}$. The polynomial $p(t) = t - x$ is non-zero, has coefficients $1, -x \in \mathbb{Q}$, and $p(x) = 0$. So x is algebraic.

0.E.14 $p(t) = t^2 - 2$ is non-zero over $\mathbb{Q}$ and $p(\sqrt{2}) = 2 - 2 = 0$.

0.E.15 Let $x = \sqrt{2} + \sqrt{3}$. Then $x^2 = 5 + 2\sqrt{6}$, so $(x^2 - 5)^2 = 24$, i.e. $x^4 - 10x^2 + 1 = 0$. So x is a root of the non-zero polynomial $t^4 - 10t^2 + 1$ over $\mathbb{Q}$.

0.E.16 By Exercise 0.66, $x + yi$ is a root of $t^2 - 2xt + (x^2 + y^2)$, whose coefficients $1, -2x, x^2 + y^2$ are rational when $x, y \in \mathbb{Q}$ (Theorem 0.16), and which is non-zero. So $x + yi$ is algebraic.

0.E.19 True: $0 = 0 \cdot n$ for every $n \in \mathbb{Z}$ (Exercise 0.38).

0.E.20 True: $n^2 = n \cdot n$ with $n \in \mathbb{Z}$, so n divides n^2 .

0.E.21 True: if $x \in \mathbb{Q}$ then $x^2 = x \cdot x \in \mathbb{Q}$ by Theorem 0.16.

0.E.23 False: $a = 0, b = 5$: $0 = 0 \cdot 5 + 0$, remainder 0, which is not less than $a = 0$. (Also any negative a : $-3 = (-1) \cdot 5 + 2$ and $2 > -3$.)

0.E.24 False: x^2 has discriminant 0 and its only root is 0 (if $\gamma^2 = 0$ then $\gamma = 0$), so it does not have two distinct complex roots.

0.E.25 Always. Since $n < b_1 < b_2$, condition (ii) allows n itself as a single digit in both bases, and $n = n \cdot b^0$ with $n \neq 0$, so the base- b_1 expansion and the base- b_2 expansion of n are both the one-digit string n .

0.E.26 Never. Suppose the two expansions are the same string $d_r \dots d_0$. Since $n > b_1$, the base- b_1 expansion has at least two digits, so $r \geq 1$ and $d_r \neq 0$. Then $n = \sum_{i \leq r} d_i b_1^i$ and also $n = \sum_{i \leq r} d_i b_2^i$. But $b_1 < b_2$ gives $d_r b_1^r < d_r b_2^r$ and $d_i b_1^i \leq d_i b_2^i$ for the other terms, so $\sum d_i b_1^i < \sum d_i b_2^i$, i.e. $n < n$: contradiction.

0.E.31 Sometimes: $x = 1$ has $x^2 = 1 \in \mathbb{Q}$ and $x \in \mathbb{Q}$; $x = \sqrt{2}$ has $x^2 = 2 \in \mathbb{Q}$ but $x \notin \mathbb{Q}$.

0.E.32 Always. From $x^2 + 1 \in \mathbb{Q}$ we get $x^2 \in \mathbb{Q}$, hence $x^4 = (x^2)^2 \in \mathbb{Q}$ (Theorem 0.16); from $x^5 + 1 \in \mathbb{Q}$ we get $x^5 \in \mathbb{Q}$. If $x = 0$ then $x \in \mathbb{Q}$. If $x \neq 0$ then $x^4 \neq 0$ and $x = \frac{x^5}{x^4} \in \mathbb{Q}$ by Theorem 0.16.

0.E.33 Always. Since $u + vi$ with $v \neq 0$ is not real, $p(x)$ has a non-real root, so $\Delta < 0$ (by Exercise 0.61 and Theorem 0.60, $\Delta \geq 0$ gives only real roots). By Exercise 0.65 the roots are $\frac{-b \pm \sqrt{|\Delta|}i}{2a}$, which are of the form $u' \pm v'i$ with $u' = \frac{-b}{2a}$, $v' = \frac{\sqrt{|\Delta|}}{2a}$. One of them is $u + vi$, so $u = u'$ and $v = \pm v'$, and the other root is $u - vi$.

B CHAPTER 1

1.1.6 p : “John is a mathematician”; q : “John lives in Pittsburgh”; the statement is $p \wedge q$.

1.1.13 Assume $(p \wedge q) \wedge r$. By Strategy 1.1.10 we may assume $p \wedge q$ (and r). By Strategy 1.1.10 again, from $p \wedge q$ we may assume q . Hence q is true.

1.1.14 Proof tree: from p and q conclude $p \wedge q$ ($\wedge$ I); from r and s conclude $r \wedge s$ ($\wedge$ I); from $p \wedge q$ and $r \wedge s$ conclude $(p \wedge q) \wedge (r \wedge s)$ ($\wedge$ I). With p : “2 is even”, q : “2 is prime”, r : “3 is odd”, s : “3 is prime”, a proof has four parts (prove 2 even, prove 2 prime, prove 3 odd, prove 3 prime), combined two at a time.

1.1.23 Since $n = 8$ or $n = 9$, we split into cases. Case $n = 8$: $8 = 2 \cdot 4$, so n is even. Case $n = 9$: $9 = 3^2$, so n is a perfect square. In both cases, n is even or a perfect square.

1.1.26 The discriminant is $\Delta = a^2 - 4b \neq 0$. If $\Delta > 0$, Theorem 0.60 gives $\alpha = \frac{-a + \sqrt{\Delta}}{2}$, $\beta = \frac{-a - \sqrt{\Delta}}{2}$ (in some order), so $\alpha - \beta = \pm \sqrt{\Delta}$; take $c = \pm \sqrt{\Delta} \in \mathbb{R}$. If $\Delta < 0$, Exercise 0.65 gives the roots $\frac{-a \pm \sqrt{|\Delta|}i}{2}$, so $\alpha - \beta = \pm \sqrt{|\Delta|}i = ci$ with $c = \pm \sqrt{|\Delta|} \in \mathbb{R}$. Either way such a c exists.

1.1.31 Suppose α is a root of $p(x)$ and $a \in \mathbb{C}$. Let $q(x) = (x - a)p(x)$. Then $q(\alpha) = (\alpha - a)p(\alpha) = (\alpha - a) \cdot 0 = 0$, so α is a root of $q(x)$.

1.1.64 In the proof of Proposition 1.1.63, the law of excluded middle is used to split into the cases “ a is even” and “ a is odd” (odd means not even). Proof by contradiction is used inside the second case: assuming b odd we derived that ab is odd, which contradicts the hypothesis that ab is even; the contradiction is “ ab is even and ab is not even”. *Version without LEM*: suppose ab is even, and suppose towards a contradiction that it is not the case that a or b is even; then (de Morgan) a is odd and b is odd, so $a = 2k + 1$, $b = 2\ell + 1$ and $ab = 2(2k\ell + k + \ell) + 1$ is odd, contradicting ab even. *Version with LEM and no contradiction*: by LEM, a is even or odd, and b is even or odd. If a is even or b is even we are done. In the remaining case both are odd, and then $ab = 2(2k\ell + k + \ell) + 1$ leaves remainder 1 on division by 2, so by the uniqueness clause of the division theorem ab is not of the form $2m$; this case is therefore excluded by the hypothesis that ab is even. (Any proof about “odd” numbers must use negation somewhere, because odd is defined as not even.)

1.2.8 (a) “ b divides a .” (b) “ x is rational.” (c) “The only natural numbers that divide n are 1 and n ” (true for $n = 1$ and for primes). (d) “There is no largest positive real number” (every positive real is exceeded by a positive real).

1.2.14 Let $x \in \mathbb{R}$. Every object is equal to itself (the basic assumption mentioned in Definition 0.3), so $x = x$. Since x was arbitrary, $\forall x \in \mathbb{R}, x = x$.

1.2.16 Let $n \in \mathbb{Z}$. Then $n = \frac{n}{1}$ with $n, 1 \in \mathbb{Z}$ and $1 \neq 0$, so $n \in \mathbb{Q}$ by Definition 0.15.

1.2.17 Let $p(x) = ax + b$ be a linear polynomial over $\mathbb{Q}$, so $a, b \in \mathbb{Q}$ and $a \neq 0$. Define $\gamma = -\frac{b}{a}$; then $\gamma \in \mathbb{Q}$ by Theorem 0.16, and $p(\gamma) = -b + b = 0$. So $p(x)$ has a rational root.

1.2.27 Define $x = \sqrt{2}$. Then x is irrational (Assumption 0.49) and $x^2 = 2 = \frac{2}{1}$ is rational.

1.2.34 The existence part gives at least one x with $p(x)$; the uniqueness part says any two elements satisfying

p are equal, so there cannot be two different ones; together: exactly one. Alternative definitions: Proposition 1.2.38's form $\exists x \in X, (p(x) \wedge \forall y \in X, (p(y) \Rightarrow y = x))$; or " $\exists x, p(x)$ and $\neg \exists a, \exists b, (a \neq b \wedge p(a) \wedge p(b))$ ".

1.2.37 (c) $\exists! n \in \mathbb{N}, (n \text{ has exactly one positive divisor})$. *Existence*: $n = 1$: its positive divisors are the positive d with $1 = qd$, and then $d = 1$ (if $d \geq 2$ then $qd \neq 1$ for $q \in \mathbb{Z}$); so exactly one. *Uniqueness*: let $n \in \mathbb{N}$ have exactly one positive divisor. By Exercise 0.38, 1 divides n , so that divisor is 1. If $n \geq 2$ then n also divides n and $n \neq 1$: two positive divisors, contradiction. If $n = 0$ then every positive integer divides 0: more than one, contradiction. Hence $n = 1$. So any two such natural numbers are both 1.

1.3.9 $\neg p$: by $(\neg I)/(\neg E)$, $\neg p$ is true exactly when p is false. $p \wedge q$: by $(\wedge I)$ it is true when both are, and by $(\wedge E_1), (\wedge E_2)$ it cannot be true when either is false. $p \vee q$: by $(\vee I)$ it is true when at least one is; if both are false, $p \vee q$ cannot be true, since by $(\vee E)$ we could then derive $\perp$ (both p and q give contradictions). $p \Rightarrow q$: by $(\Rightarrow E)$ it is false when p is true and q false; when q is true, q can be derived from anything, so $(\Rightarrow I)$ makes it true; when p is false, assuming p gives a contradiction and then q by explosion, so $(\Rightarrow I)$ makes it true.

1.3.25 (i) *Derivations*. Assume $\neg(p \vee q)$. To prove $\neg p$: assume p ; then $p \vee q$ ($\vee I_1$), contradicting $\neg(p \vee q)$; so $\neg p$. Likewise $\neg q$. Hence $(\neg p) \wedge (\neg q)$. Conversely assume $(\neg p) \wedge (\neg q)$. To prove $\neg(p \vee q)$, assume $p \vee q$ and split: p contradicts $\neg p$; q contradicts $\neg q$. So $\neg(p \vee q)$. (ii) *Truth table*: $\neg(p \vee q)$ is $\checkmark$ only in row $\times \times$, and so is $(\neg p) \wedge (\neg q)$. (iii) By part (a) applied to $\neg p, \neg q$: $\neg((\neg p) \wedge (\neg q)) \equiv (\neg \neg p) \vee (\neg \neg q) \equiv p \vee q$ (double negation). Negating both sides, $(\neg p) \wedge (\neg q) \equiv \neg \neg((\neg p) \wedge (\neg q)) \equiv \neg(p \vee q)$.

1.3.38 $\neg(p \Leftrightarrow q) \equiv \neg[(p \Rightarrow q) \wedge (q \Rightarrow p)] \equiv \neg(p \Rightarrow q) \vee \neg(q \Rightarrow p) \equiv (p \wedge \neg q) \vee (q \wedge \neg p)$, using Definition 1.1.40, de Morgan (a) and Exercise 1.3.27.

1.3.44 (b), (d) (b) Assume $p \Rightarrow (q \Rightarrow r)$; assume $p \Rightarrow q$; assume p . Modus ponens gives q (from $p \Rightarrow q$) and $q \Rightarrow r$ (from the first assumption), then r . So the formula is derivable from no assumptions, hence a tautology. *Meaning*: an implication can be distributed over an implication. (d) By Theorem 1.3.24(a), $\neg(p \wedge q) \equiv (\neg p) \vee (\neg q)$, so by Theorem 1.3.45(b) the biconditional is a tautology. *Meaning*: to show a conjunction fails, show one conjunct fails.

1.E.1 p : n even, q : n prime, r : $n = 2$. (a) "If n is an even prime, then $n = 2$ ": true for all n (an even prime is divisible by 2, so 2 is one of its only positive divisors 1 and n , and $2 \neq 1$). (b) "If n is a prime other than 2, then n is odd": true for all n (same reason). (c) " n is odd, or n is not prime, or $n \neq 2$ ": true for every n except $n = 2$, for which all three disjuncts fail; so true for some, false for some. (d) " n is an even prime different from 2": false for all n .

1.E.2 (a) p : guinea pigs are quiet; q : they are hungry; r : they are loud: $p \wedge (q \Rightarrow r)$. (b) p : 2 is even; q : 2 is prime: $p \wedge q$. (c) p : $\sqrt{2}$ is irrational; q : $\sqrt{2}$ is an integer: $p \wedge (p \Rightarrow \neg q)$ (hence $\neg q$).

1.E.3 "A is necessary for B" means $B \Rightarrow A$; "A is sufficient for B" means $A \Rightarrow B$. (a) $p \Rightarrow \neg q$: true. (b) $\neg q \Rightarrow p$: false. (c) $\neg q \Rightarrow p$: false. (d) $p \Rightarrow \neg q$: true.

1.E.4 To prove $(p \wedge q) \Rightarrow \neg r$: assume $p \wedge q$, so assume p and q ; then prove $\neg r$ by assuming r and deriving a contradiction.

1.E.5 Assume $p \vee q$; assume r ; prove s by cases: derive s from p (and r), then derive s from q (and r).

1.E.6 Prove both implications. $(\Rightarrow)$ Assume $p \Rightarrow q$; assume $\neg p$; prove $\neg q$ by assuming q and deriving a contradiction. $(\Leftarrow)$ Assume $\neg p \Rightarrow \neg q$; assume p ; prove q . *Warning*: this formula is not a tautology (with p true and q false, $p \Rightarrow q$ is false but $\neg p \Rightarrow \neg q$ is true), so the second step cannot succeed in general: $\neg p \Rightarrow \neg q$ is the *inverse*, equivalent to the converse $q \Rightarrow p$, not to $p \Rightarrow q$.

1.E.7 Prove one of the disjuncts: either prove p and prove $\neg q$ (assume q , derive a contradiction), or prove q and prove $\neg p$. (Not a tautology: false when p, q are both true.)

1.E.8 "Between any two distinct rational numbers there is an irrational number." *Proof*. Let $a, b \in \mathbb{Q}$ and suppose $a < b$. Define $c = a + \frac{b-a}{\sqrt{2}}$. First, $\frac{b-a}{\sqrt{2}} = \frac{b-a}{2} \cdot \sqrt{2}$ is irrational: if it were rational, then $\sqrt{2} = \frac{b-a}{\frac{b-a}{2}} \cdot \frac{2}{b-a}$

would be rational (Theorem 0.16, $b - a \neq 0$), contradicting Assumption 0.49. Hence c is irrational by Proposition 1.1.53. Second, since $\sqrt{2} > 1$ we have $0 < \frac{1}{\sqrt{2}} < 1$, so $0 < \frac{b-a}{\sqrt{2}} < b - a$ and therefore $a < c < b$. So c is the required number.

1.E.9 $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, y > x$. *Proof.* Let $x \in \mathbb{Z}$. Define $y = x + 1$. Then $y \in \mathbb{Z}$ (closure) and $y > x$.

1.E.10 $\exists a \in X, \exists b \in X, (a \neq b \wedge p(a) \wedge p(b) \wedge \forall x \in X, (p(x) \Rightarrow (x = a \vee x = b)))$. *Structure:* exhibit two distinct elements satisfying p , then show every element satisfying p is one of them. *Proof for $x^2 = 1$.* Take $a = 1, b = -1$: distinct, and $1^2 = (-1)^2 = 1$. Let $x \in \mathbb{R}$ with $x^2 = 1$. Then $(x - 1)(x + 1) = 0$, so $x - 1 = 0$ or $x + 1 = 0$, i.e. $x = 1$ or $x = -1$.

1.E.12 The truth tables of $p \Rightarrow q$ and $q \Rightarrow p$ differ in the rows $(\checkmark, \times)$ and $(\times, \checkmark)$: in the first, $p \Rightarrow q$ is $\times$ and $q \Rightarrow p$ is $\checkmark$. Example: p : “ $1 = 2$ ”, q : “ $1 = 1$ ”; $p \Rightarrow q$ is true (false hypothesis) and $q \Rightarrow p$ is false.

1.E.13 The column is $\checkmark$ exactly when p, q differ: $(p \wedge \neg q) \vee (q \wedge \neg p)$ (equivalently $\neg(p \Leftrightarrow q)$, Exercise 1.3.38).

1.E.14 The column equals the column of q : the formula q (or $(p \vee \neg p) \wedge q$).

1.E.15 The column is $\checkmark$ exactly in the rows where p and q have the same value, regardless of r : $p \Leftrightarrow q$.

1.E.16 The column is $\checkmark$ exactly when r is $\checkmark$ and $(p, q) \neq (\checkmark, \times)$: $r \wedge (p \Rightarrow q)$.

1.E.17 First note $p \uparrow q \equiv \neg(p \wedge q)$: if $\neg(p \wedge q)$ then by de Morgan $\neg p$ or $\neg q$, and rule (i) or (ii) gives $p \uparrow q$; conversely if $p \uparrow q$, assume $p \wedge q$, so p and q , and rule (iii) yields $\perp$; hence $\neg(p \wedge q)$. (a) $p \uparrow p \equiv \neg(p \wedge p) \equiv \neg p$; so $(p \uparrow p) \uparrow (p \uparrow p) \equiv \neg(p \uparrow p) \equiv \neg\neg p \equiv p$. (b) $(p \uparrow p) \uparrow (q \uparrow q) \equiv \neg(\neg p \wedge \neg q) \equiv p \vee q$ (de Morgan); $(p \uparrow q) \uparrow (p \uparrow q) \equiv \neg(p \uparrow q) \equiv \neg\neg(p \wedge q) \equiv p \wedge q$. (c) $p \Rightarrow q \equiv \neg p \vee q \equiv \neg(p \wedge \neg q) \equiv p \uparrow (\neg q) \equiv p \uparrow (q \uparrow q)$.

1.E.19 $\neg\exists!x \in X, p(x) \equiv \neg[(\exists x, p(x)) \wedge (\forall a, \forall b, (p(a) \wedge p(b)) \Rightarrow a = b)] \equiv (\forall x \in X, \neg p(x)) \vee (\exists a \in X, \exists b \in X, (p(a) \wedge p(b) \wedge a \neq b))$. *Strategy:* either show that no x satisfies p , or exhibit two distinct elements that do. *Application.* Let $a \in \mathbb{R}$ with $a \neq -1$ and $q(x) = x^4 - 2ax^2 + a^2 - 1 = (x^2 - a)^2 - 1 = (x^2 - a - 1)(x^2 - a + 1)$. If $a > -1$ then $a + 1 > 0$ and $x = \sqrt{a + 1}, x = -\sqrt{a + 1}$ are two distinct real roots (distinct because $a + 1 \neq 0$). If $a < -1$ then $a + 1 < 0$ and $a - 1 < 0$, so neither $x^2 = a + 1$ nor $x^2 = a - 1$ has a real solution: no real root at all. In both cases there is not a unique real root.

1.E.20 Define $\forall!x \in X, p(x)$ to mean $\neg\exists!x \in X, \neg p(x)$. Then $\neg\forall!x, p(x) \equiv \neg\neg\exists!x, \neg p(x) \equiv \exists!x, \neg p(x)$, and $\neg\exists!x, p(x) \equiv \neg\exists!x, \neg\neg p(x) \equiv \forall!x, \neg p(x)$, which are de Morgan’s laws with $\forall!, \exists!$. (It says: “it is not the case that exactly one x fails p ”.)

1.E.21 True: Theorem 1.3.19.

1.E.22 False: Exercise 1.E.12 ($p \Rightarrow q \neq q \Rightarrow p$).

1.E.23 True: by de Morgan and double negation, $p \wedge q \equiv \neg(\neg p \vee \neg q)$; replacing every conjunction this way (innermost first) gives an equivalent formula with only disjunctions and negations.

1.E.24 False: $p \wedge q$ is a counterexample. A formula built from variables with disjunctions only is true as soon as one of its variables is true; if it contains p it is true when p is true and q false, if it contains only q it is true when q is true and p false, and if it contains neither it does not depend on p, q at all; in every case it differs from $p \wedge q$ somewhere.

1.E.25 False: for p false and r true, $p \wedge (q \vee r)$ is false while $(p \wedge q) \vee r$ is true.

1.E.26 False: for p false, q true, r false, $p \vee (q \vee r)$ is true while $(p \vee q) \wedge (p \vee r)$ is $\checkmark \wedge \times = \times$.

1.E.27 False: the correct negation is $\exists x \in X, \exists y \in Y, \neg p(x, y)$. Counterexample: $X = Y = \{0\}$ with $p(0, 0)$ true; then $\neg\forall x\forall y p$ is false but $\exists x\forall y p$ is true.

1.E.31 Sometimes: if q is true, $p \Rightarrow q$ is true; if q is false (and p true), it is false.

1.E.32 Always: assume p ; together with $\neg p$ (i.e. p false) this is a contradiction, so q follows by the principle of explosion; hence $p \Rightarrow q$ (“vacuously true”).

1.E.33 Always: assume $\forall x \forall y p(x, y)$; let $y \in Y$ and $x \in X$; then $p(x, y)$ by $(\forall E)$ twice; so $\forall y \forall x p(x, y)$; and symmetrically.

1.E.34 Sometimes. Theorem 1.2.40 gives $\exists \forall \Rightarrow \forall \exists$ always. The converse can hold (e.g. $p(x, y)$: “ $y = y$ ”, both true) and can fail (Exercise 1.2.39: $p(x, y)$: “ $x + y$ even” on $\mathbb{Z}$).

1.E.35 Never. By de Morgan’s laws for quantifiers, $\exists x \exists y \neg p(x, y) \equiv \neg \forall x \forall y p(x, y)$, so the second formula is the negation of the first; a formula and its negation always have opposite truth values, so they are never equivalent.

C CHAPTER 2

2.1.1 If $R \in R$, then by definition of R we have $R \in S$ and $R \notin R$: contradiction. If $R \notin R$, then R is a set, so $R \in S$, and $R \notin R$, so R satisfies the defining property of R , giving $R \in R$: contradiction. Neither truth value is possible: a paradox.

2.1.3 “For all $x \in X$ ” must say nothing about objects outside X , and an implication with false hypothesis is true, so $x \notin X$ never produces a counterexample. “There exists $x \in X$ ” must produce a witness that is in X and satisfies p ; with $\Rightarrow$ instead, any object outside X would be a witness. De Morgan: $\neg \forall x, (x \in X \Rightarrow p(x)) \equiv \exists x, \neg(x \in X \Rightarrow p(x)) \equiv \exists x, (x \in X \wedge \neg p(x))$, which is $\exists x \in X, \neg p(x)$; and $\neg \exists x, (x \in X \wedge p(x)) \equiv \forall x, \neg(x \in X \wedge p(x)) \equiv \forall x, (x \in X \Rightarrow \neg p(x))$ (Exercise 1.3.28), which is $\forall x \in X, \neg p(x)$.

2.1.5 With Definition 2.1.4, $R \in R \Leftrightarrow R \in \mathcal{U} \wedge R \in S \wedge R \notin R$. If $R \in \mathcal{U}$ we would get the paradox of 2.1.1 again, so $R \notin \mathcal{U}$. Then $R \in R$ is simply false (the right-hand side has the false conjunct $R \in \mathcal{U}$), and $R \notin R$ no longer implies $R \in R$. No contradiction remains: R , like $\mathcal{U}$ itself, is a collection that is not an element of the universe.

2.1.13 Two sets should be equal when they have exactly the same elements, regardless of how they are described: this is Axiom 2.1.14.

2.1.28 $\emptyset$ has no elements; $\{\emptyset\}$ has one element, namely $\emptyset$; $\{\{\emptyset\}\}$ has one element, namely $\{\emptyset\}$, which is not $\emptyset$. (a) $\emptyset \in \emptyset$: false. (b) $\emptyset \subseteq \emptyset$: true. (c) $\emptyset \in \{\emptyset\}$: true. (d) $\emptyset \in \{\{\emptyset\}\}$: false (its only element is $\{\emptyset\} \neq \emptyset$). (e) $\{\emptyset\} \subseteq \{\emptyset\}$: true. (f) $\{\emptyset\} \subseteq \{\{\emptyset\}\}$: false ($\emptyset$ is in the left set but not in the right). More: $\{\emptyset\} \in \{\{\emptyset\}\}$ true; $\{\emptyset\} \in \{\emptyset\}$ false; $\emptyset \subseteq \{\{\emptyset\}\}$ true; $\{\{\emptyset\}\} \subseteq \{\emptyset\}$ false.

2.1.33 $\mathcal{P}(\emptyset) = \{\emptyset\}$; $\mathcal{P}(\mathcal{P}(\emptyset)) = \{\emptyset, \{\emptyset\}\}$; $\mathcal{P}(\mathcal{P}(\mathcal{P}(\emptyset))) = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}$.

2.2.7 (a) $A \cap (0, \infty)$. (b) $A \cap \mathbb{Z}$.

2.2.12 $(\Leftarrow)$ Suppose $b \leq c$, and let $x \in (a, b) \cap (c, d)$. Then $x < b \leq c < x$, contradiction; so the intervals are disjoint. The case $d \leq a$ is symmetric. $(\Rightarrow)$ By contraposition: suppose $b > c$ and $d > a$. Then $\max(a, c) < \min(b, d)$ (each of a, c is less than each of b, d), so $x = \frac{\max(a, c) + \min(b, d)}{2}$ satisfies $a < x < b$ and $c < x < d$, and the intervals are not disjoint.

2.2.15 $A \cap [0, \infty) \cap \mathbb{Q}$.

2.2.20 $x \in \bigcap_{n \in \mathbb{Z}} A_n$ iff $x \in \mathbb{R}$ and $x \neq n$ for every integer n , i.e. iff x is a real number that is not an integer. So the intersection is the set of non-integer real numbers ($\mathbb{R} \setminus \mathbb{Z}$ in the notation of 2.2.38).

2.2.31 $x^4 - 5x^2 + 4 = (x^2 - 1)(x^2 - 4)$ is positive iff both factors are positive or both are negative, i.e. iff $x^2 > 4$ (then also $x^2 > 1$) or $x^2 < 1$ (then also $x^2 < 4$). Hence $X = (-\infty, -2) \cup (-1, 1) \cup (2, \infty)$.

2.2.32 For sets $X, A_0, \dots, A_{n-1}$: $X \cap (A_0 \cup \dots \cup A_{n-1}) = (X \cap A_0) \cup \dots \cup (X \cap A_{n-1})$ and $X \cup (A_0 \cap \dots \cap A_{n-1}) = (X \cup A_0) \cap \dots \cap (X \cup A_{n-1})$. *Proof of the first:* $a \in X \cap (A_0 \cup \dots \cup A_{n-1}) \Leftrightarrow a \in X \wedge \exists i \in [n], a \in A_i \Leftrightarrow \exists i \in [n], (a \in X \wedge a \in A_i) \Leftrightarrow \exists i \in [n], a \in X \cap A_i \Leftrightarrow a \in (X \cap A_0) \cup \dots \cup (X \cap A_{n-1})$. *Second:* $a \in X \cup (A_0 \cap \dots \cap A_{n-1}) \Leftrightarrow a \in X \vee \forall i \in [n], a \in A_i \Leftrightarrow \forall i \in [n], (a \in X \vee a \in A_i) \Leftrightarrow a \in (X \cup A_0) \cap \dots \cap (X \cup A_{n-1})$; the middle step is the law $p \vee \forall i q(i) \equiv \forall i (p \vee q(i))$ when p does not involve i (prove it by cases on p).

2.2.42 (a) $A = \mathbb{R} \setminus \mathbb{Z}$. (b) $A = \bigcup_{n \in \mathbb{Z}} (n, n+1)$: a non-integer real x lies in $(\lfloor x \rfloor, \lfloor x \rfloor + 1)$, and every element of some $(n, n+1)$ is a real number strictly between consecutive integers, hence not an integer.

2.2.43 For any a : $a \in X \setminus (X \setminus A) \Leftrightarrow a \in X \wedge \neg(a \in X \wedge a \notin A) \Leftrightarrow a \in X \wedge (a \notin X \vee a \in A)$ {de Morgan} $\Leftrightarrow (a \in X \wedge a \notin X) \vee (a \in X \wedge a \in A)$ {distributivity, Example 1.3.3} $\Leftrightarrow a \in X \wedge a \in A$ {the first disjunct is a contradiction} $\Leftrightarrow a \in X \cap A$. *Deduction:* by Proposition 2.2.8 (with A in the role of X and X in the role of Y), $A \subseteq X$ iff $A \cap X = A$, i.e. iff $X \cap A = A$, i.e. iff $X \setminus (X \setminus A) = A$.

2.2.55 A typical element of $\mathbb{R}^3 = \prod_{i \in [3]} \mathbb{R} = \mathbb{R} \times \mathbb{R} \times \mathbb{R}$ is a triple (x, y, z) of reals, so $\mathbb{R}^3 = \mathbb{R} \times \mathbb{R} \times \mathbb{R}$ (Notation 2.2.54 defines $\mathbb{R}^3$ as this product). A typical element of $\mathbb{R}^2 \times \mathbb{R}$ is a pair $((x, y), z)$ whose first term is a pair, and $(\mathbb{R} \times \mathbb{R}) \times \mathbb{R} = \mathbb{R}^2 \times \mathbb{R}$ since $\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$. A typical element of $\mathbb{R} \times \mathbb{R}^2 = \mathbb{R} \times (\mathbb{R} \times \mathbb{R})$ is a pair $(x, (y, z))$. So the six sets fall into three equal groups: $\mathbb{R}^3 = \mathbb{R} \times \mathbb{R} \times \mathbb{R}$; $\mathbb{R}^2 \times \mathbb{R} = (\mathbb{R} \times \mathbb{R}) \times \mathbb{R}$; $\mathbb{R} \times \mathbb{R}^2 = \mathbb{R} \times (\mathbb{R} \times \mathbb{R})$. Sets from different groups are not equal: a 3-tuple is not a 2-tuple (different lengths, Definition 2.2.46), and $((x, y), z) \neq (x, (y, z))$ since their first terms (x, y) and x differ. They are all equivalent in the sense that there are one-to-one correspondences between them, e.g. $((x, y), z) \mapsto (x, y, z)$: this is the notion of *bijection* of Section 3.2 (compare Exercise 3.2.47).

2.E.1 (a) $n^2 < 20$ for an integer n iff $-4 \leq n \leq 4$ (as $4^2 = 16 < 20 < 25 = 5^2$): $\{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$. (b) $\{3, 7, 11, 15, 19, \dots\}$. (c) An odd multiple of six is $6m$ with m odd: $\{6(2k+1) \mid k \in \mathbb{Z}\} = \{12k+6 \mid k \in \mathbb{Z}\}$. (d) $\{n^2 + 1 \mid n \in \mathbb{N}\}$.

2.E.2 Take $X_n = \{k \in \mathbb{N} \mid k \geq n\}$. Then $X_{n+1} \subseteq X_n$ (if $k \geq n+1$ then $k \geq n$) and $n \in X_n \setminus X_{n+1}$, so $X_{n+1} \subsetneq X_n$. No X_n can be empty, whatever the choice of sets: if $X_n = \emptyset$ then $X_{n+1} \subsetneq \emptyset$, but a proper subset $U \subsetneq \emptyset$ would satisfy $U \subseteq \emptyset$, hence $U = \emptyset$ (an element of U would be an element of $\emptyset$), contradicting $U \neq \emptyset$.

2.E.3 The set has the two elements $a = \emptyset$ and $b = \{\emptyset, \{\emptyset\}\}$, so its power set has the four subsets $\emptyset, \{a\}, \{b\}, \{a, b\}$: $\mathcal{P}(\{\emptyset, \{\emptyset, \{\emptyset\}\}\}) = \{\emptyset, \{\emptyset\}, \{\{\emptyset, \{\emptyset\}\}\}, \{\emptyset, \{\emptyset, \{\emptyset\}\}\}\}$.

2.E.7 Let $A \in F$; we show $A \in \mathcal{P}(F)$, i.e. $A \subseteq F$. Let $x \in A$. By the hypothesis applied to $A \in F$ and $x \in A$, $x \in F$. So $A \subseteq F$, and $F \subseteq \mathcal{P}(F)$.

2.E.10 By Exercise 2.E.9, $X \triangle X = (X \setminus X) \cup (X \setminus X) = \emptyset \cup \emptyset = \emptyset$ (an element of $X \setminus X$ would be both in and not in X). And $X \triangle \emptyset = (X \setminus \emptyset) \cup (\emptyset \setminus X) = X \cup \emptyset = X$.

2.E.11 $(\Rightarrow)$ If $X = Y$ then $X \triangle Y = X \triangle X = \emptyset$ by 2.E.10. $(\Leftarrow)$ Suppose $X \triangle Y = \emptyset$. Let $a \in X$. If $a \notin Y$ then $a \in X \setminus Y \subseteq X \triangle Y = \emptyset$, impossible; so $a \in Y$. Thus $X \subseteq Y$, and symmetrically $Y \subseteq X$. Hence $X = Y$.

2.E.12 By 2.E.9, $X \triangle Y = (X \cup Y) \setminus (X \cap Y)$. $(\Rightarrow)$ If $X \cap Y = \emptyset$ then $X \triangle Y = (X \cup Y) \setminus \emptyset = X \cup Y$. $(\Leftarrow)$ Suppose $X \triangle Y = X \cup Y$ and suppose $a \in X \cap Y$. Then $a \in X \cup Y = X \triangle Y$, so $a \in X$ or $a \in Y$ but not both; yet a is in both. Contradiction, so $X \cap Y = \emptyset$.

2.E.13 Let a be arbitrary; p : $a \in X$, q : $a \in Y$, r : $a \in Z$. Write $s \oplus t$ for “ s or t but not both”. Then

$a \in X \triangle (Y \triangle Z)$ is $p \oplus (q \oplus r)$ and $a \in (X \triangle Y) \triangle Z$ is $(p \oplus q) \oplus r$:

p	q	r	$q \oplus r$	$p \oplus (q \oplus r)$	$p \oplus q$	$(p \oplus q) \oplus r$
✓	✓	✓	×	✓	×	✓
✓	✓	×	✓	×	×	×
✓	×	✓	✓	×	✓	×
✓	×	×	×	✓	✓	✓
×	✓	✓	×	×	✓	×
×	✓	×	✓	✓	✓	✓
×	×	✓	✓	✓	×	✓
×	×	×	×	×	×	×

The two columns agree (both say: a lies in an odd number of the sets X, Y, Z), so the sets are equal by extensionality.

2.E.14 With p, q, r as above: $a \in X \cap (Y \triangle Z)$ is $p \wedge (q \oplus r)$ and $a \in (X \cap Y) \triangle (X \cap Z)$ is $(p \wedge q) \oplus (p \wedge r)$. If p is false both are false. If p is true, $p \wedge q \equiv q$ and $p \wedge r \equiv r$, so both reduce to $q \oplus r$. Hence they agree for all a , and the sets are equal.

2.E.17 $(\Rightarrow)$ Let $a \in U$ and take $\delta > 0$ with $(a - \delta, a + \delta) \subseteq U$; then $u = a - \delta$, $v = a + \delta$ satisfy $u < a < v$ and $(u, v) \subseteq U$. $(\Leftarrow)$ Let $a \in U$ and take $u < a < v$ with $(u, v) \subseteq U$. Put $\delta = \min(a - u, v - a) > 0$. If $a - \delta < x < a + \delta$ then $u \leq a - \delta < x < a + \delta \leq v$, so $x \in (u, v) \subseteq U$. Hence $(a - \delta, a + \delta) \subseteq U$.

2.E.19 $(\Rightarrow)$ Suppose U is open. For each $a \in U$ choose $\delta_a > 0$ with $(a - \delta_a, a + \delta_a) \subseteq U$ (Strategy 3.2.35). Take $I = U$ and, for $i \in U$, $a_i = i - \delta_i$, $b_i = i + \delta_i$. Then $U = \bigcup_{i \in U} (a_i, b_i)$: each $a \in U$ lies in its own interval (a_a, b_a) , and each interval is contained in U . $(\Leftarrow)$ Suppose $U = \bigcup_{i \in I} (a_i, b_i)$ and let $a \in U$. Then $a \in (a_i, b_i)$ for some $i \in I$, and $(a_i, b_i) \subseteq U$. So $u = a_i$, $v = b_i$ satisfy the condition of Exercise 2.E.17, and U is open.

2.E.20 $(\subseteq)$ Let $x \in \bigcap_{i \in \mathbb{N}} A_i$ and let $j \in \mathbb{N}$. By (ii) there is $i \geq j$ with $A_i \subseteq B_j$; since $x \in A_i$, $x \in B_j$. As j was arbitrary, $x \in \bigcap_j B_j$. $(\supseteq)$ Let $x \in \bigcap_j B_j$ and let $i \in \mathbb{N}$. By (i) there is $j \geq i$ with $B_j \subseteq A_i$; since $x \in B_j$, $x \in A_i$. So $x \in \bigcap_i A_i$. (The conditions $j \geq i$, $i \geq j$ are not even needed.)

2.E.21 False. $E = \{0\}$: $\neg \forall x, x \in E$ is true (e.g. $1 \notin E$) but $E \neq \emptyset$.

2.E.22 True. $\forall x, \neg x \in E$ says E has no elements, i.e. E is empty (Definition 2.1.19), so $E = \emptyset$ by Theorem 2.1.25.

2.E.23 True: both sets have exactly the elements 1, 2, 3 (Axiom 2.1.14; order and repetition are irrelevant).

2.E.24 False: $\emptyset$ has no elements, so $\emptyset \in \emptyset$ is false.

2.E.25 True: $\{\emptyset\}$ has the single element $\emptyset$.

2.E.26 False: the only element of $\{\{\emptyset\}\}$ is $\{\emptyset\}$, and $\{\emptyset\} \neq \emptyset$ (the former has an element, the latter does not).

2.E.27 Always: both sets have exactly the elements a and b , so they are equal by Axiom 2.1.14.

2.E.29 Sometimes. $X = Y = \emptyset$: any a satisfies $a \in X \Leftrightarrow a \in Y$ (both false) and $X = Y$. $X = \{0\}$, $Y = \{1\}$, $a = 2$: $a \in X \Leftrightarrow a \in Y$ holds (both false) but $X \neq Y$.

2.E.30 Never. If $X = Y$ then $a \in X$ gives $a \in Y$, contradicting $a \notin Y$.

2.E.31 Sometimes. $X = \{0\}$, $Y = \{1\}$: $X \neq Y$ and every $a \in X$ (only $a = 0$) has $a \notin Y$. $X = \{0\}$, $Y = \{0, 1\}$: $X \neq Y$ but $0 \in X$ and $0 \in Y$.

2.E.34 Sometimes. $X = \emptyset, Y = \{\emptyset\}$: $X \in Y$ and $X \subseteq Y$. $X = \{\emptyset\}, Y = \{\{\emptyset\}\}$: $X \in Y$, but $\emptyset \in X$ and $\emptyset \notin Y$ (the only element of Y is $\{\emptyset\} \neq \emptyset$), so $X \not\subseteq Y$.

2.E.35 Always: $\emptyset \subseteq X$ (Exercise 2.1.27), so $\emptyset \in \mathcal{P}(X)$.

2.E.36 Sometimes. $\{\emptyset\} \in \mathcal{P}(X)$ iff $\{\emptyset\} \subseteq X$ iff $\emptyset \in X$. True for $X = \{\emptyset\}$; false for $X = \emptyset$ (then $\mathcal{P}(X) = \{\emptyset\}$ and $\{\emptyset\} \neq \emptyset$).

2.E.37 Always: $\emptyset \subseteq X$ and $X \subseteq X$, so both elements of $\{\emptyset, X\}$ are in $\mathcal{P}(X)$.

2.E.39 Sometimes. $X = Y = \emptyset$: $X \setminus Y = \emptyset$ and $X = Y$. $X = \emptyset, Y = \{0\}$: $X \setminus Y = \emptyset$ but $X \neq Y$. (In general $X \setminus Y = \emptyset$ iff $X \subseteq Y$.)

2.E.40 Always. First, $Y = \emptyset$: if $y \in Y$ then $y \in Z = X \setminus Y$, so $y \notin Y$, a contradiction. Hence $Z = X \setminus \emptyset = X$, and $Y \cup Z = \emptyset \cup X = X$.

2.E.42 Sometimes. $X = \{\emptyset\}, A = \emptyset$: $A \subseteq X$ and $A \in X$. $X = \{0\}, A = \{0\}$: $A \subseteq X$ but $A \notin X$ (the only element of X is $0 \neq \{0\}$).

D CHAPTER 3

3.1.3 Only (a) has the form $\forall x \in X, \exists! y \in Y, p(x, y)$: “for each real number a , the equation $x^2 + 2ax + a^2 = 0$ has exactly one real solution x ” is $\forall a \in \mathbb{R}, \exists! x \in \mathbb{R}, x^2 + 2ax + a^2 = 0$. Since $x^2 + 2ax + a^2 = (x + a)^2$, the unique solution is $x = -a$, so the function is $\mathbb{R} \rightarrow \mathbb{R}, a \mapsto -a$. Statement (b) begins with $\exists! a \in \mathbb{R}$ and (c) with $\exists! n \in \mathbb{N}$, so neither has the required form.

3.1.22 By Definition 3.1.17, $g \circ f$ requires the codomain of f to equal the domain of g , and $Y \neq Z$. Let $i: Y \rightarrow Z$ be the inclusion, $i(y) = y$ (well-defined since $Y \subseteq Z$). Then $h = g \circ i \circ f = (g \circ i) \circ f$: for $x \in X$, $(g \circ (i \circ f))(x) = g(i(f(x))) = g(f(x)) = h(x)$, and the domains and codomains agree ($X \rightarrow W$).

3.1.28 Let $x \in X$. (b) If $x \in A$ and $x \in B$: right side $1 + 1 - 1 = 1$, and $x \in A \cup B$ so left side 1. If $x \in A, x \notin B$: $1 + 0 - 0 = 1$ and $x \in A \cup B$. If $x \notin A, x \in B$: symmetric. If $x \notin A, x \notin B$: $0 + 0 - 0 = 0$ and $x \notin A \cup B$. In all cases $\chi_{A \cup B}(x) = \chi_A(x) + \chi_B(x) - \chi_A(x)\chi_B(x)$. (c) If $x \in A$ then $x \notin X \setminus A$, so $\chi_{X \setminus A}(x) = 0 = 1 - 1$; if $x \notin A$ then $x \in X \setminus A$ (as $x \in X$), so $\chi_{X \setminus A}(x) = 1 = 1 - 0$.

3.1.30 For $x \in X$, by Theorem 3.1.27: $\chi_{X \setminus (A \cup B)} = 1 - \chi_{A \cup B} = 1 - \chi_A - \chi_B + \chi_A \chi_B = (1 - \chi_A)(1 - \chi_B) = \chi_{X \setminus A} \chi_{X \setminus B} = \chi_{(X \setminus A) \cap (X \setminus B)}$, so $X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B)$ by Strategy 3.1.26. Likewise $\chi_{(X \setminus A) \cup (X \setminus B)} = (1 - \chi_A) + (1 - \chi_B) - (1 - \chi_A)(1 - \chi_B) = 2 - \chi_A - \chi_B - (1 - \chi_A - \chi_B + \chi_A \chi_B) = 1 - \chi_A \chi_B = 1 - \chi_{A \cap B} = \chi_{X \setminus (A \cap B)}$, giving $X \setminus (A \cap B) = (X \setminus A) \cup (X \setminus B)$.

3.1.34 If $y \in f[\emptyset]$ then $y = f(x)$ for some $x \in \emptyset$, which is impossible. So $f[\emptyset]$ has no elements: $f[\emptyset] = \emptyset$.

3.1.42 Let $A = f^{-1}[\{1\}] = \{x \in X \mid f(x) = 1\}$. Both f and χ_A have domain X and codomain $\{0, 1\} = [2]$. Let $x \in X$. If $f(x) = 1$ then $x \in A$, so $\chi_A(x) = 1 = f(x)$. If $f(x) \neq 1$ then $f(x) = 0$ (the only other element of the codomain) and $x \notin A$, so $\chi_A(x) = 0 = f(x)$. Hence $f = \chi_A$ by function extensionality.

3.2.7 $f: \mathbb{N} \rightarrow \mathbb{Z}, f(n) = n^2$: injective. If $m, n \in \mathbb{N}$ and $m^2 = n^2$ then $(m - n)(m + n) = 0$; if $m + n = 0$ then $m = n = 0$; otherwise $m - n = 0$. Either way $m = n$. $g: \mathbb{Z} \rightarrow \mathbb{N}, g(n) = n^2$: not injective, $g(-1) = 1 = g(1)$. $h(x, y, z) = 2^x 3^y 5^z$: injective, by uniqueness of prime factorisation (the fundamental theorem of arithmetic, Theorem 7.2.12, which the book also uses in Example 3.2.24): if $2^x 3^y 5^z = 2^{x'} 3^{y'} 5^{z'}$ then the exponents of each prime agree, so $(x, y, z) = (x', y', z')$.

3.2.13 Take $V = f[X]$ and $g: X \rightarrow f[X], g(x) = f(x)$ (well-defined since $f(x) \in f[X]$). For $y \in f[X]$ there is $x \in X$ with $f(x) = y$ (Definition 3.1.31), so $g(x) = y$: g is surjective.

3.2.15 Take $Z = f[X]$, $p : X \rightarrow f[X]$, $p(x) = f(x)$ (surjective by Exercise 3.2.13) and $i : f[X] \rightarrow Y$, $i(y) = y$ (injective: $i(y) = i(y')$ means $y = y'$). Then $(i \circ p)(x) = i(f(x)) = f(x)$ for all $x \in X$, so $f = i \circ p$.

3.2.21 $(\Rightarrow)$ Suppose f injective and let $y \in f[X]$. Existence: $y = f(x)$ for some $x \in X$ by definition of $f[X]$. Uniqueness: if $y = f(x) = f(x')$ then $x = x'$ by injectivity. $(\Leftarrow)$ Suppose the condition holds and let $a, b \in X$ with $f(a) = f(b) =: y$. Then $y \in f[X]$, and both a and b are elements x with $y = f(x)$; by uniqueness $a = b$.

3.2.22 Injective: if $2^m 3^n = 2^{m'} 3^{n'}$ then $m = m'$ and $n = n'$ by uniqueness of prime factorisation (Theorem 7.2.12). (Directly: if $m < m'$, dividing by 2^m gives $3^n = 2^{m'-m} 3^{n'}$ with left side odd and right side even, impossible; so $m = m'$, then $3^n = 3^{n'}$ gives $n = n'$.) Decoding: $1 = 2^0 3^0 \mapsto (0, 0)$; $24 = 2^3 \cdot 3 \mapsto (3, 1)$; $7776 = 6^5 = 2^5 3^5 \mapsto (5, 5)$; $59049 = 3^{10} \mapsto (0, 10)$; $396718580736 = 2^{10} \cdot 3^{18} \mapsto (10, 18)$.

3.2.28 Let $x \in X$. Then $x = g(f(x))$, so x is the value of g at $f(x) \in Y$. Hence g is surjective.

3.2.31 Let $g : Y \rightarrow X$ with $f \circ g = \text{id}_Y$ and let $y \in Y$. Then $x = g(y) \in X$ satisfies $f(x) = f(g(y)) = y$. So f is surjective.

3.2.39(c) Every $k \in \mathbb{N}$ has $k + 1 \geq 1$, and every positive integer can be written uniquely as $2^m q$ with $m \in \mathbb{N}$ and q odd (fundamental theorem of arithmetic: m is the exponent of 2 and q the product of the odd prime powers; directly: if $2^m q = 2^{m'} q'$ with q, q' odd and $m < m'$ then $q = 2^{m'-m} q'$ is even, contradiction, so $m = m'$ and $q = q'$). Write $q = 2n + 1$ and define $h^{-1} : \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$ by $h^{-1}(k) = (m, n)$ where $k + 1 = 2^m(2n + 1)$. Then $h^{-1}(h(m, n)) = h^{-1}(2^m(2n + 1) - 1) = (m, n)$ by uniqueness, and $h(h^{-1}(k)) = 2^m(2n + 1) - 1 = (k + 1) - 1 = k$. So h^{-1} is a two-sided inverse.

3.E.1 (a) Yes: $x + y = 1 \Leftrightarrow y = 1 - x$; $f(x) = 1 - x$. (b) No: for $x = 0$ both $y = 1$ and $y = -1$ satisfy the equation, so no single value $f(0)$ works (and for $x = 2$ no y at all). (c) No: for $x = 0$ the equation “ $x = 0$ ” is true for every y , but $y = f(0)$ holds for only one y . (d) Yes: $f(x) = 0$ for all x . (e) Yes: $1 + x^2 \neq 0$, so $(1 + x^2)y = 1 \Leftrightarrow y = \frac{1}{1+x^2}$. (f) No: for $x = 1$ every y satisfies $(1 - 1)y = 0$.

3.E.3 Let $f : X \rightarrow \emptyset$ be a function. If X had an element a , then $f(a) \in \emptyset$, impossible; so $X = \emptyset$. Any two functions $\emptyset \rightarrow \emptyset$ have the same domain and codomain and (vacuously) the same values, so they are equal by Axiom 3.1.4. Hence there is exactly one function with empty codomain, the empty function $\emptyset \rightarrow \emptyset$; its domain is $\emptyset$.

3.E.5 $f(-x) = (-x)^n = (-1)^n x^n$. If n is even, $(-1)^n = 1$ so $f(-x) = f(x)$: f is even; if n is odd, $(-1)^n = -1$ so $f(-x) = -f(x)$: f is odd. Conversely, if f is even then $f(-1) = f(1)$ gives $(-1)^n = 1$, so n is even (for odd n , $(-1)^n = -1 \neq 1$); if f is odd then $(-1)^n = -1$, so n is odd.

3.E.6 If f is even and odd then for all x , $f(x) = f(-x) = -f(x)$, so $2f(x) = 0$, $f(x) = 0$. Thus f is the zero function, which is indeed even and odd ($0 = 0 = -0$). So exactly one such function exists.

3.E.7 Write $-U = \{-u \mid u \in U\}$; note $x \in -U \Leftrightarrow -x \in U$ (if $x = -u$ with $u \in U$ then $-x = u \in U$; conversely $x = -(-x)$). (a) χ_U even $\Leftrightarrow \forall x, \chi_U(-x) = \chi_U(x) \Leftrightarrow \forall x, (-x \in U \Leftrightarrow x \in U) \Leftrightarrow \forall x, (x \in -U \Leftrightarrow x \in U) \Leftrightarrow -U = U$. (b) χ_U odd $\Leftrightarrow \forall x, \chi_U(-x) = -\chi_U(x) \Leftrightarrow \forall x, (-x \in U \Leftrightarrow x \notin U) \Leftrightarrow \forall x, (x \in -U \Leftrightarrow x \in \mathbb{R} \setminus U) \Leftrightarrow -U = \mathbb{R} \setminus U$.

3.E.8 Put $g(x) = \frac{f(x)+f(-x)}{2}$ and $h(x) = \frac{f(x)-f(-x)}{2}$. Then $g(-x) = g(x)$, $h(-x) = -h(x)$ and $g(x) + h(x) = f(x)$. Uniqueness: if also $f = g' + h'$ with g' even, h' odd, then $g - g' = h' - h$; the left side is even and the right side is odd, so this function is both, hence zero by Exercise 3.E.6. So $g = g'$ and $h = h'$.

3.E.9 For $n = 0$: $[0] = \emptyset$, and $\theta_0 : \emptyset \rightarrow \emptyset$ is the empty function, which is $\text{id}_\emptyset$. For $n \geq 1$: first, $\theta_1 = \text{id}_{[1]}$, since $[1] = \{0\}$ has only one function to itself (3.E.39). Fix $k \in [n]$ and let $f : [1] \rightarrow [n]$, $f(0) = k$. Then $f \circ \theta_1 = \theta_n \circ f$ gives $k = f(0) = f(\theta_1(0)) = \theta_n(f(0)) = \theta_n(k)$. As k was arbitrary, $\theta_n = \text{id}_{[n]}$.

3.E.10 $\chi_{U \triangle V} = \chi_U + \chi_V - 2\chi_U \chi_V$. Check by cases: x in both: $1 + 1 - 2 = 0$ and $x \notin U \triangle V$; in exactly one: $1 + 0 - 0 = 1$ and $x \in U \triangle V$; in neither: 0 . (Equivalently $(\chi_U - \chi_V)^2$.)

3.E.11 On $\mathbb{R} \setminus \{0\} = (-\infty, 0) \cup (0, \infty)$, a function F with $F'(x) = \frac{1}{x}$ satisfies $(F(x) - \log|x|)' = 0$ separately on each of the two intervals, so $F - \log|x|$ is constant on $(-\infty, 0)$ (say a) and constant on $(0, \infty)$ (say b), and these constants are independent. Hence $F(x) = \log|x| + a\chi_U(x) + b\chi_V(x)$ with $U = (-\infty, 0)$, $V = (0, \infty)$; conversely every such F has derivative $\frac{1}{x}$.

3.E.12 $f[\mathbb{R}] = [1, \infty)$. ($\subseteq$) $1 + x^2 \geq 1$ so $\sqrt{1 + x^2} \geq 1$. ($\supseteq$) For $y \geq 1$ put $x = \sqrt{y^2 - 1} \in \mathbb{R}$; then $f(x) = \sqrt{1 + y^2 - 1} = y$.

3.E.14 $f(1) = \{0\}$, $f(2) = \{0, 1\}$, and in general $f(n) = [n] = \{0, 1, \dots, n-1\}$ (induction: $f(n+1) = [n] \cup \{n\} = [n+1]$). So $f[\mathbb{N}] = \{[n] \mid n \in \mathbb{N}\} = \{\emptyset, \{0\}, \{0, 1\}, \{0, 1, 2\}, \dots\}$.

3.E.15 $f[\mathbb{R}^{\mathbb{R}}]$ is the set of even functions $\mathbb{R} \rightarrow \mathbb{R}$. ($\subseteq$) $f(h)(-x) = h(|-x|) = h(|x|) = f(h)(x)$, so every $f(h)$ is even. ($\supseteq$) Let g be even. Then $f(g)(x) = g(|x|) = g(x)$ for all x (if $x \geq 0$ then $|x| = x$; if $x < 0$ then $g(|x|) = g(-x) = g(x)$), so $f(g) = g$ and $g \in f[\mathbb{R}^{\mathbb{R}}]$.

3.E.16 $\sqrt{1+x^2} \in (-5, 5]$ iff $\sqrt{1+x^2} \leq 5$ (it is always > 0) iff $1+x^2 \leq 25$ iff $x^2 \leq 24$ iff $-2\sqrt{6} \leq x \leq 2\sqrt{6}$. So $f^{-1}[(-5, 5]] = [-2\sqrt{6}, 2\sqrt{6}]$.

3.E.18 $f^{-1}[\{\emptyset\}] = \{(A, B) \in \mathcal{P}(\mathbb{N}) \times \mathcal{P}(\mathbb{N}) \mid A \cap B = \emptyset\}$, the set of pairs of disjoint subsets of $\mathbb{N}$.

3.E.20 *Reading:* $v: A \rightarrow f[X]$ should satisfy $v(a) = i(a)$ for all $a \in A$. Let $q: X \rightarrow f[X]$, $q(x) = f(x)$, and $j: f[X] \rightarrow Y$, $j(y) = y$; j is injective and $j \circ q = f$. By (iii) there is a unique $u: A \rightarrow f[X]$ with $j \circ u = i$, i.e. $u(a) = i(a)$ for all $a \in A$. So $v = u$ is the only candidate (uniqueness). v is injective: $v(a) = v(b) \Rightarrow i(a) = i(b) \Rightarrow a = b$ by (i). v is surjective: let $y \in f[X]$, say $y = f(x)$; by (ii) $y = i(p(x)) = v(p(x))$. So v is the unique bijection with $v(a) = i(a)$.

3.E.21 This is Exercise 3.1.40 (worked in Section A).

3.E.22 (a) $z \in (g \circ f)[U] \Leftrightarrow \exists x \in U, z = g(f(x))$; and $z \in g[f[U]] \Leftrightarrow \exists y \in f[U], z = g(y) \Leftrightarrow \exists x \in U, z = g(f(x))$ (unfold $y \in f[U]$). (b) $x \in (g \circ f)^{-1}[W] \Leftrightarrow g(f(x)) \in W \Leftrightarrow f(x) \in g^{-1}[W] \Leftrightarrow x \in f^{-1}[g^{-1}[W]]$.

3.E.23 ($\subseteq$) Let $(x, z) \in p[S]$; then $(x, z) = p(x, y, z)$ for some $(x, y, z) \in S$, so $y = f(x)$ and $z = g(y) = g(f(x)) = (g \circ f)(x)$; hence $(x, z) \in \text{Gr}(g \circ f)$. ($\supseteq$) Let $(x, (g \circ f)(x)) \in \text{Gr}(g \circ f)$. Put $y = f(x)$, $z = g(y)$; then $(x, y, z) \in S$ and $p(x, y, z) = (x, z) = (x, (g \circ f)(x))$.

3.E.26(a) Define $f: \mathbb{Z} \rightarrow \mathbb{N}$ by $f(n) = 2n$ if $n \geq 0$ and $f(n) = -2n - 1$ if $n < 0$ (values in $\mathbb{N}$: $2n \geq 0$; for $n \leq -1$, $-2n - 1 \geq 1$). Inverse $g: \mathbb{N} \rightarrow \mathbb{Z}$: $g(m) = \frac{m}{2}$ if m even, $g(m) = -\frac{m+1}{2}$ if m odd. Check: $g(f(n)) = g(2n) = n$ for $n \geq 0$; for $n < 0$, $f(n) = -2n - 1$ is odd and $g(f(n)) = -\frac{-2n-1+1}{2} = n$. And $f(g(m)) = 2 \cdot \frac{m}{2} = m$ for even m ; for odd m , $g(m) = -\frac{m+1}{2} < 0$ and $f(g(m)) = -2(-\frac{m+1}{2}) - 1 = m$. So f is a bijection (Strategy 3.2.41).

3.E.26(c) Let $A = \{0, 1\} \cup \{\frac{1}{n} \mid n \geq 2\} \subseteq [0, 1]$ and $A' = \{\frac{1}{n} \mid n \geq 2\} \subseteq (0, 1)$. Define $f: [0, 1] \rightarrow (0, 1)$ by $f(0) = \frac{1}{2}$, $f(1) = \frac{1}{3}$, $f(\frac{1}{n}) = \frac{1}{n+2}$ for $n \geq 2$, and $f(x) = x$ for $x \notin A$ (then $0 < x < 1$). Define $g: (0, 1) \rightarrow [0, 1]$ by $g(\frac{1}{2}) = 0$, $g(\frac{1}{3}) = 1$, $g(\frac{1}{n}) = \frac{1}{n-2}$ for $n \geq 4$, and $g(y) = y$ for $y \notin A'$. Then $g(f(x)) = x$ and $f(g(y)) = y$ in every case (f maps A onto A' by the shift $0, 1, \frac{1}{2}, \frac{1}{3}, \dots \mapsto \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots$, and is the identity off A , where $[0, 1] \setminus A = (0, 1) \setminus A'$). So f is a bijection.

3.E.26(d) Compose bijections (Exercise 3.2.20): $[a, b] \rightarrow [0, 1]$, $x \mapsto \frac{x-a}{b-a}$ (inverse $t \mapsto a + (b-a)t$); then $[0, 1] \rightarrow (0, 1)$ from (c); then $(0, 1) \rightarrow (c, d)$, $t \mapsto c + (d-c)t$ (inverse $y \mapsto \frac{y-c}{d-c}$).

3.E.27 Write $T(k) = \binom{k+1}{2} = \frac{k(k+1)}{2}$, so $f(a, b) = T(a+b) + b$ and $T(k+1) = T(k) + (k+1)$. *Claim:* for every $n \in \mathbb{N}$ there is a unique $k \in \mathbb{N}$ with $T(k) \leq n < T(k+1)$. *Existence:* the set $\{k \in \mathbb{N} \mid T(k+1) > n\}$ is non-empty ($T(n+1) \geq n+1 > n$), so it has a least element k (well-ordering); then $T(k) \leq n$ (if $k = 0$, $T(0) = 0 \leq n$; if $k > 0$, minimality gives $T(k) \leq n$). *Uniqueness:* T is strictly increasing, so if $k < k'$ then $T(k+1) \leq T(k')$, and $n < T(k+1) \leq T(k') \leq n$ is impossible. *Surjective:* given n , take this k , put $b = n - T(k)$ and $a = k - b$. Then $0 \leq b < T(k+1) - T(k) = k+1$, so $b \leq k$ and $a \in \mathbb{N}$; and $f(a, b) = T(k) + b = n$. *Injective:*

if $f(a, b) = f(a', b') = n$, then with $k = a + b$ we have $T(k) \leq n = T(k) + b \leq T(k) + k < T(k + 1)$, so k is the unique number of the claim; likewise $a' + b' = k$. Then $b = n - T(k) = b'$ and $a = k - b = a'$.

3.E.28 Let $Y = \{x \in X \mid e(x) = x\}$. Define $f : X \rightarrow Y$ by $f(x) = e(x)$: well-defined since $e(e(x)) = e(x)$ shows $e(x) \in Y$. Define $g : Y \rightarrow X$ by $g(y) = y$. Then $g(f(x)) = e(x)$, so $g \circ f = e$; and for $y \in Y$, $f(g(y)) = e(y) = y$, so $f \circ g = \text{id}_Y$.

3.E.29 False: $(1, 1)$ and $(1, -1)$ are both in G , so uniqueness fails at $x = 1$ (Theorem 3.1.9).

3.E.30 True: for each $x \in \mathbb{Z}$ the unique $y \in \mathbb{N}$ with $y^2 = x^2$ is $y = |x|$ (if $y^2 = x^2$ then $y = \pm x$, and only one of $x, -x$ is in $\mathbb{N}$ unless $x = 0$). So $G = \text{Gr}(x \mapsto |x|)$.

3.E.31 False: the empty function $\emptyset \rightarrow \{0\}$ (Definition 3.1.13) has empty domain and non-empty codomain.

3.E.32 True: this is Exercise 3.E.3.

3.E.33 False: $f : \{0\} \rightarrow \{1\}$, $f(0) = 1$, has image $\{1\} \not\subseteq \{0\}$. (The image is a subset of the *codomain*.)

3.E.36 True. A left inverse g of f is surjective by Exercise 3.2.28. If g is a right inverse of f ($f \circ g = \text{id}_Y$) then f is a left inverse of g , so g is injective by Exercise 3.2.25.

3.E.39 Always: $f(x) = 0$ for all $x \in X$ is well-defined, and any function $X \rightarrow \{0\}$ must take the value 0 everywhere, so by function extensionality it equals f .

3.E.40 Sometimes. If $X = \emptyset$, the empty function $\emptyset \rightarrow \emptyset$ is the unique function (3.E.3). If $X \neq \emptyset$ there is no function $X \rightarrow \emptyset$ at all (3.E.3), so “a unique function” is false.

3.E.42 Never. The three values $f(1), f(2), f(3)$ lie in $\{1, 2\}$, so two of them coincide: if $f(1) = f(2)$ then $(f(1), 1), (f(1), 2) \in G'$; otherwise $\{f(1), f(2)\} = \{1, 2\}$ and $f(3) = f(k)$ for some $k \in \{1, 2\}$, giving $(f(3), 3), (f(3), k) \in G'$ with $k \neq 3$. In all cases some first coordinate occurs twice in G' , so G' is not a graph (Theorem 3.1.9).

3.E.43 Always: pick $x \in U$; then $f(x) \in f[U]$.

3.E.44 Sometimes. $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$: $V = \{-1\}$ gives $f^{-1}[V] = \emptyset$; $V = \{1\}$ gives $f^{-1}[V] = \{-1, 1\} \neq \emptyset$.

3.E.45 Sometimes: $f(1) = 1, f(2) = 2$ is injective; $f(1) = f(2) = 1$ is not.

3.E.46 Never: $f[\{1, 2\}] = \{f(1), f(2)\}$ has at most two elements, so at least one of 1, 2, 3 is not a value of f . (Precisely: if $f(1) = f(2)$ two of the three are missed; otherwise exactly the third element is missed.)

E CHAPTER 4

4.1.9 Define $m^0 = 1$ and $m^{n+1} = m^n \cdot m$ for all $n \in \mathbb{N}$. Then $2^2 = 2^{1+1} = 2^1 \cdot 2 = (2^0 \cdot 2) \cdot 2 = (1 \cdot 2) \cdot 2$. Now $1 \cdot 2 = 1 \cdot (1 + 1) = (1 \cdot 1) + 1$ and $1 \cdot 1 = 1 \cdot (0 + 1) = (1 \cdot 0) + 1 = 0 + 1 = 1$ (using $0 + 1 = 0 + s(0) = s(0 + 0) = s(0) = 1$), so $1 \cdot 2 = 1 + 1 = 2$. Hence $2^2 = 2 \cdot 2 = 4$ by Proposition 4.1.8.

4.1.16 Each entry is the sum of the two above it: 1 6 15 20 15 6 1 and 1 7 21 35 35 21 7 1.

4.2.12 Elements of $\prod_{k \in [n+1]} X_k$ are $(n+1)$ -tuples $(a_0, \dots, a_n)$ with $a_k \in X_k$; elements of $(\prod_{k \in [n]} X_k) \times X_n$ are pairs $((a_0, \dots, a_{n-1}), a_n)$ with the same conditions (Definition 2.2.49). Define $\varphi(a_0, \dots, a_n) = ((a_0, \dots, a_{n-1}), a_n)$ and $\psi((a_0, \dots, a_{n-1}), a_n) = (a_0, \dots, a_n)$; both are well-defined, and $\psi \circ \varphi$ and $\varphi \circ \psi$ are identities by the equality rule for tuples (Definition 2.2.46), so φ is a bijection (Strategy 3.2.41). The book asks for induction on n : the base case $n = 0$ is $X_0 \rightarrow \{()\} \times X_0$, $x \mapsto ((), x)$, and the induction step is the same re-bracketing one coordinate further; the induction is what makes the “...” in the tuples precise.

4.2.14 For $n \geq 1$; $p(n)$: $\sum_{i=0}^n (-1)^i \binom{n}{i} = 0$. *Base case*: $\binom{1}{0} - \binom{1}{1} = 1 - 1 = 0$. *Step*: let $n \geq 1$ and suppose $p(n)$. Using $\binom{n+1}{0} = 1$ and $\binom{n+1}{i} = \binom{n}{i-1} + \binom{n}{i}$ for $i \geq 1$ (Definition 4.1.15):

$$\sum_{i=0}^{n+1} (-1)^i \binom{n+1}{i} = 1 + \sum_{i=1}^{n+1} (-1)^i \binom{n+1}{i} = 1 + \sum_{i=1}^n (-1)^i \left(\binom{n}{i-1} + \binom{n}{i} \right) + (-1)^{n+1} \binom{n+1}{n+1} = 1 - \sum_{j=0}^n (-1)^j \binom{n}{j} + \left(\sum_{i=0}^n (-1)^i \binom{n}{i} - 1 \right).$$

The first sum is 0 by $p(n)$; the last sum is $\sum_{i=0}^n (-1)^i \binom{n}{i} + (-1)^{n+1} \binom{n}{n+1} = 0 + 0$. Total: $1 - 0 + (0 - 1) = 0$. The result follows by induction.

4.3.22 Suppose $\sqrt{n}$ is not an integer but is rational, say $\sqrt{n} = \frac{a}{b}$ with $a, b \in \mathbb{N}$, $b \geq 1$. Let $m = \lfloor \sqrt{n} \rfloor$, so $0 < \sqrt{n} - m < 1$. Let $X = \{b \in \mathbb{N} \mid b \geq 1, b\sqrt{n} \in \mathbb{Z}\}$; X is non-empty since $b\sqrt{n} = a \in \mathbb{Z}$. Let $b \in X$ and put $b' = b(\sqrt{n} - m) = b\sqrt{n} - bm \in \mathbb{Z}$. Then $0 < b' < b$, so $b' \in \mathbb{N}$ and $b' \geq 1$; and $b'\sqrt{n} = bn - mb\sqrt{n} \in \mathbb{Z}$. So $b' \in X$ with $b' < b$: X has infinite descent, contradicting the well-ordering principle (Strategy 4.3.19). Hence $\sqrt{n}$ is irrational. (The hint's expression $\sqrt{n} = \frac{a(\sqrt{n}-m)}{b(\sqrt{n}-m)} = \frac{bn-ma}{b'}$ exhibits the smaller denominator.)

4.E.1 $1 + 3 = 1 + s(2) = s(1 + 2) = s(1 + s(1)) = s(s(1 + 1)) = s(s(1 + s(0))) = s(s(s(1 + 0))) = s(s(s(1))) = s(s(2)) = s(3) = 4$.

4.E.2 $0 + 5 = s(0 + 4) = s(s(0 + 3)) = s(s(s(0 + 2))) = s(s(s(s(0 + 1)))) = s(s(s(s(s(0 + 0))))) = s(s(s(s(s(0)))))) = 5$, using $0 + s(n) = s(0 + n)$ five times and $0 + 0 = 0$.

4.E.3 $2 \cdot 3 = 2 \cdot (2 + 1) = (2 \cdot 2) + 2 = 4 + 2$ (Proposition 4.1.8) $= 4 + s(1) = s(4 + 1) = s(4 + s(0)) = s(s(4 + 0)) = s(s(4)) = s(5) = 6$.

4.E.4 $0 \cdot 5 = 0 \cdot (4 + 1) = (0 \cdot 4) + 0 = 0 \cdot 4$, since $m + 0 = m$. Repeating, $0 \cdot 4 = 0 \cdot 3 = 0 \cdot 2 = 0 \cdot 1 = (0 \cdot 0) + 0 = 0 \cdot 0 = 0$.

4.E.5 With $m^{n+1} = m^n \cdot m$: $2^3 = 2^2 \cdot 2 = 4 \cdot 2$ (Exercise 4.1.9) $= 4 \cdot (1 + 1) = (4 \cdot 1) + 4 = ((4 \cdot 0) + 4) + 4 = (0 + 4) + 4 = 4 + 4$ (as $0 + 4 = 4$, computed like 4.E.2) $= 4 + s(3) = s(4 + 3) = s(s(4 + 2)) = s(s(s(4 + 1))) = s(s(s(s(4 + 0)))) = s(s(s(s(4)))) = 8$.

4.E.6 $\exists^0 x \in X$, $p(x)$ means $\neg \exists x \in X$, $p(x)$. Given $\exists^n$, define $\exists^{n+1} x \in X$, $p(x)$ to mean $\exists a \in X$, $(p(a) \wedge \exists^n x \in X, (p(x) \wedge x \neq a))$: some a satisfies p , and exactly n elements other than a do.

4.E.7 $\binom{4}{2} = \binom{3}{1} + \binom{3}{2}$. $\binom{3}{1} = \binom{2}{0} + \binom{2}{1} = 1 + \left(\binom{1}{0} + \binom{1}{1} \right) = 1 + \left(1 + \left(\binom{0}{0} + \binom{0}{1} \right) \right) = 1 + (1 + (1 + 0)) = 3$. $\binom{3}{2} = \binom{2}{1} + \binom{2}{2} = 2 + \left(\binom{1}{1} + \binom{1}{2} \right) = 2 + \left(1 + \left(\binom{0}{1} + \binom{0}{2} \right) \right) = 2 + (1 + 0) = 3$. So $\binom{4}{2} = 6$.

4.E.8 The number of trailing zeros in base b is the largest r with $b^r \mid n$. (a) $10^r \mid 41!$ iff $2^r \mid 41!$ and $5^r \mid 41!$. The factor 5 occurs in 5, 10, ..., 40 (eight multiples) and once more in 25: exponent 9. The exponent of 2 is larger (below), so $41!$ has 9 trailing decimal zeros. (b) The exponent of 2 in $41!$: 20 multiples of 2, 10 of 4, 5 of 8, 2 of 16, 1 of 32: $20 + 10 + 5 + 2 + 1 = 38$. So $41!$ has 38 trailing zeros in binary.

4.E.9 $(\Rightarrow)$ If (N, z, s) satisfies Peano's axioms, apply the recursion theorem (4.1.2) with $h(n, x) = f(x)$: there is a unique $h : N \rightarrow X$ with $h(z) = a$ and $h(s(n)) = f(h(n))$, i.e. $f \circ h = h \circ s$. $(\Leftarrow)$ Assume the universal property. (iii) Let $A \subseteq N$ with $z \in A$ and $s[A] \subseteq A$. Apply the property to $X = A$, $a = z$, $f = s|_A : A \rightarrow A$: get $h : N \rightarrow A$ with $h(z) = z$ and $s(h(n)) = h(s(n))$. Composing with the inclusion $A \rightarrow N$ gives $h' : N \rightarrow N$ with $h'(z) = z$, $s \circ h' = h' \circ s$; but id_N has the same properties, so $h' = \text{id}_N$ by uniqueness (property with $X = N$, $a = z$, $f = s$). Hence $n = h(n) \in A$ for all n : $N \subseteq A$. (i) Apply the property to $X = \{0, 1\}$, $a = 0$, f the constant function 1: $h(z) = 0$ and $h(s(n)) = 1$ for all n . If $z = s(n)$ then $0 = h(z) = h(s(n)) = 1$, contradiction. (ii) Apply the property to $X = N \times N$, $a = (z, z)$, $f(x, y) = (y, s(y))$: get h with $h(z) = (z, z)$, $h(s(n)) = f(h(n))$. Writing $h(n) = (g(n), t(n))$: $t(z) = z$ and $t(s(n)) = s(t(n))$, so $t = \text{id}_N$ by uniqueness; then $h(s(n)) = f(g(n), n) = (n, s(n))$ gives $g(s(n)) = n$. So if $s(m) = s(n)$ then $m = g(s(m)) = g(s(n)) = n$.

4.E.22 Induction on n . *Base*: $(\cos \theta + i \sin \theta)^0 = 1 = \cos 0 + i \sin 0$. *Step*: if the formula holds for n then $(\cos \theta + i \sin \theta)^{n+1} = (\cos n\theta + i \sin n\theta)(\cos \theta + i \sin \theta) = (\cos n\theta \cos \theta - \sin n\theta \sin \theta) + i(\sin n\theta \cos \theta + \cos n\theta \sin \theta) = \cos((n+1)\theta) + i \sin((n+1)\theta)$ by the angle addition formulae.

4.E.23 Write $I_n = \int_0^{\pi/2} \sin^n x dx$. (a) Integrate by parts with $u = \sin^{n+1} x$, $dv = \sin x dx$: $I_{n+2} = [-\sin^{n+1} x \cos x]_0^{\pi/2} + (n+1) \int_0^{\pi/2} \sin^n x \cos^2 x dx = 0 + (n+1)(I_n - I_{n+2})$, so $(n+2)I_{n+2} = (n+1)I_n$. (b) Induction: $I_0 = \frac{\pi}{2} = \frac{\pi}{2!} \binom{0}{0}$. Step: $I_{2n+2} = \frac{2n+1}{2n+2} I_{2n} = \frac{2n+1}{2(n+1)} \cdot \frac{\pi}{2^{2n+1}} \binom{2n}{n}$, and $\binom{2n+2}{n+1} = \frac{(2n+2)(2n+1)}{(n+1)^2} \binom{2n}{n} = \frac{2(2n+1)}{n+1} \binom{2n}{n}$ (Theorem 4.2.15), so $\frac{\pi}{2^{2n+3}} \binom{2n+2}{n+1} = \frac{\pi(2n+1)}{2^{2n+2}(n+1)} \binom{2n}{n} = I_{2n+2}$. (c) $I_1 = 1$ and $I_{2n+3} = \frac{2n+2}{2n+3} I_{2n+1}$; by induction $I_{2n+1} = \frac{2^{2n}(n!)^2}{(2n+1)!}$ (base $n = 0$: 1; step: $\frac{2n+2}{2n+3} \cdot \frac{2^{2n}(n!)^2}{(2n+1)!} = \frac{2^{2n+1}(n+1)(n!)^2}{(2n+3)(2n+1)!} = \frac{2^{2n+2}((n+1)!)^2}{(2n+3)!}$, using $(2n+3)! = (2n+3)(2n+2)(2n+1)!$).

F CHAPTER 5

5.1.3 For example: “ $x < y$ ” on $\mathbb{R}$ (homogeneous); “ x is a parent of y ” on the set of all people (homogeneous); “the line L passes through the point P ” from the set of lines in the plane to the set of points (not homogeneous); “ a leaves remainder r on division by 3” from $\mathbb{Z}$ to $[3]$ (not homogeneous).

5.1.35 $(\Rightarrow)$ Let $(a, b) \in \text{Gr}(R)$, so $a R b$. By symmetry $b R a$, and then by antisymmetry $a = b$; so $(a, b) = (a, a) \in \Delta_X$. $(\Leftarrow)$ Suppose $\text{Gr}(R) \subseteq \Delta_X$. Symmetric: if $a R b$ then $(a, b) \in \Delta_X$, so $a = b$, and $b R a$ is the true statement $a R a$. Antisymmetric: if $a R b$ (and $b R a$) then $(a, b) \in \Delta_X$ gives $a = b$. Deduction: if R is also reflexive then $(a, a) \in \text{Gr}(R)$ for all a , i.e. $\Delta_X \subseteq \text{Gr}(R)$; so $\text{Gr}(R) = \Delta_X = \text{Gr}(=)$ (Exercise 5.1.18), and R is the equality relation by Theorem 5.1.14.

5.2.5 Reflexive: $\text{id}_U : U \rightarrow U$ is a bijection (3.2.19), so $U \cong U$. Symmetric: if $f : U \rightarrow V$ is a bijection then $f^{-1} : V \rightarrow U$ is a bijection (3.2.45), so $V \cong U$. Transitive: if $f : U \rightarrow V$ and $g : V \rightarrow W$ are bijections then $g \circ f : U \rightarrow W$ is a bijection (3.2.20).

5.2.10 $a \equiv b \pmod{0}$ iff $0 \mid b - a$ iff $b - a = q \cdot 0 = 0$ for some q iff $a = b$: congruence mod 0 is equality. $a \equiv b \pmod{1}$ always, since $b - a = (b - a) \cdot 1$ is divisible by 1: congruence mod 1 is the discrete relation.

5.2.14 For $r \in \{0, \dots, 9\}$, $[r]_{\approx} = \{n \in \mathbb{N} \mid n \equiv r \pmod{10}\} = \{r, r+10, r+20, \dots\}$, the natural numbers whose last decimal digit is r (Example 5.2.8). Every $n \in \mathbb{N}$ lies in the class of its last digit, so $\mathbb{N}/\approx = \{[0], [1], [2], [3], [4], [5], [6], [7], [8], [9]\}$.

5.2.16 By Example 5.2.15, $[a]_{\sim_f} = f^{-1}[\{f(a)\}]$, which always contains a . $(\Rightarrow)$ If f is injective and $x \in [a]$ then $f(x) = f(a)$, so $x = a$: $[a] = \{a\}$ has a unique element. $(\Leftarrow)$ Suppose every class has a unique element, and let $f(a) = f(b)$. Then $b \in [a]$ and $a \in [a]$, so $a = b$ by uniqueness. Hence f is injective.

5.2.20 (For $n \neq 0$; for $n = 0$ the statement fails, since $[0] = \emptyset$ while $\mathbb{Z}/0\mathbb{Z} = \{\{a\} \mid a \in \mathbb{Z}\}$ by 5.2.10.) *Injective*: let $r, s \in [n]$ with $[r]_n = [s]_n$. Then $r \equiv s \pmod{n}$ (Theorem 5.2.17), so r and s have the same remainder on division by n (5.2.9). But $0 \leq r, s < |n|$, so each is its own remainder ($r = 0 \cdot n + r$); hence $r = s$. *Surjective*: let $[a]_n \in \mathbb{Z}/n\mathbb{Z}$ and let r be the remainder of a , $a = qn + r$ with $r \in [n]$. Then $a - r = qn$, so $r \equiv a \pmod{n}$, and $[r]_n = [a]_n$ by Theorem 5.2.17; hence $[a]_n = i(r)$.

5.E.1 On $X = \{0, 1, 2\}$, give the graph. None: $\{(0, 1), (1, 2)\}$ (no $(0, 0)$; no $(1, 0)$; no $(0, 2)$). Reflexive only: $\Delta_X \cup \{(0, 1), (1, 2)\}$. Symmetric only: $\{(0, 1), (1, 0)\}$ (no $(0, 0)$; transitivity would need $(0, 0)$). Transitive only: $\{(0, 1)\}$ (vacuously transitive). Reflexive and symmetric only: $\Delta_X \cup \{(0, 1), (1, 0), (1, 2), (2, 1)\}$ (no $(0, 2)$). Reflexive and transitive only: $\leq$ on $\mathbb{R}$. Symmetric and transitive only: the empty relation on X (vacuous; not reflexive). All three: $=$.

5.E.2 This is the $(\Rightarrow)$ direction of Exercise 5.1.35 together with 5.1.18: $\text{Gr}(R) \subseteq \Delta_X = \text{Gr}(=)$.

5.E.3 Let $x \in X$. Since R is left-total there is $y \in X$ with $x R y$. By symmetry $y R x$. By transitivity $(x R y \text{ and } y R x) \Rightarrow x R x$. So R is reflexive.

5.E.5 Let S be the transport of $\sim$ along f . *Symmetric, always*: if $c S d$ via a, b ($f(a) = c, f(b) = d, a \sim b$) then $b \sim a$, so $d S c$ via b, a . *Reflexive iff f is surjective*: $c S c$ needs some a with $f(a) = c$ (then $a \sim a$ works),

so reflexivity on all of Y holds exactly when every $c \in Y$ is a value of f . *Transitive: not in general.* Take $X = \{0, 1, 2, 3\}$ with $\sim$ having classes $\{0, 1\}$, $\{2, 3\}$, and $f(0) = c$, $f(1) = f(2) = d$, $f(3) = e$. Then $c S d$ (via $0 \sim 1$) and $d S e$ (via $2 \sim 3$), but $c S e$ would need $0 \sim 3$, which is false. Transitivity does hold under the extra condition “ $f(b) = f(b') \Rightarrow b \sim b'$ ”: if $c S d$ via $a \sim b$ and $d S e$ via $b' \sim a'$ with $f(b) = f(b') = d$, then $a \sim b \sim b' \sim a'$ gives $c S e$. *Conclusion:* S is an equivalence relation on Y when f is surjective and $f(b) = f(b') \Rightarrow b \sim b'$ (for instance when f is a bijection, or when f is the quotient function $q_\sim$); in general only symmetry is guaranteed.

5.E.7 Here $a R b$ or $b R a$ means $b = a + n$ or $a = b + n$, i.e. $b - a = \pm n$. If $x \sim_R y$ via a chain $x = a_0, \dots, a_k = y$, then $y - x = \sum_{i \in [k]} (a_{i+1} - a_i)$ is a sum of k terms each equal to $\pm n$, hence a multiple of n : $x \equiv y \pmod{n}$. Conversely if $y - x = mn$, take the chain $x, x + n, x + 2n, \dots, x + mn = y$ if $m \geq 0$ (each step $a_i R a_{i+1}$) or $x, x - n, \dots, y$ if $m < 0$ (each step $a_{i+1} R a_i$); for $m = 0$ take $k = 0$. So $\sim_R$ and $\equiv \pmod{n}$ relate the same pairs and are equal.

5.E.8 Let $U, V \subseteq X$. Then $U \subseteq X$ and $V \subseteq X$, so the chain U, X, V ($k = 2$) has $U R X$ and $V R X$ (the second step in the reverse direction). Hence $U \sim_R V$.

5.E.9 ($\Leftarrow$) If $x = y$ take the chain of length 0. If $\{x, y\} = \{a, b\}$, the chain a, b or b, a works since $a R b$. ($\Rightarrow$) Let $x = a_0, \dots, a_k = y$ be a chain. Each step has $a_i R a_{i+1}$ or $a_{i+1} R a_i$, and the only related pair is (a, b) , so every step is between a and b . If $k = 0$ then $x = y$. If $k \geq 1$ then $x, y \in \{a, b\}$, so $x = y$ or $\{x, y\} = \{a, b\}$.

5.E.10 Reflexive: the chain x alone ($k = 0$) gives $x \sim_R x$. Symmetric: reverse the chain $(a_k, \dots, a_0)$; each step still satisfies “ R in one direction or the other”. Transitive: concatenate a chain from x to y with a chain from y to z .

5.E.11 If $x R y$ then x, y is a chain with $k = 1$, so $x \sim_R y$.

5.E.12 Let $\approx$ be an equivalence relation with $x R y \Rightarrow x \approx y$, and let $x \sim_R y$ via $x = a_0, \dots, a_k = y$. For each i , $a_i R a_{i+1}$ gives $a_i \approx a_{i+1}$, and $a_{i+1} R a_i$ gives $a_{i+1} \approx a_i$, hence $a_i \approx a_{i+1}$ by symmetry. By transitivity along the chain (Proposition 5.1.42) $a_0 \approx a_k$, i.e. $x \approx y$; for $k = 0$ use reflexivity.

5.E.13 By 5.E.11, $x R y \Rightarrow x \sim_R y$. By 5.E.12 applied with $\approx = R$ (an equivalence relation containing R trivially), $x \sim_R y \Rightarrow x R y$. Same domain, same pairs: $\sim_R = R$ (Axiom 5.1.4).

5.E.14 (a) $\sim$ is $\sim_D$ for the function $D : C^1(\mathbb{R}) \rightarrow (\text{functions } \mathbb{R} \rightarrow \mathbb{R})$, $D(F) = F'$, so it is an equivalence relation by Exercise 5.2.4. (b) $G \in [F]_\sim$ iff $G' = F'$ iff $(G - F)' = 0$ iff $G - F$ is a constant function c (a function on $\mathbb{R}$ with zero derivative is constant), i.e. $G(x) = F(x) + c$ for all x . (c) $\int f(x) dx = F(x) + c$ is a description of the whole equivalence class $[F]_\sim$ of one antiderivative F : the indefinite integral is the $\sim$ -class of antiderivatives, and “ $+c$ ” is exactly the content of (b).

5.E.15 False: $\leq$ on $\mathbb{R}$ is reflexive but $0 \leq 1$ and $1 \not\leq 0$.

5.E.16 True: by Exercise 5.1.35 a reflexive, symmetric, antisymmetric relation on $\mathbb{N}$ is the equality relation, so there is exactly one.

5.E.17 False: on $\{0, 1, 2\}$ the relation with graph $\{(0, 1), (1, 0), (1, 2)\}$ is not symmetric ($1 R 2$ but not $2 R 1$) and not antisymmetric ($0 R 1$, $1 R 0$, $0 \neq 1$).

5.E.19 True: $\emptyset \subseteq \mathbb{N} \times \mathbb{N}$, so by Theorem 5.1.14 it is the graph of a unique relation on $\mathbb{N}$, the empty relation $\emptyset_{\mathbb{N}, \mathbb{N}}$.

5.E.20 True: $\text{Gr}(f) \subseteq X \times X$, so it is the graph of a relation on X by Theorem 5.1.14 (namely R_f , Exercise 5.1.11).

5.E.21 False: for $X = \{0\}$ the empty relation has graph $\emptyset$, but the graph of a function $\{0\} \rightarrow \{0\}$ must contain a pair $(0, y)$ (Theorem 3.1.9). (The discrete relation on $\{0, 1\}$ is another counterexample: $(0, 0)$ and $(0, 1)$ both in the graph.)

Appendix T

One-page proof toolbox

Contents

- A** Reading off the first line of a proof
Standard unfoldings

Planning

Use this page while doing the exercises: find the outermost symbol of what you must prove, copy the first line, unfold the definitions, repeat.

A READING OFF THE FIRST LINE OF A PROOF

goal looks like	first line(s)	last line
$\forall x \in X, p(x)$	Let $x \in X$.	... so $p(x)$. Since x was arbitrary, ...
$\exists x \in X, p(x)$	Put $x = \dots$ (then check $x \in X$).	... so $p(x)$.
$\exists! x$	Existence: as above. Uniqueness: suppose x, y both satisfy p .	... so $x = y$.
$P \Rightarrow Q$	Suppose P . (or: We prove the contrapositive; suppose $\neg Q$.)	... hence Q . (... hence $\neg P$.)
$P \wedge Q$	First we prove P Next we prove Q .	both done.
$P \vee Q$	Suppose $\neg P$.	... hence Q .
$P \Leftrightarrow Q$	$(\Rightarrow)$ Suppose P ... $(\Leftarrow)$ Suppose Q ...	both directions done.
$\neg P$, “there is no”, “irrational”, $X = \emptyset$	Suppose, for a contradiction, that P (that such an object exists; that $a \in X$).	... contradiction. Hence $\neg P$.
$X \subseteq Y$	Let $a \in X$.	... so $a \in Y$.
$X = Y$	$(\subseteq)$ Let $a \in X$... $(\supseteq)$ Let $a \in Y$... or a $\Leftrightarrow$ -chain.	both inclusions done.
f injective	Let $a, b \in X$ with $f(a) = f(b)$.	... so $a = b$.
f surjective	Let $b \in Y$. Put $a = \dots$	... so $f(a) = b$.
$\forall n \geq b, p(n)$	Base case $n = b$: ... Induction step: let $n \geq b$, suppose $p(n)$.	... so $p(n+1)$. By induction ...
R reflexive / symmetric / transitive	Let a . / Let a, b with aRb . / Let a, b, c with aRb, bRc .	aRa / bRa / aRc .
$F([a]) = h(a)$ well-defined	Let $a \sim b$.	... so $h(a) = h(b)$.

Using an assumption: $\exists$ or “ $b \mid a$ ” or “ $b \in f[U]$ ” $\rightarrow$ “Fix k with ...”. $P \vee Q$ or “ $a \in X \cup Y$ ” $\rightarrow$ cases. $P \wedge Q$ or “ $a \in X \cap Y$ ” $\rightarrow$ two facts. $\forall \rightarrow$ plug in what you need. $P \Rightarrow Q$ and $P \rightarrow Q$.

Standard unfoldings

$b \mid a$	$\exists q \in \mathbb{Z}, a = qb$	$a \in \bigcap_{i \in I} X_i$	$\forall i \in I, a \in X_i$
n even / odd	$n = 2k$ / $n = 2k + 1, k \in \mathbb{Z}$	$a \in \bigcup_{i \in I} X_i$	$\exists i \in I, a \in X_i$
$x \in \mathbb{Q}$	$x = \frac{a}{b}, a, b \in \mathbb{Z}, b \neq 0$	$(a, b) \in X \times Y$	$a \in X \wedge b \in Y$
$a \in \{x \in X \mid p(x)\}$	$a \in X \wedge p(a)$	$b \in f[U]$	$\exists a \in U, f(a) = b$
$b \in \{e(x) \mid x \in X\}$	$\exists x \in X, b = e(x)$	$a \in f^{-1}[V]$	$f(a) \in V$
$X \subseteq Y$	$\forall a \in X, a \in Y$	$g \circ f = \text{id}_X$	$\forall a \in X, g(f(a)) = a$
$U \in \mathcal{P}(X)$	$U \subseteq X$	$b \in [a]_{\sim}$	$a \sim b$
$a \in X \cap Y$	$a \in X \wedge a \in Y$	$[a] = [b]$	$a \sim b$ (Lemma)
$a \in X \cup Y$	$a \in X \vee a \in Y$	$a \equiv b \pmod{n}$	$n \mid a - b$
$a \in X \setminus Y$	$a \in X \wedge a \notin Y$		